BSJ
BERKELEY
SCIENTIFIC JOURNAL
THE JOURNAL OF YOUNG SCIENTISTS

PERSPECTIVES

Volume 23, Issue 2

STAFF

Editor-in-Chief
Yana Petri

Managing Editor
Aarohi Bhargava-Shah

Layout Editor
Katherine Liu

Features Editors
Sanika Ganesh
Shivali Baveja

Interviews Editors
Elena Slobodyanyuk
Nikhil Chari

Research & Blog Editors
Whitney Li
Susana Torres-Londono

Publicity Editors
Michelle Verghese
Yizhen Zhang

Features Writers
Jonathan Kuo
Shane Puthuparambil
Nachiket Girish
Mina Nakatani
Matt Lundy
Candy Xu
Saahil Chadha
Ashley Joshi

Interviews Team
Shevya Awasthi
Matt Colbert
Doyel Das
Melanie Russo
Elettra Preosti
Cassidy Hardin
Saumi Shokraee
Michelle Lee
Rosa Lee
Emily Harari

Research & Blog Team
Ethan Ward
Devina Sen
Nicole Xu
Andrea He
Arjun Chandran
Sharon Binoy
Isabelle Chiu
Katheryn Zhou
Andreana Chou
Meera Aravinth
Xandria Ortiz

Layout Interns
Jonathan Kuo
Isabelle Chiu
Melanie Russo

EDITOR'S NOTE

Yana Petri

Aarohi Bhargava-Shah

The path to scientific success is often far less linear than the scientific method suggests. Instead, science is frequently driven by emotion, stimulated by creativity and collaboration, and accelerated by conversation and debate. Inherent to these drivers is the diversity of perspective, which allows us to see things in a new light, or in a manner that we might not have previously considered. Approaching scientific concepts from new vantage points, both from within and outside the traditional boundaries of science, gives us remarkable insight into the progress we have made, while pointing us in new directions moving forward.

In this issue, our staff explores how the newest frontiers of research do exactly that. Organoids, for example, promise to elevate traditional 2D tissue-culture to the third dimension, allowing us to predict the *in vivo* effect of new therapeutics more accurately than ever before. Meanwhile, many engineers are innovating in reverse, shifting away from traditional 3D materials and delving into the two-dimensional world of those that are just a single molecular layer thick. Recognizing the importance of examining scientific discourse in the context of its human impact, our writers consider economic and geopolitical perspectives on climate change, deforestation, and mining. Finally, the spirit of lively interdisciplinary debate is also a central theme in this issue, which notably features a cross-discipline panel discussion on drug policy as well as coverage of the annual Genomics of Energy and Environment Meeting hosted by the Department of Energy Joint Genome Institute.

Listening to diverse perspectives is an important component of ongoing innovation, and BSJ itself is no exception. This semester, our staff had the distinct pleasure of hearing from Caroline Kane, professor emeritus and BSJ faculty advisor; Tinsley Davis, executive director of the National Association of Science Writers; and Sara ElShafie, science storytelling pioneer and science communication strategist; each of whom provided a fresh take on the future of science journalism. With new ideas in hand, BSJ is proud to maintain its place as a stalwart of undergraduate science journalism, while continuously evolving in new directions. We are excited to present yet another captivating issue of the *Berkeley Scientific Journal*.

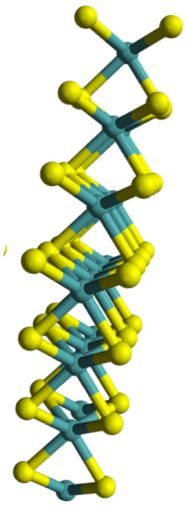
Aarohi Bhargava-Shah
Managing Editor

TABLE OF CONTENTS

Features

4. Organoids: The Future Of Medicine Jonathan Kuo
8. Symmetry Breaking And Asymmetry In The Universe Nachiket Girish
11. Thinking Smaller: The Future Of 2D Materials Mina Nakatani
14. Clinical Oracle: Machine Learning In Medicine Saahil Chadha
18. Renewable Energy Goals In The Face Of Climate Change Matt Lundy
22. On The Brink of Disconnection Candy Xu
26. Satellite Imagery Combats Eco-Destructive Activity In The Amazon Shane Puthuparambil

Interviews



29. Insights From The JGI User Meeting: Using Genomics To Tackle Environmental Problems (Dr. Mary Firestone, Dr. Mary Wildermuth, Dr. Arturo Casadevall, Dr. Dan Jacobson)
Emily Harari, Matt Colbert, Nikhil Chari

34. Biological Insights Into Single-Molecule Imaging (Professor Eric Betzig)
Shevya Awasthi, Doyel Das, Emily Harari, Elettra Preosti, Saumi Shokraee, Elena Slobodyanyuk

40. Drug Use And Policy: A Cross-Discipline Dialogue (Dr. Veronica Miller, David Showalter, Breanna Ford, Dr. Johannes 'Han' De Jong)
Shevya Awasthi, Matt Colbert, Doyel Das, Emily Harari, Cassidy Hardin, Rosa Lee, Michelle Lee, Elettra Preosti, Melanie Russo, and Saumi Shokraee

44. Bridging Science And Buddhism: Toward An Expanded Understanding Of Mind (Professor David Presti)
Shevya Awasthi, Doyel Das, Cassidy Hardin, Rosa Lee, Melanie Russo, Elena Slobodyanyuk

49. Bringing A New Perspective To Gene Regulation (Professor Elçin Ünal)
Matt Colbert, Emily Harari, Michelle Lee, Elettra Preosti, Saumi Shokraee, Nikhil Chari

54. Bioengineering Technology With A Social Responsibility (Professor Luke Lee)
Matt Colbert, Cassidy Hardin, Michelle Lee, Rosa Lee, Melanie Russo, Nikhil Chari

Research

59. Bird Health in California's Central Coast: Interactions Between Agricultural Land Use and Avian Life History Victoria Marie Glynn
68. Introducing an Anti-Terminator Paralog Gene to Induce Production of Natural Products in *Clostridium* Species Alexander Leung

ORGANOIDS: THE FUTURE OF MEDICINE

BY: JONATHAN KUO

On Friday, December 13, 1799, the end of the century brought with it the end of the first president of the United States: George Washington would die in less than 48 hours. His treatment? Molasses, vinegar, butter, sage tea, calomel, various natural salves, and the removal of nearly 2.5 liters of his blood, or what amounted to about half his total blood volume.[1,2] While such a treatment regimen may seem crude by modern medicinal standards, such was the norm since the times of Ancient Greece, when Hippocrates first theorized that the imbalance of four bodily fluids, called "humors," causes disease. Draining an excess humor soon became a common medical treatment, leading to the phenomenon of bloodletting and one basis of nearly 2,000 years of medicine.[3]

▶ ***Image: Cerebral organoids expressing green fluorescent protein.*** *(Dr. Abed Mansour, Gage Lab, The Salk institute).*

c. 400 BC: *Hippocrates began formulating humorism, which was developed further by Galen in c. 100 AD.*

c. 300 BC: *Miasma theory stemmed from humorism; air contaminated with "miasma" was thought to cause disease.*

c. 300 BC: *Aristotle formalized the theory of spontaneous generation, claiming that living organisms could be derived from nonliving matter.*

1665: *Robert Hooke discovered the first microorganisms (fungi).*

1668: *Francesco Redi began disproving spontaneous generation through works with maggots, meat, and nets.*

1673: *Anton van Leeuwenhoek began writing about his improvements to Hooke's microscopes in 50 years of letters to the Royal Society of London.*

1854: *John Snow identified the origin of the London Broad Street cholera outbreak, contributing to his skepticism of miasma theory.*

1862: *Louis Pasteur was recognized for disproving spontaneous generation through his swan-necked flask experiment, leading to the advent of the germ theory of disease.*

1890: *Koch published his four postulates, which set criteria for determining whether a bacterium causes a certain disease.*

1902: *William Castle began studying mice for genetic work; mice are increasingly used for biomedical research.*

1952: *Henrietta Lacks' cells were the first human cell line to be cultured, immortalized in memory as the HeLa cell line.*

1981: *Mina Bissell at Lawrence Berkeley National Laboratory published a landmark paper leading to the rise of 3-D cell culture.*

Figure 1: A brief timeline of the history of disease and disease modeling.

But as humanity's understanding of the world around us—and within us—expanded throughout the centuries, so too did our understanding of disease. Through advancements in microscopy, scientists uncovered a world of microbial organisms concealed within the environment, a world largely invisible to the naked eye. With the discovery of this microbial world, humorism and other theories were replaced by the germ theory of disease. Understanding microorganisms, however, solved only part of the problem. Efforts during the past century sought to reveal how these disease-causing agents interacted with and infected humans, leading to the development of cell culture and animal modeling. Yet the fight did not stop there. As humanity began to make progress against the foes of the microbial worlds, other public health problems emerged. Today, rather than succumbing to microorganisms, people die more frequently from genetic diseases, cancer, and lifestyle disorders, in part due to increased human life expectancy. Fortunately, by serving as miniaturized models of human organs, organoids may prove to be the gold standard for the future of studying disease.

DISEASES IN THE LABORATORY

Currently, diseases are roughly studied in two contexts: cell culture and animal models. In cell culture, diseases are modeled by infecting or mutating cells and growing them in Petri dishes. Infection is used when the disease is caused by some external pathogen, like a virus, while mutation is used when the disease is caused by mistakes in the genetic code, like cancer.[4] Animal models operate under a similar basis, but animals such as mice or rats are grown in cages rather than Petri dishes.[5] Treatments, such as novel pharmaceutical drugs, can then be administered to the diseased cells or animals. Theoretically, if treatments are effective in these contexts, they are more likely to be effective in humans, generating a treatment development pipeline from cells to animals to clinical trials to clinical application.

Practically, this isn't quite the case. Both cellular and animal models suffer from several issues that prevent them from accurately forecasting success in clinical trials. Experimental cell lines are different from human cells because replicating cells introduces errors in the genome, so modified cell lines that replicate continuously are more similar to cancer cells than human cells. Cell cultures are also particularly simplistic compared to the human body because cultured cells usually consist of only one cell type (such as kidney cells) and exist in 2-D.[4] On the other hand, while animals simulate more complex conditions, animals and humans exist on different time scales of life and differ in factors such as their physical size, metabolic rate, and diet, factors that introduce variation scientists are unable to control.[6,7] In short, both cell cultures and animal models do not accurately model the conditions of the human body, so it isn't surprising that treatments affect them in different ways than they affect humans. Organoids, however, may be an improvement over existing non-human models by resolving limitations in both types of models.

HOW TO GROW A BRAIN ORGANOID

Organoids have been defined as "containing several cell types that develop from stem cells or organ progenitors and self-organize through cell sorting and spatially restricted lineage commitment, similar to the process in vivo."[8] Put simply, organoids are grown from stem cells, and have three key features: (a) they contain multiple cell types specific to the organ they originate from, (b) they can replicate a function of an organ, such as neural activity, and (c) they are structurally and spatially similar to an organ.[8,9] But how exactly are organoids grown and applied in research? Tracing the development of a brain organoid from its "birth" may provide insight into this question.

A typical brain organoid, like any other organoid, is first grown from stem cells, specifically pluripotent stem cells (PSCs). These stem cells have the potential to grow into multiple different types of cells, hence the name "pluripotent." These stem cells are seeded into a liquid medium and develop into 3-D spherical embryoid bodies, which are then transported to a firm matrix made of gel. This matrix, which is similar to the human body's extracellular matrix, contains many factors that initiate signalling pathways within the embryoid bodies, causing further development and specialization of the PSCs. Finally, physical agitation of the embryoid bodies causes them to form cavities that resemble ventricles, the neural cavities that contain cerebrospinal fluid (CSF). This culture can be maintained for several months, during which it begins to resemble a brain even more closely: after 6 months, functional synapses can be seen, while after more than 9 months, higher-level organization of features such as active neuronal networks can be observed. Notably, however, whether these networks are similar to those of humans is still uncertain, and further research needs to be done to compare the two.[10,11,12,13]

Cultured brain organoids can model a wide variety of different categories of diseases. The causes of some neurological diseases are easy to identify, like Sandhoff disease, which is caused by a mutation in the HEXB gene leading to a buildup of fats in the central nervous system, eventually leading to death. Scientists from the NIH and the University of Massachusetts have grown two sets of brain organoids; one with the HEXB mutation, the other without. They found that the HEXB-mutated organoids showed a buildup of fats similar to that in Sandhoff disease and were able to monitor how the disease modulated the development and differentiation of the stem cells.[14] Other diseases, such as Alzheimer's Disease (AD), have causes that are more complex. Some features of AD are known, such as a characteristic buildup of amyloid plaques and neurofibrillary tangles. Scientists who have grown brain organoids afflicted with AD have also observed a similar buildup of amyloid plaques and neurofibrillary tangles, demonstrating the potential utility of these organoids in accurately testing therapies against AD.[15] And, of course, cancers, which can affect most organs in the body and are notoriously difficult to treat, have been modeled by organoids. By inducing mutations in stem cells using technologies like CRISPR-Cas9,[16] scientists have created brain organoids with features that resemble cancers like glioblastomas.[17] These models of disease are significantly cheaper to produce, easier to organize, and less labor-intensive than animal models. Perhaps most importantly, these models more accurately resemble humans compared to cell culture and animal models, demonstrating their superiority to current models of disease.[18] And as advanced microscopy methods and big data image analytics develop further, permitting better visualization of organoids, organoids will play a major role in general live tissue 4-D cell biology of the future.[19]

Organoid technology, however, isn't quite the perfect model of diseases and treatments yet. Tissues in the human body are surrounded by an extensive vascular network consisting of arteries, veins, capillaries, and lymphatic vessels that help feed nutrients to cells and remove waste products. The complexity of current vascularization networks created for organoids pale in comparison, and the size of organoids are limited in part because they can not be vascularized well enough.[20] Body tissues are also bathed in extracellular fluids that create particular biochemical microenvironments for

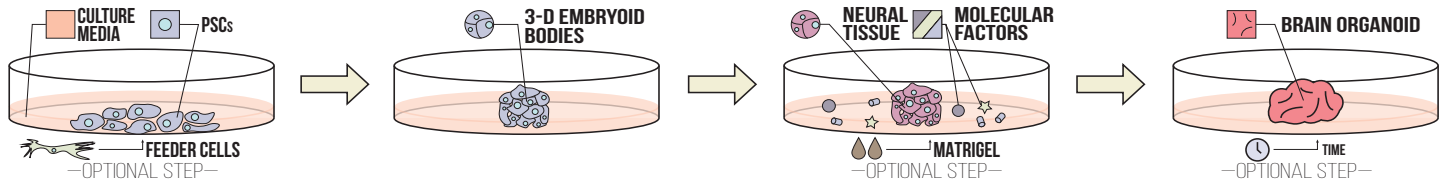


Figure 2: General brain organoid culture protocol. *Growing brain organoids can be generalized to about four steps. First, human pluripotent stem cells (PSCs) are cultured in appropriate culture media. These cells can be cultured with feeder cells that provide the stem cells with additional nutrients, although feeder-free media such as mTeSR1 have been developed. After some time and the addition of certain enzymes that break the physical linkages between the PSCs and the plate they are growing on, the PSCs aggregate into 3-D spheres called embryoid bodies (EBs). Next, the embryoid bodies begin differentiating into neural tissue after specific molecular factors are added. Any remaining steps serve to further refine the structure of the differentiated EBs. For instance, growing the EBs in Matrigel, a gelatinous matrix produced by mouse cancer cells, promotes self-organization of the organoids into layers similar to those in the human brain. Additionally, growing the organoids in bioreactors that periodically agitate the organoids allow more nutrients to perfuse throughout the organoids, increasing the size limits observed in other types of cultures. The protocol described in this paper is only one of many, many ways to culture a brain organoid.*

"By inducing mutations in stem cells using technologies like CRISPR-Cas9, scientists have created brain organoids with features that resemble cancers like glioblastoma."

tissue types, but organoids sometimes lack these environments. The brain, for instance, is surrounded by a bath of CSF, but organoids that create CSF have not been yet been described in literature. And finally, certain developmental cues that guide organ development in humans have not been replicated—such as the growth of tissues in specific directions on specific axes—so organoids do not quite perfectly resemble organ superstructure.[21]

Interestingly enough, organoids can be derived from patient cells. That is, cells can be collected from a patient suffering from a genetic condition and grown into organoids, allowing testing of medicine without significant risk to the patient.[13,14] Organoids could therefore possibly be used in a future of personalized medicine, when treatment regimens are not only created based on specific patient phenotypes, but are also tested and modulated before ever being used on patients. Such applications would reduce some of the inherent risks of medicine—rather than hoping that treatments will be effective for patients, why not just grow some organoids and test on those instead?

Acknowledgements: I would like to acknowledge postdoctoral fellows Dr. Johannes Schöneberg (UC Berkeley) and Dr. Aparna Bhaduri (UCSF) for their detailed feedback and fruitful discussions about organoids and language during the writing process.

REFERENCES

1. Kline, C. L. (n.d.). The Tobias Lear Journal: an account of the death of George Washington. *Historical Society of Pennsylvania.* Retrieved from https://hsp.org/history-online/digital-history-projects/tobias-lear-journal-account-death-george-washington
2. Morens, D. M. (1999). Death of a president. *New England Journal of Medicine, 341*(24), 1845-1850. doi: 10.1056/NEJM199912093412413
3. Greenstone, G. (2010). The history of bloodletting. *British Columbia Medical Journal, 52*(1), 12-14. Retrieved from https://www.bcmj.org/premise/history-bloodletting
4. Ryan, S.-L., Baird, A.-M., Vaz, G., Urquhart, A. J., Senge, M., Richard, D. J., . . . Davies, A. M. (2016). Drug discovery approaches utilizing three-dimensional cell culture. *ASSAY and Drug Development Technologies, 14*(1), 19-28. doi: 10.1089/adt.2015.670
5. Perlman, R. L. (2016). Mouse models of human disease: An evolutinoary perspective. *Evolution, Medicine, and Public Health, 2016*(1), 170-76. doi: 10.1093/emph/eow014
6. Bracken, M. B. (2009). Why animal studies are often poor predictors of human reactions to exposure. *Journal of the Royal Society of Medicine, 102*(3), 120-122. doi: 10.1258/jrsm.2008.08k033
7. Jucker, M. (2010). The benefits and limitations of animal models for translational research in neurodegenerative diseases. *Nature Medicine, 16*(11), 1210. doi:10.1038/nm.2224
8. Fatehullah, A., Tan, S. H., & Barker, N. (2016). Organoids as an *in vitro* model of human development and disease. *Nature Cell Biology, 18*(3), 246-254. doi: 10.1038/ncb3312
9. Lancaster, M. A., & Knoblich, J. A. (2014). Organogenesis in a dish: modeling development and disease using organoid technologies. *Science, 345*(6194). doi: 10.1126/science.1247125
10. Wang, H. (2018). Modeling neurological diseases with human brain organoids. *Frontiers in Synaptic Neuroscience, 10*. doi: 10.3389/fnsyn.2018.00015
11. Quadrato, G., Nguyen, T., Macosko, E. Z., Sherwood, J. L., Yang, S. M., Berger, D., . . . Arlotta, P. (2017). Cell diversity and network dynamics in photosensitive human brain organoids. *Nature, 545*(7652), 48-53. doi: 10.1038/nature22047
12. Lancaster, M. A., Renner, M., Martin, C.-A., Wenzel, D., Bicknell, L. S., Hurles, M. E., . . . Knoblich, J. A. (2013). Cerebral organoids model human brain development and microcephaly. *Nature, 501*(7467), 373-379. doi: 10.1038/nature12517
13. Yin, X., Mead, B. E., Safaee, H., Langer, R., Karp, J. M., & Levy, O. (2016). Stem cell organoid engineering. *Cell Stem Cell, 18*(1), 25-38. https://doi.org/10.1016/j.stem.2015.12.005
14. Allende, M. L., Cook, E. K., Larman, B. C., Nugent, A., Brady, J. M., Golebiowski, D., . . . Proia, R. L. (2018). Cerebral organoids derived from Sandhoff Disease-induced pluripotent stem cells exhibit impaired neurodifferentiation. *Journal of Lipid Research, 59*(3), 550-563. doi: 10.1194/jlr.M081323
15. Gerakis, Y., & Hetz, C. (2019). Brain organoids: a next step for humanized Alzheimer's Disease models? *Molecular Psychiatry, 24*(4), 474-478. doi: 10.1038/s41380-018-0343-7
16. Doudna, J. A., & Charpentier, E. (2014). The new frontier of genome engineering with CRISPR-Cas9. *Science, 346*(6213). doi: 10.1126/science.1258096
17. Ogawa, J., Pao, G. M., Shokhirev, M. N., & Verma, I. M. (2018). Glioblastoma model using human cerebral organoids. *Cell Reports, 23*(4), 1220-1229. doi: 10.1016/j.celrep.2018.03.105
18. Ho, B. X., Pek, N. M. Q., & Soh, B.-S. (2018). Disease modeling using 3-D organoids derived from human induced pluripotent stem cells. *International Journal of Molecular Sciences, 19*(4). doi: 10.3390/ijms19040936
19. Schöneberg, J., Dambournet, D., Liu T-L., Forster, R., Hockemeyer, D., Betzig, E., Drubin D. G. (2018). 4-D cell biology: big data image analytics and lattice light-sheet imaging reveal dynamics of clathrin-mediated endocytosis in stem cell–derived intestinal organoids. *Molecular Biology of the Cell, 29*(24). doi: 10.1091/mbc.E18-06-0375
20. Kelava, I., & Lancaster, M. A. (2016). Dishing out mini-brains: current progress and future prospects in brain organoid research. *Developmental Biology, 420*(2), 199-209. doi: 10.1016/j.ydbio.2016.06.037
21. Shen, H. (2018). Core concept: organoids have opened avenues into investigating numerous diseases. But how well do they mimic the real thing? *Proceedings of the National Academy of Sciences of the United States of America, 115*(14), 3507-3509. doi: 10.1073/pnas.1803647115

SYMMETRY BREAKING AND ASYMMETRY IN THE UNIVERSE

BY NACHIKET GIRISH

The big bang theory postulates that all matter that currently exists in the universe was created at one moment, nearly 14 billion years ago, at the birth of the universe itself. All the fundamental particles we know, from the massive Higgs boson to the miniscule neutrino, were formed in that one moment. This primordial particle soup then interacted with itself and combined in various ways. Out of the primeval chaos emerged the universe we live in today.[1]

However, there is a nagging problem with this narrative: prima facie, the universe should have no reason to discriminate between matter and antimatter. Antimatter has all the exact properties of regular matter, with only the signs of its charges reversed. Why should the universe favor one over the other? And yet, if matter and antimatter had been created in exactly equal amounts, they would have precisely annihilated each other, leaving the universe full of energy from their explosive demise but nothing else—no atoms, no molecules, no stars and galaxies, and no us.[2]

Evidently, the clear symmetry between matter and antimatter, known as C symmetry, should result in a universe devoid of matter. But a quick glance at their surroundings should convince any skeptic that the universe is in fact not empty; moreover, it seems to have a great deal more matter in it than antimatter.

How do we explain this puzzling matter-antimatter asymmetry? The answer to this question lies, surprisingly, in the most fundamental symmetries of nature, leading science to question its long held beliefs about physical symmetry. It is in the universe's departure from perfect symmetry that we may find the answer to one of the most fundamental questions of them all—why do we exist in the first place?

SYMMETRIES IN PHYSICS

"How nice it would be if we could only get through into looking glass house! I'm sure it's got, oh, such beautiful things in it! Let's pretend there's a way of getting through into it, somehow."

Through the Looking-Glass, by Lewis Carroll

"The mirror world of classical physics was, in fact, once a fairly boring place, its monotone enforced by the principle of parity symmetry, which demands that the mirror world be exactly identical to the real one."

The story of matter-antimatter asymmetry begins with one of the most basic symmetries of nature: mirror symmetry. Our macroscopic world is in no way universally mirror symmetric. Most of the objects we see in everyday life would look very different through a mirror. Things

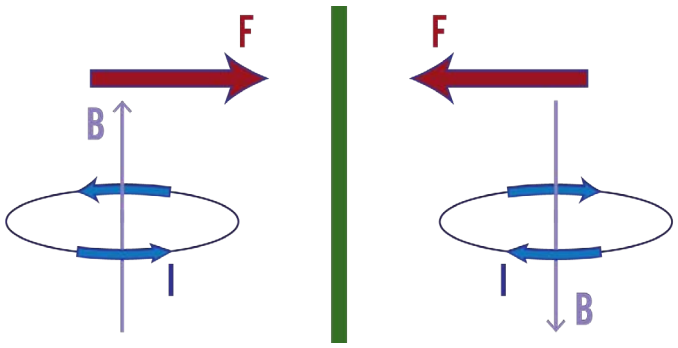


Figure 2: Magnetic field does depend on the left-right orientation of atomic currents. *B denotes the magnetic field created due to current flowing along the direction of I. A left-right reversal of the spin of the currents causes the magnetic field to flip from up to down. However, even in this case, while the magnetic field might gain an up-down flip which would cause its mirror image to appear different from itself, all its observable effects, such as the exerted force, are still flipped in the left-right direction only, preserving parity symmetry.*

get much more interesting (or much less interesting, depending on your point of view) when we analyze the basic phenomena of classical physics, such as fields and particle interactions. The mirror world of classical physics is governed by the principle of parity symmetry, which demands that the mirror world be exactly identical to the real one. To define it more formally, classical physics demands that a system be unchanged when the coordinate axes we use to measure it are reversed, which is called a parity transformation. Granted, what is right for a person is left for their mirror self, but the concept of left and right is itself a relative one, and without arbitrarily deciding on one direction as "right" or "left," there is no way we can differentiate between these two worlds. In this sense, all the laws of physics must work the same way in the mirrorscape as they do in "real" life. The mirror world (which is to say a "parity-transformed world") is no less "real" than the normal one we live in.

At least until the twentieth century, all of the known fundamental forces of nature were understood to be invariant under a mirror reflection. Gravity, for instance, would work the same way in a mirrored world as it does in ours. The force of gravity is entirely described by the relative orientations of the interacting bodies. It does not depend on any absolute sense of a left or right direction, and thus is unaffected by a parity transformation. The universal obedience of physical systems to the principle of parity symmetry would lead physicists to declare parity conservation to be a fundamental property of nature.

SYMMETRY BREAKING

While the principle of parity symmetry is an irontight rule in classical physics, quantum physics presents a whole new story. The development of quantum mechanics was motivated, after all, by a series of particle phenomena which threw a series of monkey wrenches into the exquisitely built structures of classical physics. In 1957, a stunning experiment by experimental physicist Chien-Shiung Wu at the University of Columbia showed that parity symmetry is, in fact, broken in the radioactive decay of particles, which is governed by the then newly-discovered weak nuclear force. Professor Wu and her group aligned the spin (and thus the "atomic currents" and individual magnetic fields) of supercooled cobalt nuclei along an external magnetic field, and found that when the nuclei decayed radioactively, they emitted electrons preferentially opposite to the direction of their magnetic field. Magnetic fields are produced by spinning currents as shown in Figure 2; therefore, if the axes of rotation of the currents were aligned vertically, a purely left-right reflection of the direction of their spin would cause the decay electrons to go up rather than down. The emission of electrons, as an effect of the weak nuclear force, is thus antiparallel to the magnetic field, which is in contrast to the observed effects of the magnetic field itself, which always appear at right angles to the field as discussed in the figure. This is significant, because if we have two mirror reflections of the Wu apparatus, we can tell which one is in the real world and which is in the mirrorscape! Even though observers in both worlds may agree that the current flows from left to right in their reference frames (due to lateral inversion), the left-right reflection does not affect the up-down orientation of the observers and thus there will be a glaring difference between the two worlds in the direction of motion of the electrons. In the real world we only see electrons going downwards for this orientation of currents, therefore the world where the electrons go up must be the mirror world. The weak nuclear force has thus given us a universal standard for left and right, and equally significantly, shattered one of the core beliefs of physics.[3,4]

Having been suddenly deprived of their anchor of parity symmetry, adrift physicists now searched for the correct law for the weak nuclear force. Fortunately, there seemed to be a solution ready at hand—scientists found that if the weak nuclear force acted on matter particles spinning one way, it would act on the corresponding antimatter particle spinning the opposite way. Back to the Wu experiment: if we take the mirror image of the experimental setup and replaced all the atoms with antimatter atoms, then it would again be impossible to differentiate between the two copies of the experiment. This new symmetry was dubbed "CP" symmetry, which is a combination of C symmetry ("charge conjugation" symmetry, which is to say replacing matter with antimatter), and P or parity symmetry.[3]

This solution, however, only held up for so long. In 1964, scientists James Cronin and Val Fitch at the Brookhaven National Laboratory experimentally confirmed a violation of CP symmetry in the weak nuclear force—which won them the Nobel Prize in 1980 and threw the question of symmetry conservation wide open.[5,6]

If the discovery of P-symmetry breaking taught physicists that nature differentiates between left and right, the discovery it led to, that of CP violation, led physicists to a much more surprising conclusion—the

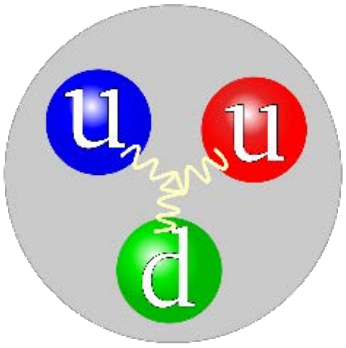


Figure 3: The quark structure of the proton.[13] *The proton is composed of three quarks bound by the strong nuclear force. The Strong CP problem refers to the strange lack of CP symmetry breaking in the strong nuclear force. There is no physical reason why the strong force would not show CP breaking, and its presence would go a long way in solving the matter-antimatter asymmetry puzzle. Observationally, however, not a single case of CP violation in the strong force has been found to this day.*

universe does discriminate between matter and antimatter. There exist nuclear reactions which produce more matter than antimatter, just as there exist reactions which produce particles of one spin more than particles of the opposite spin. This seems like exactly what was needed to solve the matter-antimatter asymmetry problem with the big bang theory. Sure enough, in 1966, physicist Andrei Sakharov proposed a recipe for a matter-antimatter asymmetrical big bang, establishing CP violation as an essential requirement to obtain a non-empty universe. The "Sakharov conditions" allow an eventual matter-antimatter asymmetry to arise out of a big bang where matter and antimatter particles were initially produced equally.[7,8]

Sakharov's conditions are but a set of requirements any theory of the formation of matter (baryogenesis) must fulfill, however. We are far from having solved the problem of matter antimatter asymmetry. In fact, one big problem physicists face today is to explain why there isn't enough asymmetry! For while Sakharov's conditions demand the existence of CP symmetry breaking to explain the existence of a matter dominated universe, none of the physical models we have today can provide enough symmetry breaking to account for the observed ratio between the amount of matter and antimatter.[9] This is known as the "strong CP problem." Finding new sources of CP symmetry violation is in fact a very active research field, with a CP breaking interaction being discovered in new particles as recently as 2019, which could possibly hint at a solution to this question. Matter antimatter asymmetry still remains a mystery though, and is considered one of the biggest unanswered questions of physics today.[10]

PERFECTION IN IMPERFECTION

Ancient natural philosophers were convinced that the heavens were absolute in their perfection. As long ago as the fourth century BC, Plato insisted that the celestial bodies were made in the most perfect and uniform shape.[11] Through the works of Kepler and Galileo, however, it was shown that the universe did not conform to humanity's conception of perfection. Planetary orbits were ellipses, not perfect circles. The other planets of the solar system had craters, "blemishes and scars," the same way the earth did.[12]

Perhaps our search for symmetries in the physical world is but an attempt to create order out of the apparent chaos we see around us. And yet, the most fundamental studies of reality have taught us that it is from the lack of symmetry, from the tiniest imperfection, that all that we see around us came to be. It is because the universe is just imperfect enough, that it is perfect for our existence.

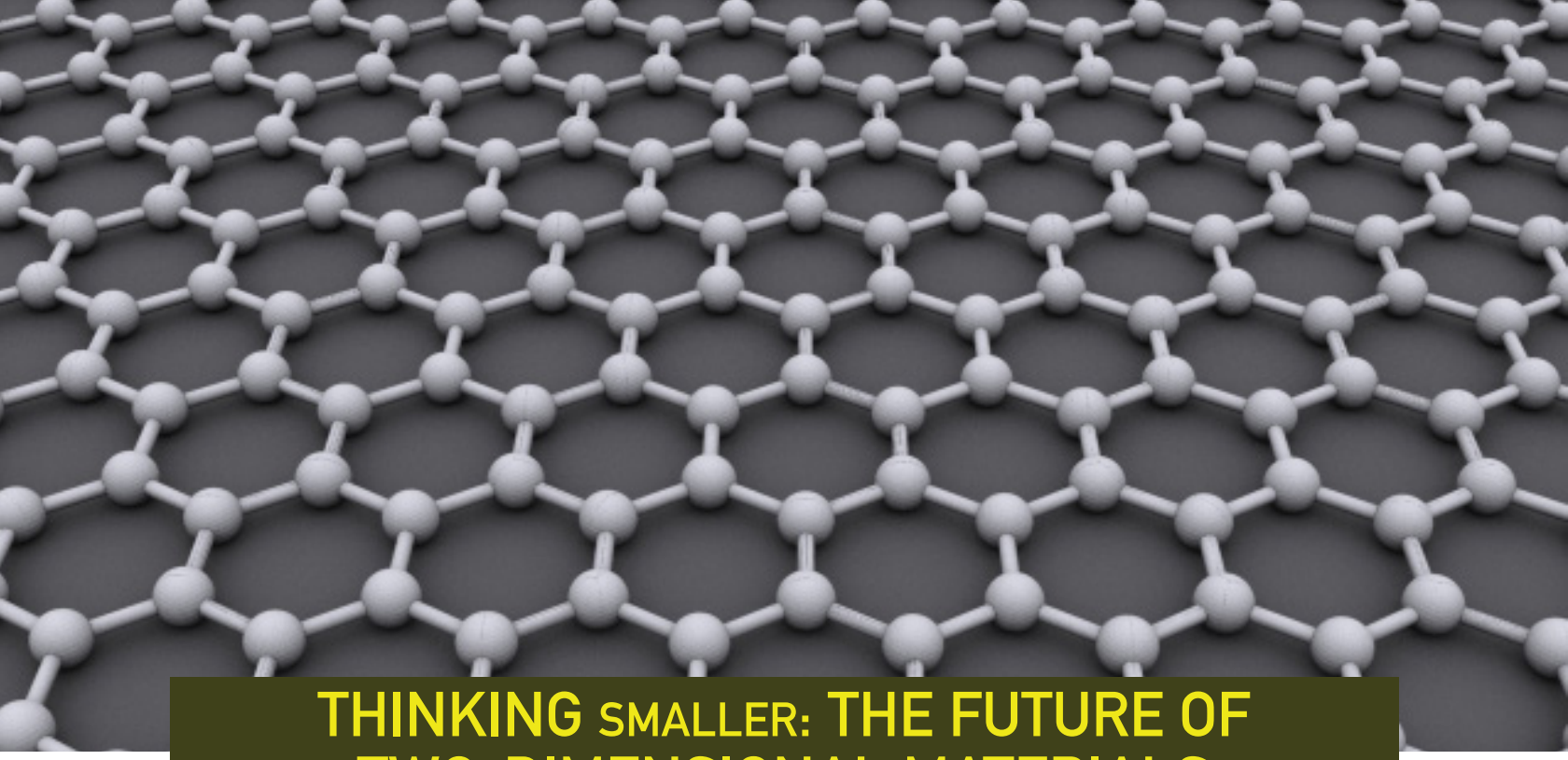
Acknowledgements: I would like to acknowledge Kishore Patra (PhD candidate in Astrophysics at UC Berkeley) for his input and valuable feedback during the writing process.

REFERENCES

1. The universe's primordial soup flowing at CERN. (2016, February 9). *Niels Bohr Institute.* Retrieved from https://phys.org/news/2016-02-universe-primordial-soup-cern.html
2. Garbrecht, B. (2018). Why is there more matter than antimatter? Calculational methods for leptogenesis and electroweak baryogenesis. arXiv preprint arXiv:1812.02651
3. Feynman, R., Leighton, R., & Sands, M. (1963). *The Feynman lectures on physics: volume 1.* (2nd ed.). Retrieved from http://www.feynmanlectures.caltech.edu/I_52.html
4. Garvey, G. T., & Seestrom, S. J. (1993). Parity violation in nuclear physics: signature of the weak force. *Los Alamos Science, 21.*
5. Christenson, J. H., Cronin, J. W., Fitch, V. L., Turlay, R. (1964). Evidence for the 2pi decay of the K20 meson. *Physical Review Letters, 13*(4), 138. doi: 10.1103/PhysRevLett.13.138
6. Gardner, S., & Shi, J. (2019). Patterns of CP violation from mirror symmetry breaking in the eta —>pi+pi-pi0 Dalitz plot. arXiv preprint arXiv:1903.11617
7. Balazs, C. (2014). Baryogenesis: A small review of the big picture. arXiv preprint arXiv:1411.3398.
8. Steigman, G., & Scherrer, R. J. (2018). Is The Universal Matter-Antimatter Asymmetry Fine Tuned? arXiv preprint arXiv:1801.10059
9. Mavromatos, N. E., & Sarkar, S. (2018). Spontaneous CPT Violation and Quantum Anomalies in a Model for Matter-Antimatter Asymmetry in the Cosmos. *Universe, 5*(1), 5. doi: 10.3390/universe5010005
10. LHCb collaboration. (2019). Observation of CP violation in charm decays. arXiv preprint arXiv:1903.08726
11. Zeyl, D., & Sattler, B. (2005). Plato's Timaeus. *The Stanford Encyclopedia of Philosophy*. Retrieved from https://plato.stanford.edu/archives/sum2019/entries/plato-timaeus/
12. Pasachoff, J. M., & Filippenko, A. (2013). *The Cosmos.* (4th ed.).Berkeley, CA: Cambridge University Press.

IMAGE REFERENCES

13. Rybak, J. (2018, January). *The quark structure of the proton* [digital image]. Retrieved from https://commons.wikimedia.org/wiki/File:Proton_quark_structure.svg



THINKING SMALLER: THE FUTURE OF TWO-DIMENSIONAL MATERIALS

BY MINA NAKATANI

Los Angeles, Chicago, New York. Huge metropolitan cities with skylines defined by towering skyscrapers and blinking lights that stretch high above the heads of those walking the streets. People crane their necks to see the literal heights of human accomplishment, because building higher in three-dimensional space—overcoming the limits of gravity—has long been an incredible feat of engineering and technology. But what if the path to advancement exists in two dimensions rather than three?

The field of two-dimensional materials is a fairly new one, focusing on materials that are only a single molecular layer thick. These materials are not like the physical matter that people come into contact with in daily life, which typically consist of many layers of atoms stacked on top of each other, forming a three-dimensional object. Instead, the structure of two-dimensional materials is a bit like that of sticky notes: if a three-dimensional material like graphite is like the block of notes purchased at a store, graphene is like a single note pulled off the block. Like the flexible single sheet compared to the rigid block of sticky notes, two-dimensional materials have different properties—electrical, chemical, and physical—from their three-dimensional counterparts, giving them enormous potential, despite the challenges they still need to overcome.

One such challenge for two-dimensional materials arises largely from the recency of their development. In the field of electronics, their structure is often incompatible with the shapes and structures used for three-dimensional materials. The process of conforming them to a certain geometry would most likely damage such thin sheets, ruining their electronic functions in the process.[1] Beyond that, recent studies show that, in terms of performance, two-dimensional materials such as molybdenum disulfide (MoS_2) can compete with silicon in electronic devices, but only by utilizing a delicate fabrication process involving the use of nanometer-wide wires in order to achieve particular geometries.[2] This need for structural changes also complicates the issue of synthesizing two-dimensional materials. Though it is true that merely peeling single-layers off of graphite using tape—a process first used in the early 2000s and later involved in Nobel Prize-winning research in 2010—does yield graphene, the shape and presence of defects cannot be reliably controlled.[3] For this reason, working

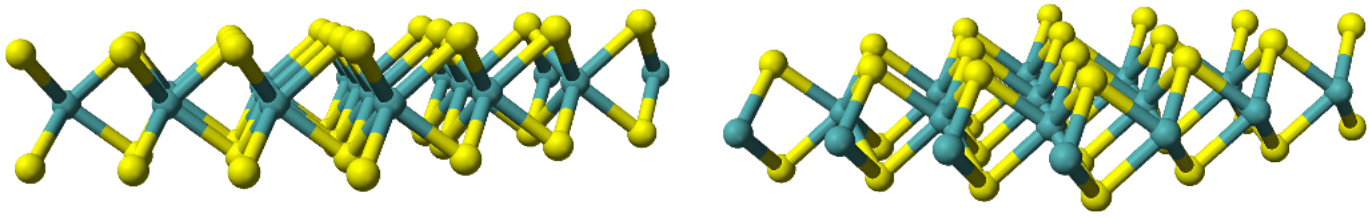


Figure 1: Structure of Molybdenum Disulfide (MoS_2). *Molybdenum disulfide is one type of two-dimensional material which exhibits unique electrical properties, allowing it to possibly be utilized in a number of different applications, from electronics to gas sensors.*

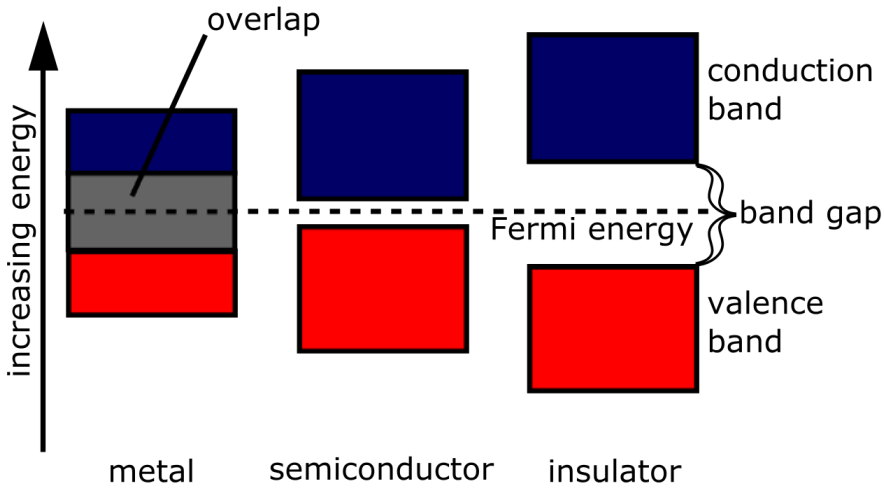


Figure 2: Band Gaps.[12] *Electricity flows through a material via movement of electrons from the valence band (red) to the conduction band (blue). Insulators have a big gap which electrons cannot cross—no electricity can flow. Metals have no gap, making them conductors. Semiconductors have a small gap that electrons can cross, but not as easily as in metals.*

with two-dimensional materials is more complicated and time-consuming. Creating a two-inch sheet of graphene, a relatively simple structure consisting only of carbon atoms arranged in hexagons, under controlled conditions can take upwards of seven hours. Creating more complex two-dimensional materials takes greater time and control over environmental conditions, no match for the relatively easy industrial production of silicon-based devices.[4]

Nonetheless, two-dimensional materials offer unique benefits that three-dimensional materials cannot. Materials such as MoS_2 exhibit properties of semiconductors; thus, they are similar to silicon in that their band gaps are fairly small (Fig. 2).[5] These properties imply that materials such as MoS_2 are theoretically capable of replacing silicon (Fig. 3) while offering even greater utility. As a two-dimensional structure, MoS_2 is able to withstand deformation, capable of bending and stretching without breaking. MoS_2 is flexible in ways that three-dimensional silicon is not.[5] This phenomenon is much like the way in which paper can be folded without tearing. As such, two-dimensional materials are attractive not only for the creation of stretchable and wearable electronics,[6] but also for their tunability, or the ability of a given material to have slightly different properties depending upon the intended use. MoS_2 in particular does have a bandgap indicative of a semiconductor, but the actual size of that gap is dependent on the strain experienced by the material. Therefore, physically stretching the material changes how well it is able to conduct electricity, and thus allows the material to be adapted to a variety of different needs. In fact, sufficiently stretching a layer of MoS_2 forces it to conduct electricity as if it were a metal, a versatility that silicon cannot match.[5]

"By running energy through the material and monitoring conductivity of only two to four layers of MoS_2, NO gas can be detected even at low concentrations of 0.8 ppm, less than one ten-thousandth of a percent."

Adjusting the chemical structure of these materials—by adding in other molecules or combining multiple kinds of two-dimensional layers—also allows for further control of their properties. Again, MoS_2 exhibits this ability well. By binding different functional groups to the surface—sticking new molecules onto the sheets—or changing the actual geometry of the bonds in MoS_2 itself, the size of the bandgap is theorized to change by 25%.[7] Though that may seem small, given the size of the electron itself, a change of 25% is significant. This change allows MoS_2 to fulfill a wide range of needs in electronics.

The tunability that comes from the sensitivity of MoS_2 to the presence of other molecules also allows it to function as a low-energy, highly effective gas sensor.[8] Considering that the conduction of MoS_2 should change with the introduction of other molecules, researchers found this principle to remain true with nitric oxide (NO) gas, a chemical linked to acid rain and the depletion of the ozone layer. By running energy through a MoS_2-based gas sensor and monitoring its conductivity, NO gas could be detected even at low concentrations of 0.8 ppm, or less than one ten-thousandth of a percent.[8] Furthermore, the unique tunability of two-dimensional materials also plays a part in their performance as gas sensors. By merely changing the thickness of the MoS_2 within the sensor from 18 nanometers to 2 nanometers—about the width of a strand of DNA—the sensor's sensitivity to nitrogen dioxide (NO_2) gas increased from 0.8% to 7%.[9] Beyond that, by binding platinum ions to the surface of the sensor, its sensitivity sensor is tripled, allowing it to detect NO_2 down to a concentration of 2 ppb, or 0.0000002 of a percent.[9]

"In fact, sufficiently stretching a layer of MoS_2 forces it to conduct electricity as if it were a metal, a versatility that silicon cannot match."

The flexibility of two-dimensional materials and their ability to respond to a number of different chemical environments also allows these gas sensors to detect both air pollutants and organic molecules. Namely, certain molecules such as toluene, ethanol, and acetone, are known markers of lung cancer, existing in the

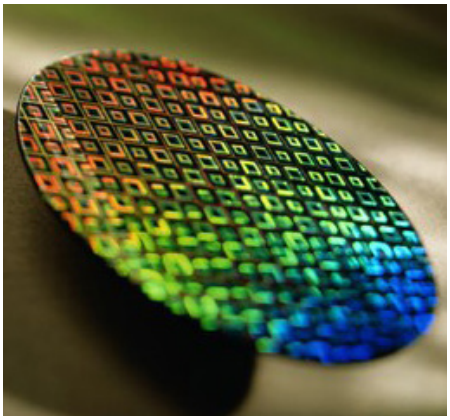


Figure 3: Silicon Computer Chip. *Silicon is the most widely used semiconductor, used to produce silicon chips for computers. As a semiconductor, it exhibits a small band gap, allowing it to be turned "on and off," acting as either a pure conductor or insulator—or something in between—depending on its environment. This variability in conductivity is the reason semiconductors are so useful to electronic devices.*

breath of diseased patients.[10] MoS_2-based sensors can detect these molecules much like they can detect NO or NO_2 gas; by binding a molecule called mercaptoundecanoic acid to the surface of MoS_2, the sensor can give unique responses to each different molecule, doing so to a limit of 1 ppm.[10] As such, this two-dimensional gas sensor can be used for more than pollution detection, also functioning as a simple breath analyzer for medical diagnosis.[10]

Much of modern technological advancement is exemplified by the huge structures of metropolitan cities. But some of the most remarkable feats of technology have come from making things smaller. Computers that fit in backpacks and phones that slide into back pockets are the most well-known examples, and two-dimensional materials are merely an extension of that same concept. They present the idea of thinking in new dimensions, looking for answers in places previously unexplored. The way to advancement—in all fields, not only technology—comes from new perspectives and courage to change the way things have always been done.

Acknowledgements: I would like to acknowledge Dr. Shan Wu (postdoctoral researcher in Professor Robert Birgeneau's lab at UC Berkeley) for offering her expertise on this topic and advising me on the content of my article.

REFERENCES

1. Duan, X., Wang, C., Pan, A., Yu, R., & Duan, X. (2016). ChemInform Abstract: Two-Dimensional Transition Metal Dichalcogenides as Atomically Thin Semiconductors: Opportunities and Challenges. *ChemInform, 47*(6). doi:10.1002/chin.201606232
2. Liu, Y. et al. (2016). Pushing the Performance Limit of Sub-100 nm Molybdenum Disulfide Transistors. *Nano Letters, 16*(10), 6337-6342. doi:10.1021/acs.nanolett.6b02713
3. Dresselhaus, M. S., & Araujo, P. T. (2010). Perspectives on the 2010 Nobel Prize in Physics for Graphene. *ACS Nano, 4*(11), 6297-6302. doi:10.1021/nn1029789
4. Yu, J., Li, J., Zhang, W., & Chang, H. (2016). ChemInform Abstract: Synthesis of High Quality Two-Dimensional Materials via Chemical Vapor Deposition. *ChemInform, 47*(2). doi:10.1002/chin.201602214
5. Roldán, R., Castellanos-Gomez, A., Cappelluti, E., & Guinea, F. (2015). Strain engineering in semiconducting two-dimensional crystals. *Journal of Physics: Condensed Matter, 27*(31), 313201. doi:10.1088/0953-8984/27/31/313201
6. Kim, S. J., Choi, K., Lee, B., Kim, Y., & Hong, B. H. (2015). Materials for Flexible, Stretchable Electronics: Graphene and 2D Materials. *Annual Review of Materials Research, 45*(1), 63-84. doi:10.1146/annurev-matsci-070214-020901
7. Tang, Q., & Jiang, D. (2015). Stabilization and Band-Gap Tuning of the 1T-MoS2 Monolayer by Covalent Functionalization. *Chemistry of Materials, 27*(10), 3743-3748. doi:10.1021/acs.chemmater.5b00986
8. Li, H., et al. (2012). Layered Nanomaterials: Fabrication of Single- and Multilayer MoS2 Film-Based Field-Effect Transistors for Sensing NO at Room Temperature (Small 1/2012). *Small, 8*(1), 2-2. doi:10.1002/smll.201290004
9. He, Q., et al. (2012). Fabrication of Flexible MoS2Thin-Film Transistor Arrays for Practical Gas-Sensing Applications. *Small, 8*(19), 2994-2999. doi:10.1002/smll.201201224
10. Kim, J., Yoo, H., Choi, H. O., & Jung, H. (2014). Tunable Volatile Organic Compounds Sensor by Using Thiolated Ligand Conjugation on MoS2. *Nano Letters, 14*(10), 5941-5947. doi:10.1021/nl502906a

Figure 4: Nitrogen dioxide (NO_2) gas.[13] *MoS_2-based gas sensors are capable of detecting gases such as NO_2 and toluene, even at low concentrations.*

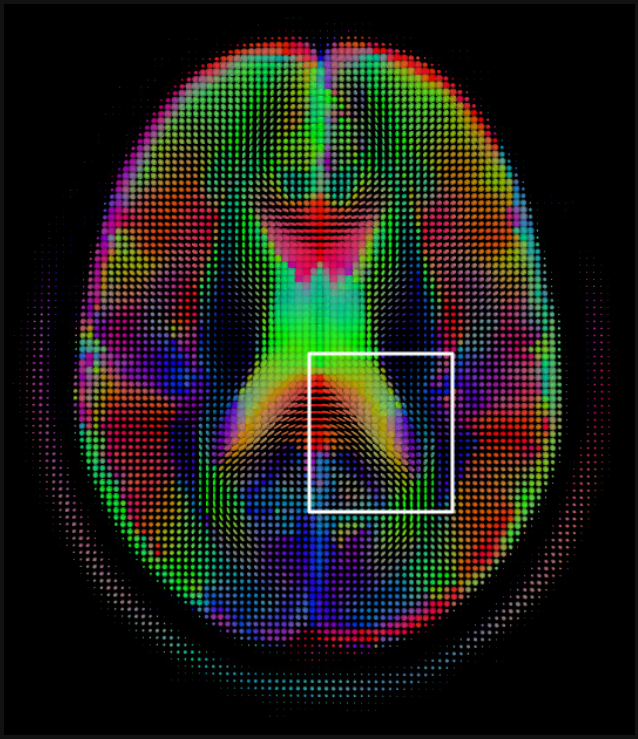
IMAGE REFERENCES

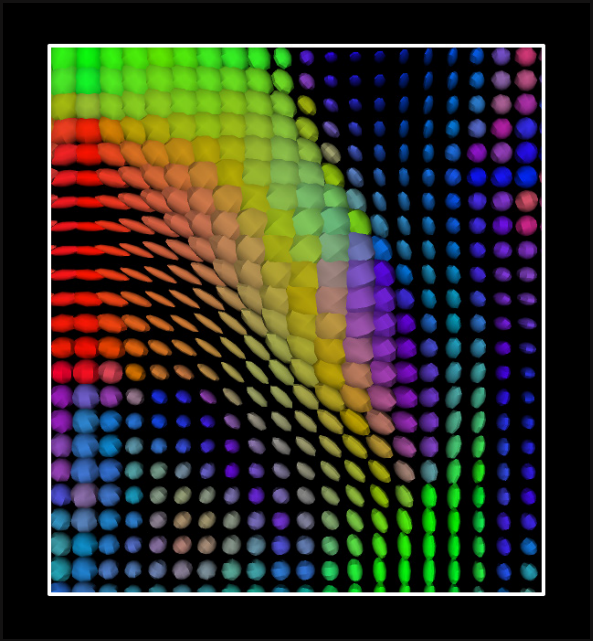
11. Banner image: AlexanderAlUS. (2010). *The ideal crystalline structure of graphene is a hexagonal grid* [digital image]. Retrieved from https://commons.wikimedia.org/wiki/File:Graphen.jpg
12. Inductiveload. (2006). *A comparison of the band gaps of metals, insulators and semiconductors* [digital image]. Retrieved from https://commons.wikimedia.org/wiki/File:Band_gap_comparison.svg
13. Eframgoldberg. (2013). *Nitrogen dioxide at different temperatures* [digital image]. Retrieved from https://commons.wikimedia.org/wiki/File:Nitrogen_dioxide_at_different_temperatures.jpg

CLINICAL ORACLE:

MACHINE LEARNING IN MEDICINE

BY: SAAHIL CHADHA





Radiology is the glue that holds every hospital together. Without X-rays, CT scans, and MRIs, it would be very difficult to glean much more than a surface-level understanding of a patient case. We put our faith in radiologists to correctly interpret our medical images so that we know what is wrong with our bodies, and we expect them to be right.

Contrary to what one would hope, radiologist errors are all too common, an estimated day-to-day rate of 3-5%.[1] And these mistakes can be very costly. Of course, misdiagnosis is most damaging to patients—late identification of a disease like cancer can be fatal. However, misdiagnosis can also cost radiologists and their hospitals through malpractice claims. In fact, radiologists are involved in a disproportionately large number of malpractice suits. Despite being the eighth-largest group of American physicians, they are the fourth highest in terms of cases closed against them.[2] Moreover, the most common cause of these suits is misdiagnosis, especially of breast cancer.[3] It's clear that misdiagnosis hurts all parties involved: the hospital, the physician, and most importantly, the patient. But what can be done?

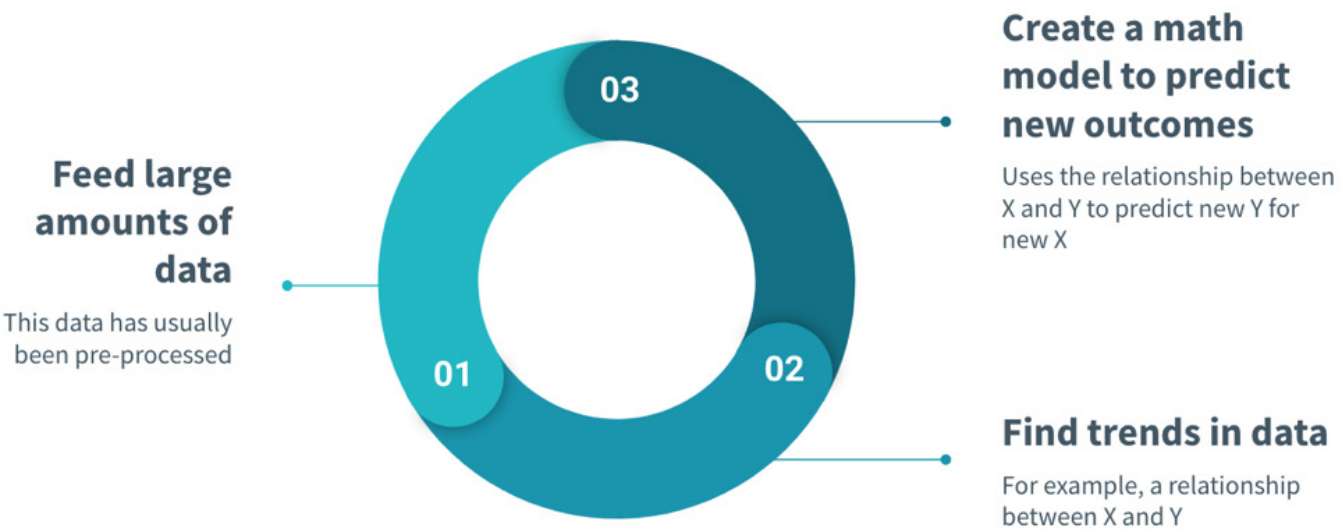


Figure 1: A general pipeline for creating an ML model. *This involves three steps: (1) pre-processed training data is fed into the ML algorithm with input features (e.g. medical images) and their associated output classes (e.g. malignant or benign), (2) the ML algorithm finds trends between the inputs and outputs, and (3) a model is created to predict the output classes for new inputs.*[9]

The field of radiology is steadily marching towards improved diagnoses, largely due to automation. In 1957, the automated film-processor substantially increased radiologists' efficacy.[4] By automating the image development process, X-ray technicians were, both literally and figuratively, brought out of the darkroom and into the light. This eliminated a tedious, mechanical task that had previously been a large part of the radiologists' daily lives, which allowed them to commit more time to interpreting scans. Additionally, this success represented the inception of a larger trend towards increased automation and precision in radiology. Modern developments take this trend a step further.

> "Studies show that receiving a second opinion on medical image screening can significantly increase the accuracy of readings."

The cutting edge of research includes applying the computer science field of machine learning (ML) to medical images to make diagnoses. A study in the journal *Radiology* showed that receiving a second opinion on medical image screening can significantly increase the accuracy of readings.[5] In fact, this study highlighted breast cancer as an area where a second reader is particularly helpful, which is important because breast cancer misdiagnosis is the most common cause of malpractice suits against radiologists. ML models use sophisticated pattern-detection to identify irregularities in medical images and then create highly accurate diagnoses. These reports could act as second opinions, supplementary to a primary radiologist. However, in order to evaluate the current state of ML within radiology, it is necessary to first develop a base-level understanding of the discipline.

THE HISTORY OF ML

The term "machine learning" was coined in 1959 by Arthur Samuel.[6] It is the subset of artificial intelligence that seeks to answer the question: "How can computers learn to solve problems without being explicitly programmed to do so?" Samuel pioneered the use of machine learning in programming a computer to play checkers. Not only was his device capable of comparing choices to pick the one that would bring it closest to winning, but, by remembering every position it had encountered before, the computer was able to learn and improve itself over time. Eventually, Samuel's computer was able to put up a good fight against amateur checker players.[7] This was revolutionary. Computers no longer acted simply as mindless calculators that used rules written into their code to perform predictable operations. Rather, these machines were now able to rival humans in matters of intellect through their own form of complex processing.

Today, ML has grown and branched off in a plethora of different directions. Impressively, ML has become ubiquitous in our daily lives.[8] Examples include autocomplete while texting, targeted advertisements on Amazon, and custom Google search results. Each of these instances of ML seeks to predict something about us: what we're going to type, what we want to buy, and what we want to know. This idea of *prediction*, creat-

ing actionable insights from large amounts of data, lies at the heart of ML. Though ML has mostly been used to personalize the customer and company relationship, its opportunities for clinical application are similarly abundant.

ML AND MEDICAL IMAGING

ML could define the future of medicine.[10] Since the price of data collection has plummeted in recent years, hospitals and insurance companies have jumped on the opportunity to gather and store huge amounts of patient information. However, much of this information remains unused. Researchers are actively investigating how to harness this information for good, and one of the most notable achievements in this area has been with breast cancer detection.

Computer-aided diagnosis (CAD) from medical images follows a two-step process. First, relevant high-level information is extracted from images; then, the extracted information is run through a previously-made ML model that outputs a prediction of which one of many classes the inputted features belong to.[11]

For example, microcalcifications (MCs) are small calcium deposits that show up brightly on mammograms, and clustered MCs are good predictors of breast cancer.[12] Thus, one possible CAD pipeline for this case would be first identifying the relative locations, sizes, brightness, and shapes of MCs, and then using that information to predict whether cancer is present.

Because ML models are able to find patterns across a vast array of different images and cases, they are much more accurate than their human counterparts. In fact, Enlictis, a startup applying deep learning to medicine, developed a tool to identify malignant lung tumors.[13] When the accuracy of diagnosis was compared to that of three expert radiologists, the company's model outperformed humans by 50%. Additionally, in an NPR interview, UCSF radiologists in training worried that "they could be replaced by machines" because "computers are awfully good at seeing patterns."[14] However, we should view ML as a tool that helps radiologists do their jobs better rather than another machine that is going to take jobs away from hard-working Americans. This is where content-based image retrieval (CBIR) comes into play.[11]

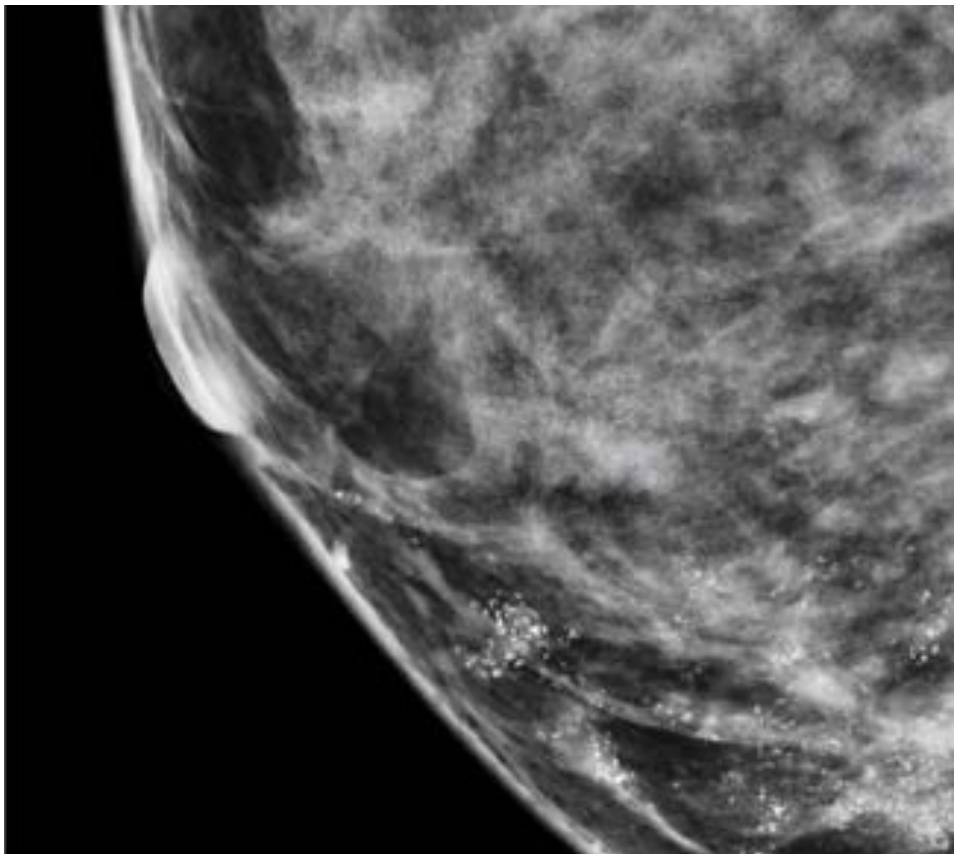


Figure 2: Microcalcifications in a breast scan.[15] *Microcalcifications are seen in this scan as the white dots in the bottom right of the image. They are a reliable early indicator of breast cancer, and radiologists usually decide whether further tests are needed based on their size, density, and distribution.*

Conventionally, ML models don't give explanations for their decisions. These models take in a patient's scans and other relevant information and output a percent likelihood value of malignancy. As a second opinion, these models are only useful if they are able to justify their outputs. CBIR bridges this gap by acting like Google's "search by image" function. [11] In addition to presenting a diagnosis, a system that includes CBIR returns relevant training data most similar to the case being considered. This function greatly improves the model's utility and allows for its use by radiologists in a clinical setting.

Misdiagnosis is a critical issue in radiology. It costs patients tremendous amounts of pain and suffering and costs physicians monetarily in the form of malpractice suits. ML presents a solution to this problem. By gleaning insights from more scans than any one physician could consider in a lifetime, these models are able to be more accurate and powerful than their human counterparts. Because of this potential as well as their novelty, many physicians have been reluctant to adopt them into their daily rou-

> "When the accuracy of ML diagnosis was compared to that of three expert radiologists, the company's model outperformed humans by 50%."

Figure 3: View of Moscow, Russia. *This photograph was edited using Deep Dream Generator (https://deepdreamgenerator.com/)—a computer vision program that allows us to see what a deep neural network is seeing when it is looking at a given image.*

tines. But, if integrated into medical practice, ML does not have to replace jobs. On the contrary, radiologists would be able to increase their relevance by specializing in cases that ML cannot effectively tackle. Thus, the new technology represents a major step in automation and has the potential to revolutionize medicine. By taking advantage of ML as a tool, radiologists can assure that patients receive the best care.

Acknowledgements: I would like to acknowledge Nathan Yan Cheng (computational biologist at the Broad Institute) for reviewing this article.

REFERENCES

1. Brady, A. P. (2017). Error and discrepancy in radiology: inevitable or avoidable?. *Insights into imaging, 8*(1), 171-182. doi: 10.1007/s13244-016-0534-1
2. Whang, J. S., Baker, S. R., Patel, R., Luk, L., & Castro III, A. (2013). The causes of medical malpractice suits against radiologists in the United States. *Radiology, 266*(2), 548-554. doi: 10.1148/radiol.12111119
3. Hamer, M. M., Morlock, F., Foley, H. T., & Ros, P. R. (1987). Medical malpractice in diagnostic radiology: claims, compensation, and patient injury. *Radiology*, 164(1), 263-266. doi: 10.1148/radiology.164.1.3588916
4. Demy, N. G., & Catlin, K. G. (1969). Automation in Diagnostic Radiology–A Critique. *Radiology, 93*(3), 698-701. doi: 10.1148/93.3.698
5. Coolen, A. M. et al. (2018). Impact of the second reader on screening outcome at blinded double reading of digital screening mammograms. *British Journal of Cancer*, 119(4), 503. doi: 10.1038/s41416-018-0195-6
6. Koza, J. R., Bennett, F. H., Andre, D., & Keane, M. A. (1996). Automated design of both the topology and sizing of analog electrical circuits using genetic programming. In *Artificial Intelligence in Design'96* (pp. 151-170). Dordrecht, Netherlands: Springer. doi: 10.1007/978-94-009-0279-4_9
7. Schaeffer, Jonathan. "One Jump Ahead: Challenging Human Supremacy in Checkers." *ICGA Journal* 20, no. 2 (1997): 93-93. doi: 10.1609/aimag.v20i1
8. Mohri, M., Rostamizadeh, A., & Talwalkar, A. (2018). Introduction. In Foundations of machine learning (pp. 3). Boston, MA: MIT press.
9. Maglogiannis, I. G. (2007). *Emerging artificial intelligence applications in computer engineering: real word ai systems with applications in ehealth, hci, information retrieval and pervasive technologies.* (Vol. 160). Ios Press.
10. Kononenko, I. (2001). Machine learning for medical diagnosis: history, state of the art and perspective. *Artificial Intelligence in medicine, 23*(1), 89-109. doi: 10.1016/S0933-3657(01)00077-X
11. Wernick, M. N., Yang, Y., Brankov, J. G., Yourganov, G., & Strother, S. C. (2010). Machine learning in medical imaging. *IEEE signal processing magazine, 27*(4), 25-38. doi: 10.1148/rg.2017160130
12. Nalawade, Y. V. (2009). Evaluation of breast calcifications. *Indian Journal of Radiology & Imaging, 19*(4). doi: 10.4103/0971-3026.57208
13. Automation and anxiety. (2016, June 25). *The Economist*. Retrieved from https://www.economist.com/special-report/2016/06/25/automation-and-anxiety
14. Silverman, L. (2017, September 04). Scanning The Future, Radiologists See Their Jobs At Risk. *NPR.* Retrieved from https://www.npr.org/sections/alltechconsidered/2017/09/04/547882005/scanning-the-future-radiologists-see-their-jobs-at-risk

IMAGE REFERENCES

15. (n.d.). *Microcalcifications in breast cancer* [digital image]. Retrieved from https://www.news-medical.net/health/Microcalcifications-in-Breast-Cancer.aspx

BY MATT LUNDY

RENEWABLE ENERGY GOALS IN THE FACE OF CLIMATE CHANGE

To address the looming threat of climate change along with other environmental and political problems surrounding energy, the world is beginning to turn towards clean and renewable energy sources. Most seem to agree that the transition from depletable fossil fuels to renewable energy should eventually be made; the only question is that of immediacy.[1] Europe tends toward an answer of "Now!" with its Renewable Energy Directive.[2] The US gives a more reluctant answer with its recent surge in oil and natural gas development, paired with a vocal hesitation to be held to higher standards than large emerging economies like China.[3, 4] But the question of whether or not the US should commit to clean energy while other countries do not assumes a lethargy on part of the latter, and such an assumption is not clearly justified without a comprehensive look. So, let's take a look at China and find out where it really stands both in terms of current emissions and in adoption of clean energy sources.

The starting point to this undertaking is a look at the statistics of China's energy usage and carbon emissions and how they compare to those of the US. Since 2000, China has roughly tripled its energy consumption, accompanied by a similarly meteoric increase in GDP. This has put China's current energy consumption slightly above the USA's with a CO_2 emission rate of almost double that of the USA (Fig. 1). While this data appears to favor the USA in terms of the ratio of carbon emission to energy consumption, it leaves out that the fact that China has a population roughly quadruple that of the US. Some consider the per capita metric to be a key to the ethical case for global emissions policy. China's per capita CO_2 emission rate (in terms of tonnes per person) is 6.59 (similar to the rate of the EU) whereas the US per capita emission rate is a whopping 15.53, well over double that of China.[3] This means that the world would be in a much lower emission state if all nations emitted at a rate similar to China, and not the US.

Another contextualizing piece of evidence in the discussion of carbon emissions is one of legacy, the historical trends in national emissions. Looking at cumulative carbon emissions over time (through 2016), the US sits at almost 400 billion tonnes and China at almost 200 billion tonnes.[5] The long-term perspective shows the massive gap between the two nations with the US at just about double China's total emissions. Given its poor track record, it seems

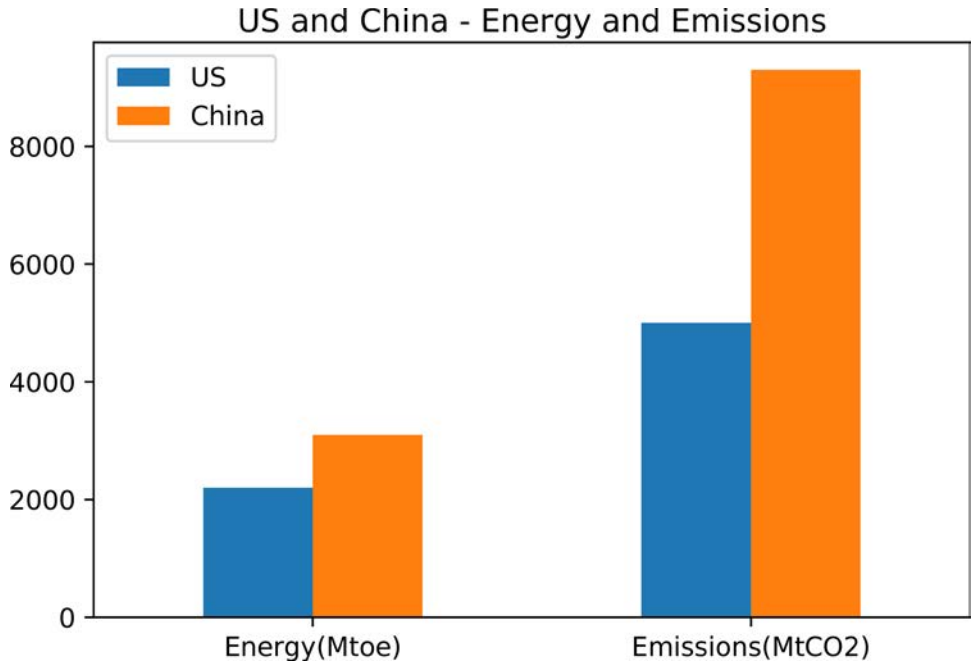


***Figure 1: US vs. China: Total Energy Consumption and CO_2 Emissions**. China's current energy consumption is around 3,100 Mtoe (million tonnes of oil equivalent) with a CO_2 emission rate of almost 9,300 $MtCO_2$ (million metric tons of CO_2) per year. In comparison, the US currently consumes roughly 2,200 Mtoe of energy and emits around 5,000 MtCO.*[2,3]

clear that the US should be trying much harder to curb additional emissions.

With such context in mind, China's current high emission rate is more sympathetic and cannot be so facilely used to condemn those in the US that wish to limit our own emissions. "China emits more than the US, so why should the US curb any of its own emissions?" becomes a much less defensible position. Despite what has been said, we still desperately want to avoid significant amounts of continued emission, regardless of its source. So enters a possible follow-up: "Even if the US curbs its emissions, it won't matter because China will keep on emitting." While much could and should be said about the fallacious nature of such a position—if someone else is doing something wrong or damaging, that does not justify one's own doing of the same thing; and, even if another nation engages in something like pollution, a shrinkage in pollution by other nations could be enough to mitigate the major damage of said pollution—this claim again makes the assumption that China will necessarily not do better in regards to emissions. But much evidence points to the contrary: arguably, China is taking a stronger stance against carbon emissions and in support of the need to combat climate change, than the US.

The justification for this argument starts with a discussion of the Paris Agreement concerning climate change: an agreement made by the United Nations Framework Convention on Climate Change (UNFCCC) to prevent a global average temperature increase of 2°C by curbing greenhouse gas emissions. As of the beginning of 2019, 195 UNFCCC members have signed the agreement.[1] The United States' current president Donald Trump has made it clear that his administration is against the current iteration of the agreement and plan to formally exit as early as the agreement allows, which would be November 4, 2020.[6] The agreement does not have any mechanism to enforce that a participating state follow any specific plan for the future. However, participation in the agreement does signal to the world a state's commitment to a future of clean energy and fighting climate change. While the US plans to pull out of the agreement, China does not.

As stated, the Paris Agreement does not have enforcement mechanisms, but it does have a way to set and keep track of goals: through Nationally Determined Contributions (NDCs).[7] NDCs are emission-related goals determined by each member state that are supposed to fall in line with what the agreement defines as "ambitious… with the view of achieving the purpose of this Agreement."[1] China's NDCs for 2030 include the following: peaking total CO_2 emissions around 2030 or earlier, lowering CO_2 emissions per unit of GDP by 60-65% of the 2005 level, increasing the non-fossil fuel share of energy consumption to roughly 20%, and increasing their total forest volume by 4.5 billion cubic meters more than their 2005 level.[7] This entire discussion of NDCs is juxtaposed with the case of the United States. The NDC Partnership website still states a singular goal set by the US to reduce "its greenhouse gas emissions by 26-28% below its 2005 level in 2025" but no continued commitment to such a goal has been stated or shown since the USA's announcement to leave the Paris Agreement.[7]

As the US is in the process of abandoning its explicit, globally-stated goals of limiting greenhouse gas emissions, its future direction, desired or realized, is ambiguous. On the other hand, China's goals can be compared with its actions to assess its commitment. While it is difficult to quantify how successful China will be in meeting the above goals, as they have not yet begun decreasing their yearly rate of carbon emission, if they do achieve their goal of peaking emission by 2030, then they will almost surely keep their emission per capita drastically lower than the US. More concretely, the forestry claims laid out in the NDCs are quite plausible because China has previously proven its commitment to that specific task. In the 1950s, China had a forest coverage of 8.6% which they increased to 21.93% by the end of 2016.[8] Obviously, the future is always uncertain, especially when it comes to executing governmental plans. That said, a clear commitment to a cleaner future, from setting specific energy goals to researching the best ways to implement renewable energy, are a promising start.[9]

Given the relative positions of the US and China, the US has ample opportunities to strengthen and reaffirm its commitments to a cleaner future, which, if taken advantage of, can mitigate the worries of climate change. Despite initiatives from California and a few regional groupings, the US currently does not have many national goals in mind when it comes to limiting carbon emissions

and transitioning to clean energy sources.[10] With the planned exit of the Paris Agreement, the US would have no NDCs. No explicit plans have been made by the current presidential administration to deal with the problem of greenhouse gas emissions and their impact on climate change, although this is not for a lack of plans or ideas available.[11, 12] Going further than simply choosing non-action on the issue of climate change, the current administration is turning backwards. Under former Administrator Scott Pruitt, the EPA put into motion a repeal of the Clean Power Plan, which was meant to help curb the effects of climate change by reducing carbon emissions from electrical power generation.[13] By abdicating its previous goals to fight emissions and actively removing previous legislature that did tangibly fight them, the US is currently not just greatly reluctant but completely antagonistic toward the idea of modernizing energy production and consumption to cleaner alternatives that would help stop climate change.

As the single largest emitters of CO_2 by far, and together comprising nearly half of the entire world's CO_2 emissions, there are no nations more important than the US and China when it comes to understanding the current and future state of carbon emissions and their consequences. Whereas China has made explicit commitments to limiting emissions and growing non-fossil fuel energy sources, the US has declined to step up in similar ways and has instead regressed. As a world leader in numerous facets, the US could be using its unique position on the global stage to facilitate and accelerate the world's transition to cleaner, safer fuels, but it must first decide for itself that such a transition is worth it. The US cannot justifiably use China as an excuse for self-imposed inaction and ignorance of its consequences.

As it is clear that China is not a legitimate excuse or reason for American inaction, what is motivating those that make such a claim? What groups would benefit from the US continuing to significantly rely on fossil fuels, but have to lie about the motivations and reasons that the US should do so? It is certainly not the clean energy companies.

Acknowledgements: I would like to acknowledge and thank Dr.

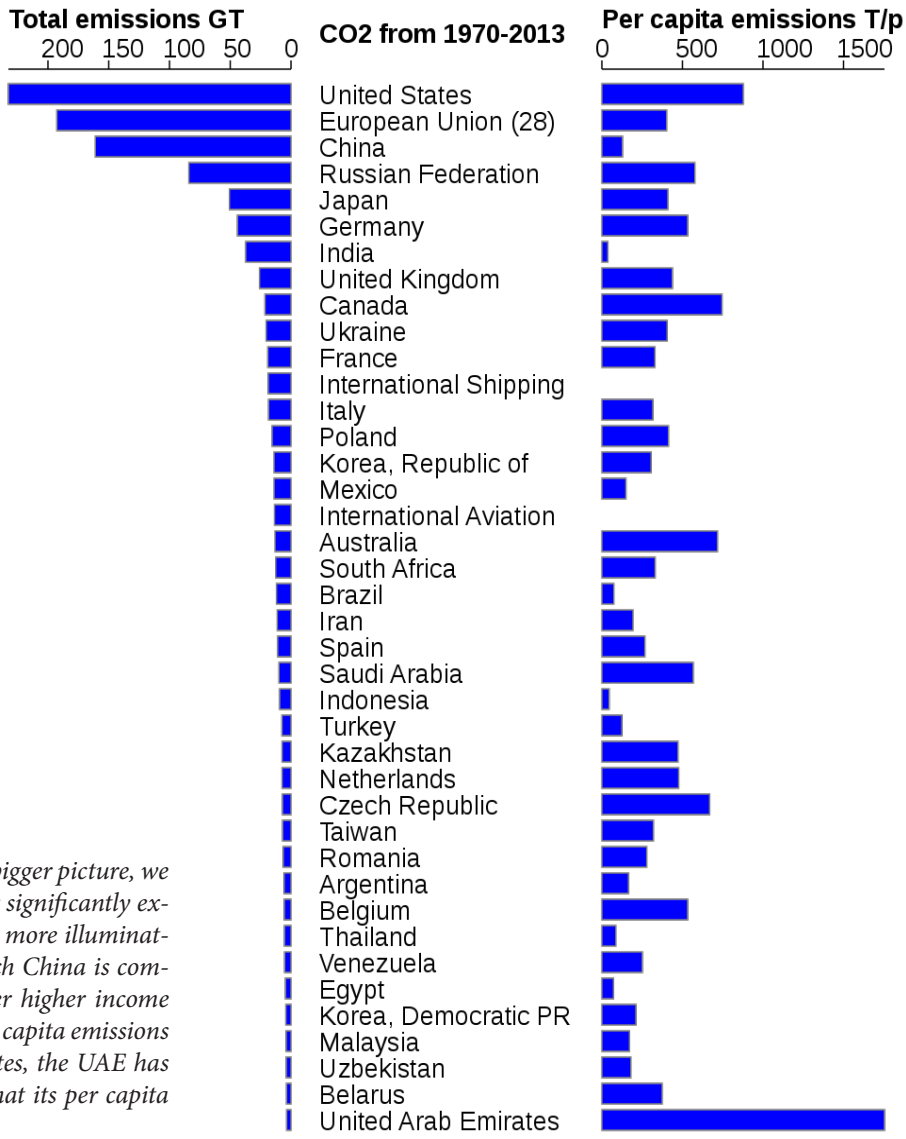


Figure 2: Cumulative CO_2 Emissions*. Looking at the bigger picture, we can see that since 1970, the US and the EU together significantly exceed China in terms of total carbon emissions. Even more illuminating is the difference in per capita emissions, in which China is completely dwarfed by the US, the EU, and many other higher income economies. While nearly every other country has per capita emissions that come nowhere close to the United Arab Emirates, the UAE has produced such a low total emission since the 70's that its per capita rate is not too worrying.*

Figure 3: Mulan Wind Farm. *This Mulan Wind Farm is "one of the first wind farms in China, raising the profile of renewable power and acting as a flagship for replication of the technology across the country." It generates roughly 25 GWh of energy each year.*

David Roland-Holst (Adjunct Professor in the Department of Agricultural & Resource Economics at UC Berkeley) for his insightful feedback that helped turn my article into a much more nuanced and relevant work.

REFERENCES

1. Paris Agreement under the United Nations Framework Convention on Climate Change. UNFCC. April 22, 2016. Retrieved from https://unfccc.int/sites/default/files/english_paris_agreement.pdf
2. European Comission. (2018). Revised Renewable Energy Directive. Retreived from https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=uriserv:OJ.L_.2018.328.01.0082.01.ENG&toc=OJ:L:2018:328:TOC
3. Enerdata. (2018). Global Energy Statistical Yearbook 2018. Retrieved from https://yearbook.enerdata.net/total-energy/world-consumption-statistics.html
4. Kessler, G. (2017, April 14). EPA administrator Scott Pruitt's claim that China and India have "no obligations" until 2030 under the Paris Accord. *The Washington Post*. Retrieved from https://www.washingtonpost.com/news/fact-checker/wp/2017/04/14/epa-administrator-scott-pruitts-claim-that-china-and-india-have-no-obligations-until-2030-under-the-paris-accord/?noredirect=on&utm_term=.9b7be4b61da7
5. Ritchie, H. M. R. (2019). CO_2 and Other Greenhouse Gas Emissions. *Our World in Data*. Retrieved from https://ourworldindata.org/co2-and-other-greenhouse-gas-emissions
6. Mooney, C. (2018, December 12). Trump can't actually exit the Paris deal until the day after the 2020 election. That's a big deal. *The Washington Post*. Retrieved from https://www.washingtonpost.com/energy-environment/2018/12/12/heres-what-election-means-us-withdrawal-paris-climate-deal/?utm_term=.c2a672595d28
7. NDC Partnership. (2018). Country Pages. Retrieved from https://ndcpartnership.org/
8. Li, W. (2018, March 29). China now has the largest amount of planted forest in the world. *GBTiimes.* Retrieved from gbtimes https://gbtimes.com/china-now-has-the-largest-amount-of-planted-forest-in-the-world
9. Xinyu, G., Bo, J., Bin, L., Kai, Y., Hongguang, Z., & Boyuan, F. (2011). Study on renewable energy development and policy in China. *Energy Procedia*, 5, 1284-1290. doi: 10.1016/j.egypro.2011.03.224
10. California Natural Resources Agency. (2019). Safegaurding California and Climate Change Adaptation Policy. Retrieved from http://resources.ca.gov/climate/safeguarding/
11. Jacobson, M. Z., Delucchi, M. A., Bazouin, G., Bauer, Z. A., Heavey, C. C., Fisher, E., ... & Yeskoo, T. W. (2015). 100% clean and renewable wind, water, and sunlight (WWS) all-sector energy roadmaps for the 50 United States. *Energy & Environmental Science, 8*(7), 2093-2117. doi: 10.1039/c5ee01283j
12. Hand, M. M., Baldwin, S., DeMeo, E., Reilly, J. M., Mai, T., Arent, D., ... & Sandor, D. (2012). Renewable Electricity Futures Study. Volume 1. Exploration of High-Penetration Renewable Electricity Futures (No. NREL/TP-6A20-52409-1). *National Renewable Energy Lab*. (NREL), Golden, CO (United States).
13. Eilperin, J. (October 10, 2017). EPA's Pruitt signs proposed rule to unravel Clean Power Plan. *The Washington Post*. Retrieved from https://www.washingtonpost.com/politics/epas-pruitt-signs-proposed-rule-to-unravel-clean-power-plan/2017/10/10/96c83d2c-add2-11e7-a908-a3470754bbb9_story.html

IMAGE REFERENCES

14. Chris55. (2015, August 9). *CO_2 cumulative emissions 1970-2013* [digital Image]. Retrieved from https://commons.wikimedia.org/wiki/File:Co2_cumulative_emissions_1970-2013.svg
15. Land Rover Our Planet. (2007, November 12). *Mulan Wind Farm* [digital image]. Retrieved from https://www.flickr.com/photos/our-planet/5371963671

ON THE BRINK OF *DISCONNECTION*

BY: CANDY XU

On April 19, 1965, Gordon Moore presented an article that revolutionized the entire computer industry. He pointed out that due to the falling cost of circuit components, we would be able to squeeze more and more of them onto silicon chips over the next several decades.[1] One effect of this increase in components is an increase in speed, a core characteristic that determines the functionality of a computer. Computing speed is highly related to the arrangement of components and communication between signals and code. These design mechanisms all belong to the field of computer architecture. Vital elements of computer architecture include the central processing unit (CPU), random access memory (RAM), read-only memory (ROM), Input and Output (I/O), and system bus.[2] Together, these components construct a path for software and hardware to communicate with each other. If a computer is like a human body, then computer architecture would be the ways in which the brain and the rest of the body, namely the mental and physical components, interact. While existing architecture has already matured greatly since its genesis, the next decade will likely bring about efforts to revise the current architectural design further so as to continue increasing computing speed.[3]

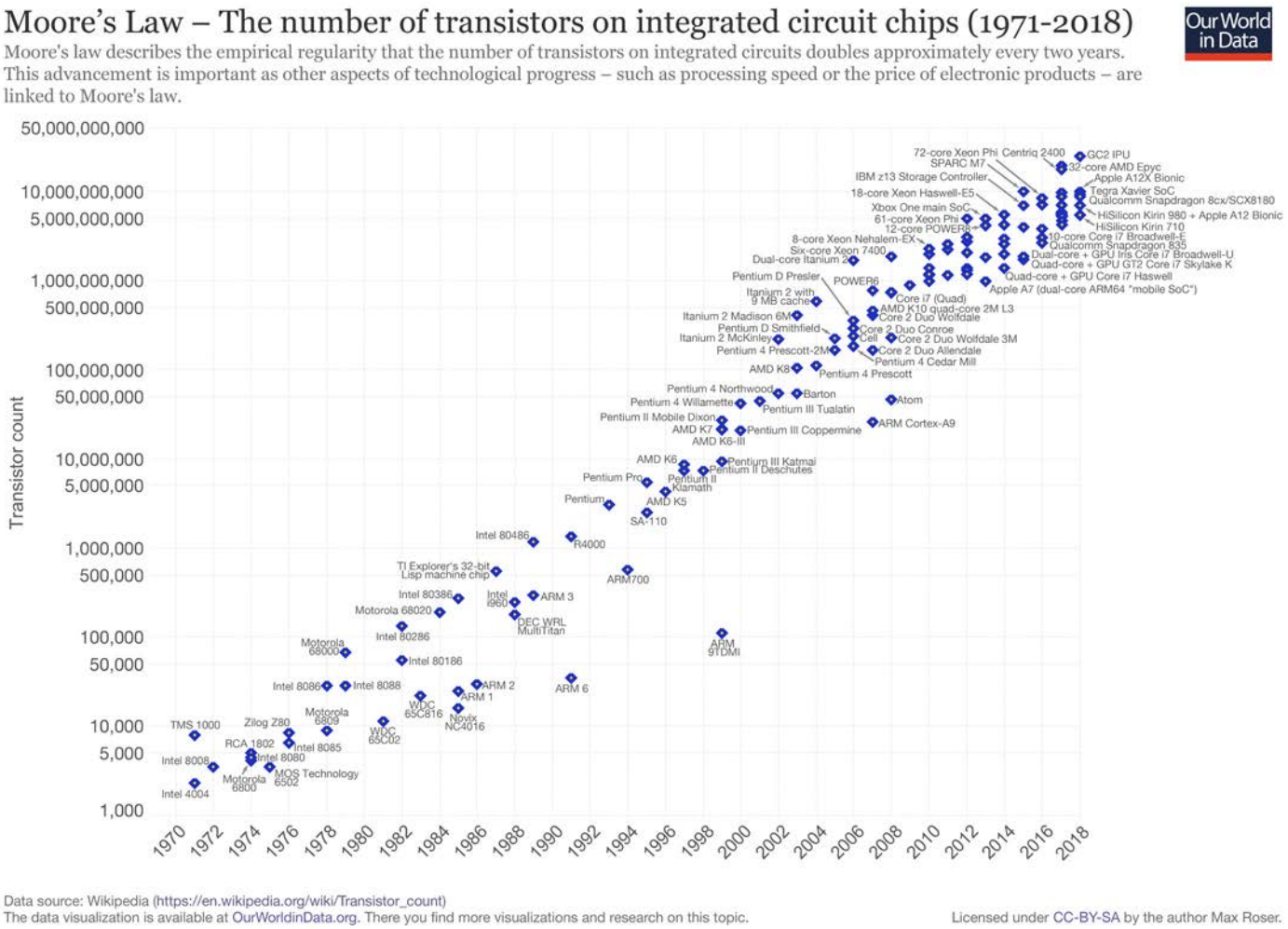


Figure 1: Transistor Count.[11] *The increase of transistors on chips started to slow down in the 21st century.*

MOORE'S LAW

Computing speed is important for computer architects to consider, and improving chips is one way to achieve speedup. Most chips depend on complementary metal-oxide-semiconductor (CMOS) transistors, a type of semiconductor used in integrated circuits and digital logic design. For more than 30 years, we enjoyed "free" computer speedup by brutally adding more and more transistors on chips.[4] This is the idea that Moore presented in his 1965 article. He predicted that the number of transistors would double every year, later revising the statement to every two years.[4]

With this wonderful physical capability, the era of the "lazy software engineer" had begun. The speed of computers did not need to rely on code efficiency or runtime. All that programmers had to do was to wait for microchips to gain more transistors in order for their computers to run faster. This effortless speedup continued until we finally approached the physical limits of the technology. The time frame in which transistors double has increased tremendously because the transistors simply cannot get any smaller (Fig. 1). Currently, the smallest dimension of CMOS on electronic device is close to 10-20 atomic diameters.[5] Any smaller than that and CMOS transistors would physically stop working.[5] As a result, computer scientists began to explore other venues to accelerate computer performance.

LANGUAGE DECODING

With a stagnation in speed, we would lose the ability to implement more powerful programs or enhance the performance of existing ones. The slowing down of Moore's Law thus poses a pressing issue to the whole

> "The slowing down of Moore's Law thus poses a pressing issue to the whole industry: how can we get computers to operate on a faster timescale?"

industry: how can we get computers to operate on a faster timescale? Now that we cannot rely solely on hardware improvements, the solution probably lies in software or the intersection of software and hardware.

Computer scientists have already discovered an area with high potential for increasing computing speed—decoding from high-level programming languages to low-level ones.[3] Languages that are more abstract can be thought of as higher level since they are more readable to humans and easier to use when programming. However, they are thus more structurally complicated and have longer runtimes. Languages that are less abstract can be thought of as lower level, as they usually refer to assembly code or machine code, which can communicate directly with hardware without the need for a complier. Python is one of the most popular examples of a high-level programming language, while the language C is a great example of a language that is less abstract and lower level than Python.

Jun and Ling once performed recursion, a function that calls on itself, to a Fibonacci series calculation using both Python and C to illustrate the significant runtime difference in the two languages: the computation took Python 3.0 about 2.5396 seconds and took C <<0.0001 seconds.[6] This hints at the great potential in speeding up high-level languages like Python by using more efficient techniques to decode them into low-level ones. Currently, JIT is working in that area by directly converting real-time Python models to machine code execution and using cache, a smaller but more accessible storage area for disk data, to temporarily store recently used data, respectively.[6]

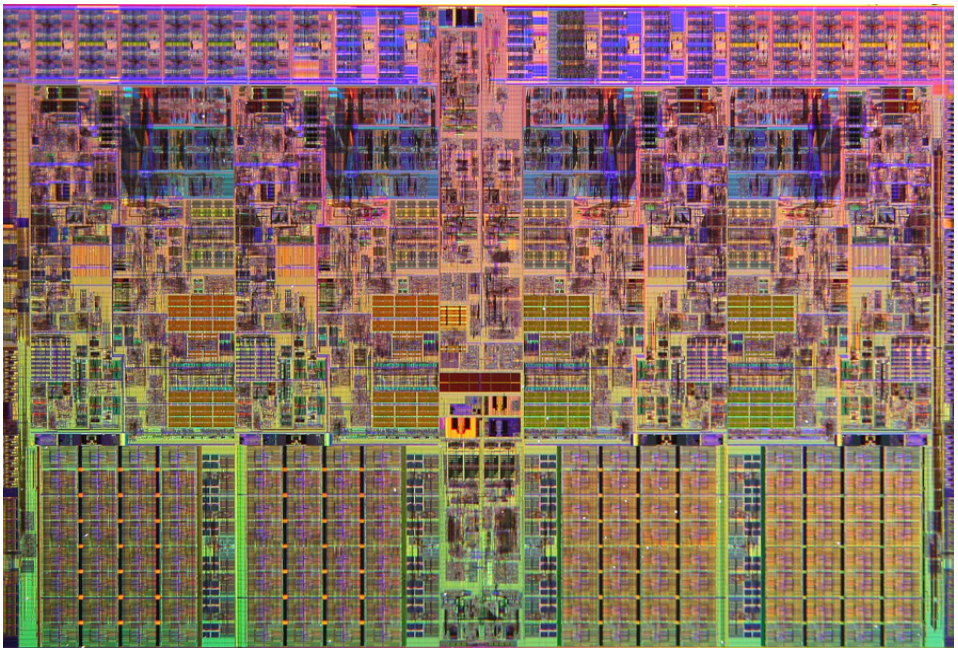


Figure 3: Intel Core i7 Processor.[13] *Intel's Core i7 is an example of a well-architected processor that is widely used in industry right now.*

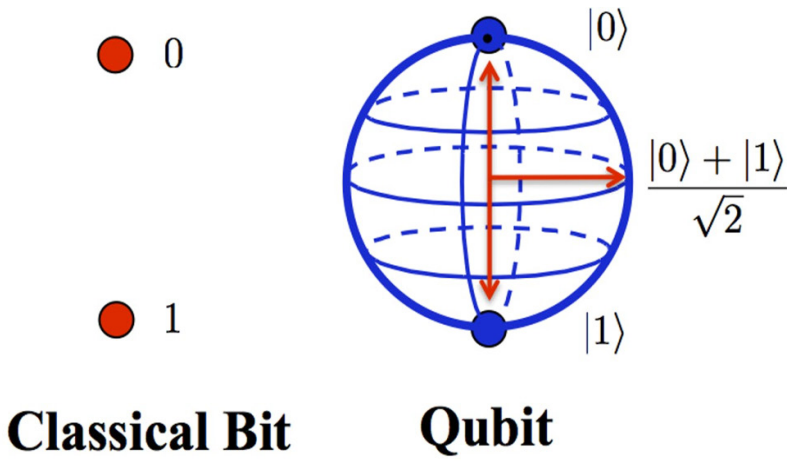


Figure 2: Classical Bit vs. Qubit.[12] *Qubit allows access to many more states than classical bits. The combination of multiple states generate quantum superpositions, which tell us that we can freely add two quantum states together and obtain another valid outcome.*[9] *However, we have to be careful when measuring a superposition state as random results might appear for certain measurements. On the other hand, entangled states cannot be separated. Their ways of combination cannot be recreated, but the two states do have perfect correlation.*[9] *When measuring one state, the other will behave exactly the same.*

DOMAIN-SPECIFIC LANGUAGES

Turing Award winner and Professor at University of California, Berkeley, David Patterson also suggests that another emerging field which may significantly increase computing speed is domain-specific languages.[3] Unlike general languages, such as Java and Python, which can be used in a variety of applications, domain-specific languages are customized for a certain field of interest. For example, Matlab is primarily for numerical computing.

Schaumont and Verbauwhede once ran a 128-bit key Advanced Encryption Standard algorithm using a completely customized program involving application-specific integrated circuit (ASIC) and Java. It turns out the customization increased the performance by a factor of nearly 3 million.[7] Therefore, in order to support domain-specific computing, we would need greater customizable computer architecture. It is not

> "These 10 years of disconnection in the speed up of computers would likely rely on re-structuring the way software languages communicate with each other and their hardware."

only easier to run domain-specific languages on such a architecture, but also cheaper. The cost of implementing all applications using ASICs with a 45nm CMOS is already exceeding $50,000,000, implying the need for another architectural method if we want to spread the use of customized computing.[7] Technologies such as CHP, a customizable heterogeneous platform that integrates customizable cores and tunes performance to a specific application's needs, are currently being researched and developed.

QUANTUM COMPUTING

Other newly emerging ideas are also trying to break through the barrier of speed limits. Quantum computing is one that gained a huge amount of attention recently. Instead of calculating information based on binary systems, which consist only of two levels (0 or 1), quantum systems can distinguish between multiple levels and enable data access to many parts of the computer simultaneously. They rely on qubits, superposition, and entanglement, which together allow us to manipulate combinations of individual states (Fig. 2).[8]

Although methods such as harnessing entanglement for computation have boosted quantum computing speed, this technology is so new and powerful that a real-world application has not been found yet. Professor Patterson and others suggest that a tangible application of quantum computing is most likely not going to take effect within the next decade.[9,10]

Thus, these ten years of disconnection in the speed up of computers would likely rely on re-architecturing the way that software languages communicate with each other and their hardware counterparts. If developed correctly, we can achieve as great of an increase in computing speed as we saw in the era of the "lazy software engineer."

Living in this era of great technological advancement, we have the opportunity to join the battle and get involved with the next golden decade of computer re-architecture. In order to reconstruct the current system, we must redefine the means of interaction between software and hardware. Collectively, such efforts could not only power development in newly emerging technologies, but could in turn push past the boundaries that are currently preventing the next generation of advancements in computing.

Acknowledgements: I would like to express my sincere appreciation and acknowledge Professor David Patterson (Professor Emeritus of Computer Science at UC Berkeley and Google Distinguished Engineer) for his support during the writing process.

REFERENCES

1. Moore, G. (1998). Cramming more components onto integrated circuits. *Proceedings of the IEEE, 86*(1), 82-85. doi:10.1109/jproc.1998.658762
2. Wells, C. J. (2009, January 28). Computer architecture. *Technology UK.* Retrieved from http://www.technologyuk.net/computing/computer-systems/architecture.shtml
3. Hennessy, J. L., & Patterson, D. A. (2019). A new golden age for computer architecture. *Communications of the ACM, 62*(2), 48-60. doi:10.1145/3282307
4. Brenner, A. (1997). Moore's law. *Science, 275*(5306), 1401-1404. Retrieved from http://www.jstor.org/stable/2893682
5. Spencer, M. (2018). The end of Moore's law. *US Black Engineer and Information Technology, 42*(1), 76-76. Retrieved from http://www.jstor.org/stable/26305580
6. Jun, L., & Ling, L. (2010, October). Comparative research on Python speed optimization strategies. In *2010 International Conference on Intelligent Computing and Integrated Systems* (pp. 57-59). IEEE.
7. Cong, J., Sarkar, V., Reinman, G., & Bui, A. (2011). Customizable domain-specific computing. *IEEE Design & Test of Computers, 28*(2), 6-15. doi: 10.1109/MDT.2010.141
8. IBM Research Editorial Staff. (2019, February 08). Quantum computing: you know it's cool, now find out how it works. *IBM*. Retrieved from https://www.ibm.com/blogs/research/2017/09/qc-how-it-works/
9. Patterson, D. (2019, February 25). Talk with Professor David Patterson [Personal interview].
10. National Academies of Sciences, Engineering, and Medicine. (2019). *Quantum Computing: Progress and Prospects.* Washington, DC: The National Academies Press. doi: 10.17226/25196

IMAGE REFERENCES

11. Roser, M. (2019). *Moore's Law—the number of transistors on integrated circuit chips (1971-2016)* [digital image]. Retrieved from https://commons.wikimedia.org/wiki/File:Moore%27s_Law_Transistor_Count_1971-2018.png
12. Zahid, H. (2016). Strengths and Weaknesses of Quantum Computing. *International Journal of Scientific and Engineering Research, 7*(9), 1526-1531.
13. Intel. (2008). *Nehalem die shot 2* [digital image]. Retrieved from https://www.intel.com/pressroom/archive/releases/2008/20081117comp_sm.htm

SATELLITE IMAGERY COMBATS ECO-DESTRUCTIVE ACTIVITY IN THE AMAZON

GOLD MINING IN PERU

The Madre de Dios (MDD) region of southern Peru is one of the most biodiverse regions in the world, home to thousands of endemic flora and fauna. Like most wild places, MDD is facing tremendous environmental degradation, primarily in the form of artisanal gold mining. This phenomenon has had seemingly irreversible impacts on the department's water quality, forest carbon stores, biodiversity, and human health.[1] All of these complications emerge from the unregulated excavation of soil along waterways, removal of trees, and heavy use of mercury to extract gold.[2]

The 2008 global recession, along with the construction of the Interoceanic Highway through the Madre de Dios, has accelerated gold mining in recent years, providing miners with more access to forested areas. Due to

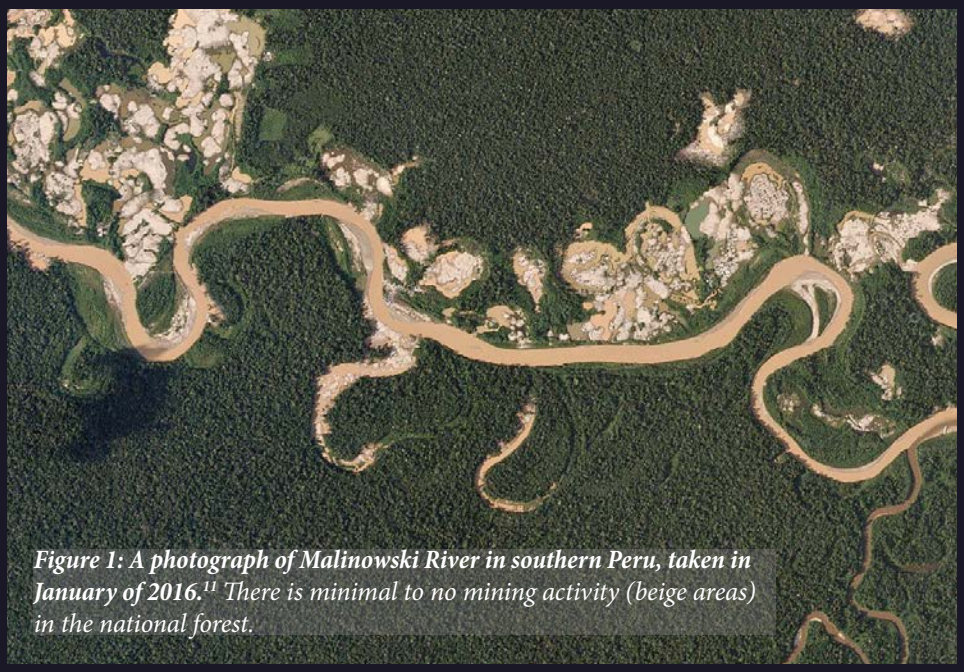


Figure 1: A photograph of Malinowski River in southern Peru, taken in January of 2016.[11] *There is minimal to no mining activity (beige areas) in the national forest.*

BY: SHANE PUTHUPARAMBIL

increasing economic demands for gold, miners increased their pace in MDD, and the total estimated gold mining area skyrocketed by 40% between 2012 and 2016.[3]

Although Peru adopted legislation in 2012 to prevent and enforce a ban on illegal gold mining, one of the biggest challenges for the government has been monitoring and detecting small-scale attempts to mine gold, especially within heavily forested areas. In order to combat the small-scale mining, scientists, environmental organizations, and the Peruvian government began using detailed software forest monitoring technologies like CLASlite. These softwares automatically convert satellite images into a raw format that makes it easier for scientists to differentiate between high-impact deforestation zones and untouched areas.[3,4] While CLASlite can pinpoint where miners are operating, raids on mining camps are often unsuccessful due to the ease of mobility of the illegal mining equipment and the sheer number of miners willing to replace those who are arrested. Regardless, CLASlite has helped scientists paint a more accurate picture of the destruction in the Peruvian Amazon. According to research conducted by CLASlite in southern Peru in 2018, previous estimates for gold-mining related damage were off by over 30%.[3,4] It is clear that the technology to monitor gold-mining is operational, but in order to halt the industry altogether, more money needs to be allocated for police enforcement. The Peruvian government also needs to make the punishment for such activity more severe. This approach has had a proven track-record of success, especially in Brazil.

Figure 2: Another photo of the same Malinowski River, this time taken in January of 2017.[12] *The photo demonstrates the pace and damage of gold mining. Mining activity infiltrated even the Tambopata National Reserve.*

PREVENTIVE MEASURES

In order to mitigate the environmental issues within the Amazon, local governments need to find ways to prevent further damage to these ecosystems. A symptom that illegal deforestation, cattle ranching, and gold mining all have in common is the severe loss of trees in the impacted regions.[5] Tracking these illegal activities is a matter of identifying the regions where trees are at risk.

The main preventative measure against deforestation is the strict enforcement of environmental legislation, specifically using technology as a tool to detect and identify forestry exploitation in real time. Adopting this methodology, the Brazilian government successfully managed to decrease deforestation by 70% between 2005 and 2013.[6,7] The sharp decline was a result of a government initiative started in 2004 known as the Plan for the Protection and Control of Deforestation in the Amazon (PPCDAm) which utilizes Real Time System for Detection of Deforestation, DETER, a satellite deforestation imaging technology. The government began monitoring the rainforest in real-time, and whenever an area was identified as an active deforestation zone, Brazilian environmental police from the Brazilian Institute for the Environmental and Renewable Natural Resources, IBAMA, would raid the region, fining and arresting the responsible perpetrators. Researchers studying Brazil's enforcement techniques noted that an increased number of law enforcement personnel with better training helped curb significant illegal activity and concluded that the increased number of fines issued decreased the amount of deforestation in the following year.[6,8] The fines create a disincentive for those planning to continue to exploit these resources.[9]

Another unique problem attributed to the prevention of ecosystem damage is that all deforestation is not the same. In the late 1990s and early 2000s, farmers in Brazil clear-cut large portions of land, yet with the increase in government crackdown, landowners realized that a better way to clear forest was to do so in smaller plots. As long as the area cleared is less than 25 hectares (or the equivalent of about 20 soccer fields), DETER and other satellite technologies cannot detect changes in forest size. In addition, different states in Brazil tend to distribute land in different ways, therefore changing how landowners clear property.[10] For example, in the state of Mato Grosso, the average property size is significantly larger so deforestation is easy to detect. On the other hand, Para state has smaller, harder-to-detect properties, which requires more accurate technologies to monitor. As a result of these differences between states and deforestation techniques, policymakers must modify their strategies between regions to be most effective.

"While CLASlite can pinpoint where miners are operating, raids on mining camps are often unsuccessful due to the ease of mobility of illegal equipment and the number of miners willing to replace the arrested."

Figure 3 (right): A dredging operation in the MDD. *The boat collects and rinses rock in search of gold, then dumps it back into the river. Photo courtesy of Dr. Donald C. Taphorn.*

Figure 4: Destruction of riparian forest can be seen as the miners gouge out the shoreline of Mazaruni River. *Photo courtesy of Dr. Donald C. Taphorn.*

Gold mining in southern Peru continues to pose a major threat to the Tambopata National Reserve, one of the most biodiverse ecological regions globally. Major steps need to be taken to combat the damage. First, the Peruvian government must adopt more restrictive legislation to prevent deforestation and gold mining. In addition, larger penalties and heavier fines need to be assigned to those who break the laws. Second, the Peruvian government must allocate more funding towards hiring forestry police. Third, the current satellite imaging technology needs to be updated so that finer details in forest composition can be traced. Hope for these areas still remains—as long as nonprofits and governments work together to not only find a solution for forestry conservation or restoration, but also discover an economic incentive for the underprivileged workers.

Acknowledgements: I would like to acknowledge Suyash Sharma (B.A. Economics from UC Berkeley) and Dr. Donald C. Taphorn (Royal Ontario Museum) for supporting me and reviewing the article.

REFERENCES

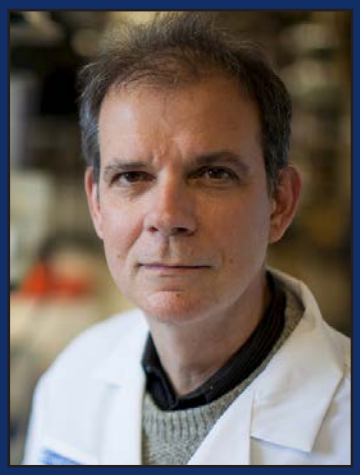
1. Hurd, L. E., Sousa, R. G., Siqueira-Souza, F. K., Cooper, G. J., Kahn, J. R., & Freitas, C. E. (2016). Amazon floodplain fish communities: habitat connectivity and conservation in a rapidly deteriorating environment. *Biological Conservation, 195*, 118-127. doi: 10.1016/j.biocon.2016.01.005
2. Coe, M. T., Brando, P. M., Deegan, L. A., Macedo, M. N., Neill, C., & Silverio, D. V. (2017). The forests of the Amazon and Cerrado moderate regional climate and are the key to the future. *Tropical Conservation Science, 10*, 1940082917720671. doi: 10.1177/1940082917720671
3. Gaworecki, M. (2018, November 29). Novel research method reveals small-scale gold mining's impact on Peruvian Amazon. *Mongabay.* Retrieved from https://news.mongabay.com/2018/11/novel-research-method-reveals-small-scale-gold-minings-impact-on-peruvian-amazon/
4. Asner, G. P., & Tupayachi, R. (2016). Accelerated losses of protected forests from gold mining in the Peruvian Amazon. *Environmental Research Letters, 12*(9), 094004. doi:10.1088/1748-9326/aa7dab
5. Soares-Filho, B. S., Nepstad, D. C., Curran, L. M., Cerqueira, G. C., Garcia, R. A., Ramos, C. A., ... & Schlesinger, P. (2006). Modelling conservation in the Amazon basin. *Nature, 440*(7083), 520. doi: 10.1038/nature04389
6. Assunção, J., Gandour, C., Pessoa, P., & Rocha, R. (2015). Strengthening Brazil's forest protection in a changing landscape. *Policy Brief*, 1-4.
7. Nepstad, D., McGrath, D., Stickler, C., Alencar, A., Azevedo, A., Swette, B., ... & Armijo, E. (2014). Slowing Amazon deforestation through public policy and interventions in beef and soy supply chains. *Science, 344*(6188), 1118-1123. doi: 10.1126/science.1248525
8. Assunção, J., Gandour, C., & Rocha, R. (2013). DETERring deforestation in the Brazilian Amazon: environmental monitoring and law enforcement. *Climate Policy Initiative*, 1-36.
9. Lynch, J., Maslin, M., Balzter, H., & Sweeting, M. (2013). Sustainability: Choose satellites to monitor deforestation. *Nature, 496*(7445), 293. doi: 10.1038/496293a
10. Cerullo, G. R., & Edwards, D. P. (2019). Actively restoring resilience in selectively logged tropical forests. *Journal of Applied Ecology, 56*(1), 107-118. doi: 10.1111/cobi.13182

IMAGE REFERENCES

11. Planet Labs, Inc. (Photographer). (2016). *Illegal Mining In Peru 2016* [digital image]. Retrieved from https://commons.wikimedia.org/wiki/File:Illegal_Mining,_Peru,_2016-01-29_by_Planet_Labs.jpg
12. Planet Labs, Inc. (Photographer). (2017). *Illegal Mining In Peru 2017* [digital image]. Retrieved from https://commons.wikimedia.org/wiki/File:Illegal_Mining,_Peru,_2017-01-20_by_Planet_Labs.jpg

INSIGHTS FROM THE JGI USER MEETING: USING GENOMICS TO TACKLE ENVIRONMENTAL PROBLEMS

BY EMILY HARARI, MATT COLBERT, AND NIKHIL CHARI



From left to right: *Dr. Mary Firestone, Dr. Mary Wildermuth, Dr. Arturo Casadevall, and Dr. Dan Jacobson.*

Soil microbiology, plant-pathogen interactions, computational biology, and big data all converge in the rising field known as genomics, which involves DNA sequencing and analysis to determine the interactions between organisms integral to biological processes.[1] The Department of Energy Joint Genomics Institute (JGI) aims to solve problems in renewable energy, ecosystem nutrient cycling, and decontamination using genomics techniques.[1] BSJ covered the 2019 JGI User Meeting where we had the opportunity to speak with four distinguished scientists working with genomics to tackle issues as wide-ranging as the fate of carbon in soil systems to the discovery of higher-order combinatorial interactions in cells. BSJ spoke with soil microbiologist Mary Firestone and plant pathologist Mary Wildermuth from UC Berkeley, immunologist Arturo Casadevall from Johns Hopkins, and computational biologist Dan Jacobson from Oak Ridge National Laboratory. Our conversations highlighted some of the widely varied genomics research taking place today.

Figure 1: Grand Prismatic Spring, Yellowstone National Park.[7] *Microbial mats give the spring its vivid colors.*

Professor **Mary Firestone** and Professor Wildermuth both spoke about their research in plant and soil microbiology on April 4th. Firestone, a renowned soil microbiologist and biogeochemist, was the day's keynote speaker, and discussed her research in using various genomics techniques to probe the fate of carbon (C) in soil systems. She spent much of her career investigating microbial relationships to nutrient cycling, especially nitrogen (N), in soils, and is more recently diving into C dynamics.

In our interview, Firestone explained that "soil microbes, in a lot of ways, are Mother Earth," and that they are as "fundamental to life on Earth as anything." Microbes in soil have long been known to be critical players in the decomposition of soil organic matter (SOM) to carbon dioxide, CO_2. But more recently, they have been recognized for their importance in the stabilization of C in soil, from C sorption to mineral surfaces to the formation of aggregates—clusters of soil particles that bind together and can sequester C within them.

Firestone explained why she chooses to trace each type of DNA found in these soil clusters. There has been little research in the soil viral genome because soil is a solid matrix—viruses are difficult to filter out and can easily sorb to minerals or organic materials. Consequently, very little is known about the importance of viruses to soil ecosystems and nutrient and C cycling. Although Firestone classifies soil viruses as a "big unknown," she considers fauna to be a "big known." Much research has been done to determine the roles of soil fauna (small Metazoa like nematodes and mites) in making essential nutrients like N and phosphorus (P) bioavailable to plants, but, according to Firestone, the relationship between faunal food webs and soil C remains under-investigated and "begging for improved molecular tools."

Among these tools is a genomics technique called primer quantitative PCR, or qPCR. Faunal quantification has typically been performed by visual identification and direct counting in soils. However, Firestone and colleague Javier Ceja-Navarro at Lawrence Berkeley National Lab found that primer qPCR yields largely similar quantitative results.[2] Firestone explained how qPCR can make quantifying fauna in soil much easier: "You simply have to take a large sample of soil to encompass faunal diversity, extract the DNA, and use specific primers to do PCR amplification of the barcode that you want to quantify. That gave us very, very similar results to the direct counting method, and confirmation that qPCR works quantitatively."

Firestone's research also tackles the impact of arbuscular mycorrhizal fungi (AMF) on soil ecosystems. AMF are unlike most fungi in that they are not saprotrophs. Where saprotrophs use dead C as an energy source, AMF are biotrophic, or as Firestone said, "they eat

Figure 2: One of Dr. Casadevall's watercolor paintings. *Reprinted with permission.*

living C by plugging directly into plant roots." The exchange is mutual—AMF also pick up essential nutrients like N and P from the surrounding soil and transport them back to plants. Firestone's recent genomics research focuses on what she calls "helper bacteria," which associate with AMF. Though Firestone stressed that "this is a story that's still unraveling," she believes helper bacteria could work to decompose organic N into ammonia—a bioavailable form that AMF can help transport into plants. By ^{13}C-labeling AMF hyphae and tracing which bacteria around the hyphae are picking up ^{13}C, Firestone and her group are working to "establish a direct link between hyphae supplying C and bacteria living around them that are consuming it."

Despite her strides in soil metagenomics, Firestone stressed that there is still much work to be done. "Soil is the most diverse microbial habitat on Earth," she explained, but a "very, very small percentage of the soil genome—less than 1%—has been sequenced." This is largely because it's difficult to extract DNA from a solid yet dynamic matrix like soil. But Firestone still believes there are "many, many lives of interesting research for young students in this area of research."

Professor **Mary Wildermuth** also studies plant-microbe interactions in soil systems, but her path to the field could not have been more different. While she conducted undergraduate research and graduated with a degree in chemical engineering from Cornell, she wasn't sure she wanted to stick with the field. "I realized as a chemical engineer that I was missing biology," Wildermuth recounted. "From there, biotech was pretty new, and I realized I didn't want to go into traditional chemical engineering." She obtained a position at biotech company Gibco-BRL and helped create the first Hepatitis B DNA detection probe (for research use) in concert with doctors at the NIH. While contemplating her future options, Wildermuth elected to travel abroad. "I had always wanted to go to Africa to get more of a perspective on what I wanted to study," she said. After teaching science for two years in Molepolole, Botswana, she returned to the US and worked at the National Center for Atmospheric Research (NCAR), studying plant gas production and climate change. She ultimately obtained her doctorate in Biochemistry at the University of Colorado, Boulder, exploring how plants make the reactive gas isoprene.

In her postdoctoral work at Massachussets General Hospital (Harvard Medical School), Wildermuth then expanded her study of plants to include microbial pathogens, with a special focus on the powdery mildew fungus. "The reason that I love it is that it's an obligate biotroph, so it can only grow on a living plant," she said. Powdery mildew is a fungal plant pathogen characterized by white spots on a plant's leaves. An individual powdery mildew species can typ-

"If you go into a hospital today, the microbes you recover from infected individuals look very different from those that would have been causing disease in the 1900s."

"I often wonder, if Darwin had written *On the Origin of Species* today, where would he have published?"

ically infect similar host plants, while different species of powdery mildew can infect different sets of plants. "The spores are windborne, and it's almost all asexually reproduced… and they'll make them in five days to a week," Wildermuth explained. This makes them a common nuisance.

Powdery mildew's interactions with the plant host are what most fascinate Wildermuth, who likens them to a mass-balance equation. In this interaction, the powdery mildew establishes a feeding structure in the upper layer of the plant in order to siphon nutrients for itself. The interaction doesn't only touch the upper layer though, as Wildermuth's lab discovered. "Below those, in the mesophyll cells in the plant, the powdery mildew induces endoreduplication," she said. "You go through replication of the DNA, but you don't go through cell division." While it may seem strange to force cells to increase their ploidy (amount of DNA), the mildew has an express goal in doing so. "What we found is that if we compromise the induction of that plant endoreduplication, the powdery mildew doesn't grow as well," Wildermuth recounted. Furthermore, the level of endoreduplication that is induced is positively correlated with the number of spores the mildew can produce.

That isn't the only aspect of the interaction that the Wildermuth lab examined; a protein called MyB3r4 also affects endoreduplication and the growth of the mildew.[3] MyB3r4 is a conserved cell cycle regulator that appears in other organisms besides plants, and Wildermuth's team noticed that infected sites demonstrated altered MyB3r4 expression. Wildermuth and her team wanted to find out what happens in the host in the absence of MyB3r4. "We found that when we knocked out that gene, we no longer see this induced endoreduplication," she says, leading to the finding that the MyB3r4 protein could play multiple roles in cell cycle regulation. Currently, Wildermuth's lab is examining how induced endoreduplication in the plant alters plant metabolism to promote the mildew's propagation. She believes her research may have broad-scale agricultural implications in controlling this fungal disease.

On April 5th, the focus of the conference shifted to computational biology. Dr. **Arturo Casadevall**, a Professor of Immunology and Microbiology at Johns Hopkins University, discussed how developments in host-microbe modeling can change the way scientists tackle broad-scale problems. "If you go into a hospital today, the microbes you recover from infected individuals look very different from those that would have been causing disease in the 1900s," Casadevall said. This is because the virulence of a pathogen changes as it mutates and interacts with its human host. But, Casadevall explained, most current microbial models fail to incorporate these adaptations. Without considering the host and other aspects of the microbe's environment, "microbe-centric theories cannot cope with these changes."

In his talk, Casadevall introduced his formula for calculating "pathogenic potential." This formula accounts for the different aspects of host-microbe interactions that are neglected in models where all microbes are assumed pathogens. Casadevall wants to tackle the nuances of immunology with a quantitative approach. "If I lived in the ancient world, I'd be called part of the cult of Pythagoras," he said. He believes that "the mathematics of the system will tell you the degree to which anything is predictable," and his current collaboration with biomathematician Aviv Bergman at Yeshiva University turns to dynamical systems as a way to predict trends within biological systems. As Casadevall explained, dynamical systems use a function to describe the time dependence of a point in space. Rather than hold the microbe constant or the host constant, as in most contemporary experimental designs, dynamical systems allow Casadevall and Bergman to look at the interactions of both organisms over time. By analyzing patterns from many simulations, they have concluded that virulence is an emergent property—a characteristic exhibited of a complex system like the host-microbe relationship, rather than an individual microbe. Casadevall explained that microbes "hedge their bets" by randomly varying their behavior in order to better infect their hosts. These small variations can perturb a biological system with dramatic consequences.

Currently, Casadevall studies immuno-compromised individuals who suffer from human immunodeficiency virus (HIV). HIV patients lack adaptive immunity, and are thus more susceptible to fungal infections. But Casadevall explained that immune-compromised individuals are not restricted to patients of disease. "Anybody who ages becomes immuno-compromised," Casadevall explained, "and medical progress is often associated with immuno-suppressed states." For older individuals or those seeking medical treatments, studying fungal infection in compromised immune systems can reveal useful insights into human health. In fact, Casadevall theorizes that fungi are responsible for the emergence of the human species, and mammalian life in general. He cites studies that demonstrate how endothermy, the ability to regulate body temperature, protects mammals and birds against many fungal infections.[4] Specifically in humans, most fungal diseases are considered "opportunistic"—associated with mutations in the host genome or bacterial infections.

"Instead of asking, 'What killed the dinosaurs?' ask, 'What kept down the reptiles, such that we did not have a second reptilian age?'" Casadevall explained how fossil evidence indicates abundant fungal growth during the time of dinosaur extinction, possibly due to the cool and moist conditions that volcanic activity and resulting ashen skies promoted. Most dinosaurs were probably killed by the cataclysm but the surviving reptiles were ectotherms—their body temperatures were dependent on the temperature of their environments—so fungal diseases from the outside environment may have been able to adapt easily to the dinosaurs' immune systems, potentially killing them. As a result, endotherms with internal body temperatures that exceed the thermal tolerance of fungi might have emerged, leading ultimately to the age of mammals and the evolution of humans. Casadevall illustrated this theory with paintings. "I urge you to draw your science," he said. "It doesn't take that long and it doesn't have to be that good, but I'm really happy with my meteorite."

In addition to our past, Casadevall also illustrated what the future condition of the human race may look like in the context of climate change. "As the planet gets warmer, some of these fungi will

"If we look at a big scary problem, we should stop saying, 'That's astronomical,'" he said, "We should say, 'That's huge. That's *biological*.'"

adapt to higher temperatures." He points out that the average fungal thermal tolerance over the past 30 years has already risen. One such example is *Candida auris*, a fungus that has independently emerged around the globe in the past five years and is killing many immuno-compromised individuals.[5]

Though he remains passionate about his ideas, Casadevall has been largely limited to publishing his theories in mini-reviews, "because thought is very difficult to get published."

"I often wonder, if Darwin had written *On the Origin of Species* today, where would he have published?" Casadevall asked. Today, he promotes scientific thought as the Editor-in-Chief of *mBio*, an open access journal sponsored by The American Society for Microbiology that publishes many different forms of science communication including mini-reviews, opinions and hypotheses, commentaries, and perspectives. Casadevall is optimistic about the potential of his research. "I went into science because I like to explore ideas," he said.

Another JGI speaker with a focus in computational modeling was Dr. **Dan Jacobson**, a computational systems biologist at Oak Ridge National Laboratory. Jacobson is a world record holder addressing the world's biggest problems with big data.

Jacobson's team developed the software for the supercomputer Summit, breaking the exascale barrier, which had previously limited computations to a billion floating point operations per second. The use of tensor cores on graphical processing units (GPUs) allowed for a massive boost in computation performance which enabled the discovery of higher-order combinatorial interactions in cells. Modeling these higher-order interactions is leading to a better mechanistic understanding of biological systems. A challenge associated with these tensor cores is the use of mixed precision (16-bit) numbers. However, tensor cores boosted the speed of calculations, allowing scientists to process much larger datasets. "For this application, by going to 16-bit numbers from 64-bit, we actually lost no accuracy whatsoever," Jacobson explained. "There was just a tremendous speed boost." Jacobson would like to increase speed even more by going down to 8- and even 1-bit numbers. He revealed that currently 11% of computing speed is attributed not to calculating, but to communication overhead associated with moving data onto and off of the GPUs. The remaining limitations of computation speed for 16-bit numbers, therefore, may lie not within the calculations themselves, but in the computer's ability to communicate them and relay their information.

Jacobson is using Summit to bring supercomputing to biology. With explainable artificial intelligence (AI), he'd like to create more comprehensive epistatic calculations. Epistasis is the phenomena by which multiple mutations interact to affect a phenotype even each individual mutation has little to no effect by itself. AI looks at many iterations of data and classifies them based on patterns the computer observes. "Most AI methods.... don't tell you the pattern that they're using," Jacobson said, "What we want for biology is that pattern that the algorithm is using for classification." Explainable AI identifies this pattern that traditional AI algorithms do not normally reveal on their own. Summit can also help geneticists fill in holes in their data by recapturing rare genetic variants. Statistically, these are convenient to ignore. However, "they often have some of the largest effects on phenotypes," according to Jacobson. Supercomputing can also help construct pan-genomes for whole populations, which will more accurately capture what an entire genome of a species looks like.

In addition to improving how scientists interpret data in the lab, Jacobson's work can also benefit agricultural scientists and farmers in the field. "If you design a new plant genotype that's going to make lots of biofuels or food, but then put it someplace where it dies, you have failed," he explains. Jacobson took data from genomes of many different plants and revealed that different climate types correlated with specific genetic adaptations. He is discovering useful genome-wide associations that plant biologists can use to engineer crops for abiotic stresses like drought, and carbon sequestration for bioenergy, both challenges associated with climate change.[6] "As [biologists] design genotypes that we want to deploy, we want to know whether they'll be successful." Jacobson uses clustering algorithms for different regions of the world to test how successfully different plant genotypes would survive projected climate conditions around the globe.

Jacobson believes these genomic computing techniques have the potential to change the way we look at macroscale biological patterns. "If we look at a big scary problem, we should stop saying, 'That's astronomical,'" he said, "We should say, 'That's huge. That's *biological*.'"

REFERENCES

1. DOE Mission Areas. (n.d.). Retrieved from https://jgi.doe.gov/our-science/doe-mission-areas/
2. Firestone, M. (2019, April). Disentangling the Web of Interactions Shaping Microbial Communities in Soil. Presented at 2019 JGI User Meeting, San Francisco, CA.
3. Wildermuth, M. (2019, April). Host Factors in Plant-Pathogen Interactions. Presented at 2019 JGI User Meeting, San Francisco, CA.
4. Casadevall, A. (2012). Fungi and the Rise of Mammals. *PLoS Pathogens,8*(8). doi:10.1371/journal.ppat.1002808.
5. Richtel, M., & Jacobs, A. (2019, April 06). A Mysterious Infection, Spanning the Globe in a Climate of Secrecy. Retrieved from https://www.nytimes.com/2019/04/06/health/drug-resistant-candida-auris.html
6. Jacobson, D. (2019, April). Exascale Biology: From Genome to Climate With a Few Stops Along the Way. Presented at 2019 JGI User Meeting, San Francisco, CA.

IMAGE REFERENCES

7. Brocken Inaglory. (Photographer). (2008). *Grand Prismatic Spring and Midway Geyser Basin* [digital image]. Retrieved from https://commons.wikimedia.org/wiki/File:Grand_Prismatic_Spring_and_Midway_Geyser_Basin_from_above.jpg

BY SHEVYA AWASTHI, DOYEL DAS, EMILY HARARI, ELETTRA PREOSTI, SAUMI SHOKRAEE, AND ELENA SLOBODYANYUK

BIOLOGICAL INSIGHTS INTO SINGLE-MOLECULE IMAGING

Interview with Professor Eric Betzig

Dr. Eric Betzig *is a Professor of Cell and Developmental Biology, Eugene D. Commins Presidential Chair in Experimental Physics, and a Howard Hughes Medical Institute Investigator at the University of California, Berkeley. He is also a Senior Fellow at the Janelia Research Campus. In 2014, Professor Betzig was awarded the Nobel Prize in Chemistry for his development of super-resolution fluorescence microscopy. Professor Betzig's research centers on advancing imaging tools for biological discovery in dynamic living systems. In this interview, we discuss the foundations of fluorescence microscopy, challenges associated with live-cell imaging, and the project of mapping nanoscale synaptic proteins across the entire Drosophila brain.*

BSJ: You have a very diverse background, ranging from physics to hydraulics engineering to optical microscopy. Can you tell us about how you came to working on developing optical imaging tools for biological research?

EB: By accident, I guess. I went to Caltech as an undergrad in the late seventies. I originally wanted to be an astronaut or an astrophysicist, but I realized that was really hard and probably beyond what I could do. I started working in a lab and I liked doing experimental work, so I switched gears toward applied physics and engineering. Back then, the only two graduate schools that had applied physics programs were Stanford and Cornell. I had had enough of California at that point, so I moved to Cornell. The department was really small; there were only about 12 professors, and two were young associate professors who had just gotten tenure. One was an electron microscopist and the other was a Raman spectroscopist. Together, they had come up with this crazy idea to use an electron beam to drill a hole into a silicon film and shine light through it, making a "nano-flashlight" that could be driven around on a sample. That was called near-field microscopy. What these two professors were doing sounded kind of nutty and fun, so I got involved in microscopy that way. I did near-field microscopy at Cornell for six years and then got a job at Bell Labs, where I did the same thing in my own lab for another six years.

BSJ: Why is diffraction-unlimited microscopy useful for studying biological systems?

EB: We've learned a lot about biology without going beyond the diffraction limit, but the main problem with the diffraction limit is that at the fundamental level, cells are made of molecules. The resolution for a normal optical microscope is 100 times too coarse to see what's going on at the molecular level. We'd like to understand how one inanimate molecule interacts with other inanimate molecules to somehow make a cell, which can move, excrete, reproduce, and is deterministic. You want a microscope that can get down to the molecular resolution if you want to understand how the cell works.

BSJ: What does it mean to image "fixed" cells? How do fixation procedures introduce artifacts in imaging?

EB: Fixation is a sort of secret home brew that's been developed over many generations. There are all sorts of recipes, but typically chemicals like formaldehyde and glutaraldehyde are used to cross-link proteins together. At that point, you have a static cell. But with the way the proteins are cross-linked, cellular structures can be distorted. This distortion isn't so bad if you're looking at tissues at low resolution. But with super-resolution microscopes, you realize that cells look like roadkill after they've been fixed. The real fun in biology is looking at living, moving things.

BSJ: Super-resolution microscopy methods center on detecting fluorescence signals from chemical compounds called fluorophores. Can you explain what fluorescence is and how it is useful for single-molecule studies?

Image: Professor Eric Betzig.[1] *Giving a talk at a conference at École Polytechnique on the theme of high-resolution imaging.*

EB: Certain molecules will absorb photons if you shine light on them. When the molecule absorbs a photon, it gets excited to a higher energy state. A few nanoseconds later, the molecule trickles down to a lower energy state and then returns to the ground state. As it does this, it emits a slightly redder color than what it absorbed (Fig. 1). In this way, you can spectrally distinguish the molecule. The beauty of fluorescence is that you can tag any cellular protein you want with a fluorophore. Before fluorescent labeling techniques, you were basically limited to visualizing cells as ghostly-looking bags that contained some bumps. Maybe you could distinguish a mitochondrion if you looked in an electron microscope, but you couldn't really know where the proteins were, and it's the proteins that drive what happens in the cell. Another advantage is that fluorescence only lights up the molecule you want to see, so you have a black background. This is particularly important at the single-molecule level.

BSJ: What makes an optimal fluorescent label?

EB: There is no such thing. The disadvantage to fluorescent labels is that they are not intrinsic to proteins. Instead, they are like a bowling ball that you stick on the side of a protein. One of the unmet holy grails is to get protein-specific contrast without having to attach a large, non-native molecule to the protein. No one knows how to do that, but that is one nut we would like to crack someday.

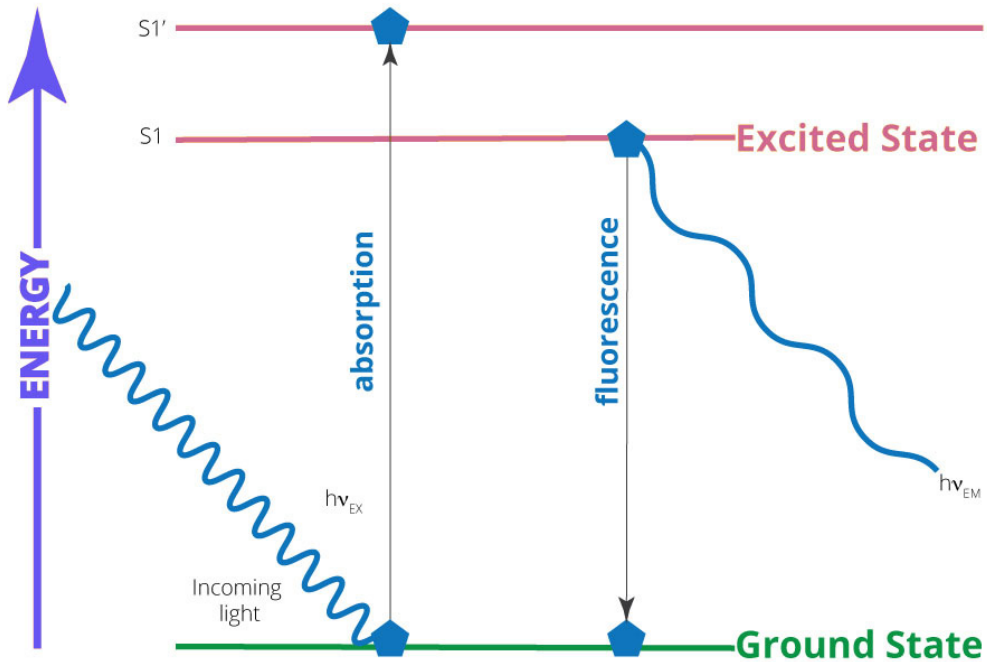


Figure 1: Electronic state diagram illustrating the principle of fluorescence.[2] *A fluorophore absorbs a photon, which causes an electron to move to a higher energy state (ground state to S1'). The electron relaxes to a lower excited state (S1' to S1), and the fluorophore subsequently emits a photon as the electron returns to the ground state (S1 to ground state). The camera detects this emitted photon as the fluorescence signal, which has a longer wavelength than the incoming photon.*

BSJ: What is the Nyquist criterion and how does it pose a problem for labeling density?

EB: If you remember the pre-HDTV era, you know how crappy the pictures were on your TV screen. The problem is that old TVs didn't have as many pixels, so the images looked coarser. The more pixels you have, the higher the resolution. The Nyquist criterion says that if I want to see something of a given size, I need to sample at least half the size of that object. This corresponds to the size of pixels in my image (Fig. 2a). If you have lots of molecules in the sample, but only 0.1% of the molecules have fluorophores on them, you don't see a continuous image of the structure. Instead, you see a random field of dots (Fig. 2b). Getting enough fluorophores on your molecule is a huge problem in super-resolution imaging. The more fluorophores you add, the more you perturb the system, but if you don't add enough fluorophores, you can't see what you want to see. So you're always playing this trade-off.

BSJ: What are some challenges associated with non-invasive live-cell imaging?

EB: Lots of challenges! The first is introducing labels in a way that doesn't perturb the physiology of the sample. Before advances in CRISPR-Cas9 genome editing, you had to add the DNA sequence of your fluorescent protein into the cell. Typically, this makes the cell produce much greater than native amounts of the protein, which makes the cell sick. Now, you can precisely edit the genome to get an endogenous level of tagged protein, which is more physiologically relevant. The biggest problem in non-invasive imaging is that the amount of light that most microscopes put on the sample is equivalent to putting you on the planet Mercury—you're not going to be happy for very long. A lot of my work post-super-resolution was about developing new microscopes that are gentler to cells so that we can look at them for longer periods of time. Finally, many biologists have been constrained to looking at immortalized cells on cover slips. Those cells are really pathological, like the HeLa cell—that cell is such a beast. It has so many extra chromosomes and it is such an anomaly, but researchers use it as a basis of studying mammalian cells. Plus, it's in isolation. In biology, the phenotypes you see are a result of gene expression, and gene expression is controlled by the environment. So if you put a cell in a non-native environment like a cover slip, it can be like the desert to the cell, even with media around it. If you don't study cells in the organism in which they evolved, how can you trust what you're seeing? A big part of our effort has been developing adaptive optics microscopes that allow us to look at cells in a more physiological context.

> "If you don't study cells in the organism in which they evolved, how can you trust what you're seeing?"

BSJ: We read about your development of photoactivated localization microscopy (PALM), which achieves nanometer spatial resolution and led to your award of the Nobel Prize in Chemistry in 2014.[4] Can you briefly explain how PALM works?

EB: The idea behind PALM is a simple one. In a normal optical microscope, you can see a single fluorescent molecule, but it's a big, fuzzy blob. You can point to its center with high pre-

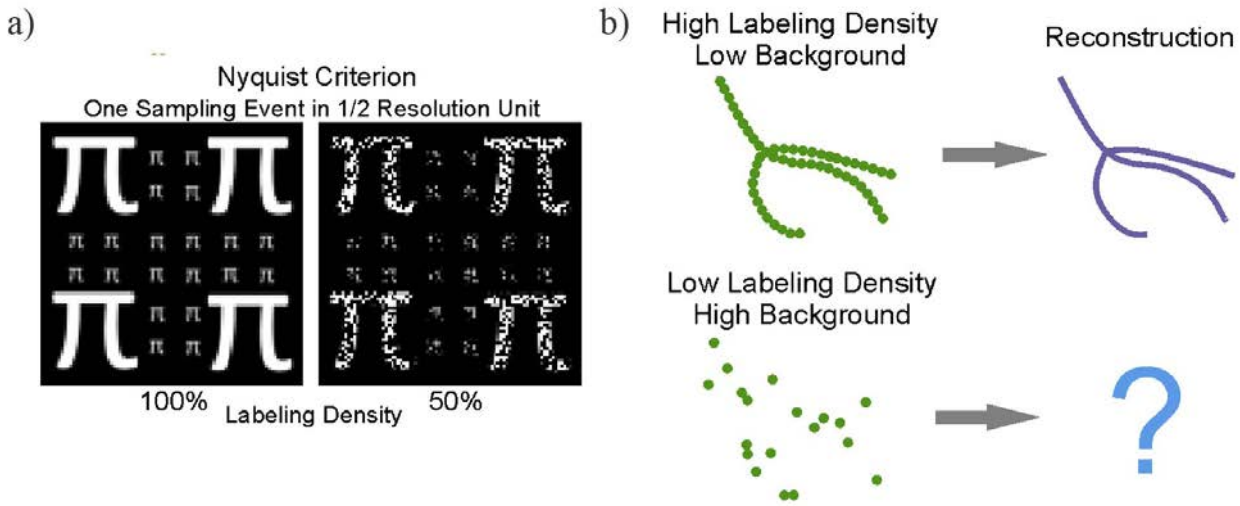


Figure 2: Importance of labeling density for achieving a clear image.[3] *(a) According to the Nyquist criterion, in order to resolve a structure (pi shape) of a particular size, the mean separation between fluorescent labels must be no greater than half of the size of the structure (labeling density must be above 50%). (b) If the structure of interest is labeled too sparsely, then its shape cannot be visualized.*

cision, but the fuzzy blobs overlap so much that you can't make any sense of them. In the early 2000s, a new type of fluorescent protein called photoactivatable green fluorescent protein was developed by my friend Jennifer Lippincott-Schwartz at the NIH. If you use photoactivation, then only a few of the fluorescent proteins light up. You can find their centers, turn off that subset, turn on another subset, and so on (Fig. 3). When my friend Harold Hess and I got that idea, we were unemployed, but it was a simple enough idea that we could do it ourselves.

BSJ: What are some current applications of PALM?

EB: PALM is used as a structural tool to complement other methods. A big area that recently got the Nobel Prize is cryo-electron microscopy, which allows you to determine the structures of individual proteins with high precision. But there can be a lot of ambiguity in how different proteins assemble into larger structures. PALM removes that ambiguity and allows you to understand exactly how things are organized in a macromolecular assembly. I'd say the most important application is in single-particle tracking PALM (sptPALM), where you photoactivate subsets of molecules and watch how they move. Our conception of how live cells work is currently undergoing a revolution. We are realizing that everything in the cell has multiple purposes and is interacting all the time. There is a "stickiness" to how different molecules come together for a tiny period of time, sometimes just hundreds of milliseconds. This is how practically every basic biochemical process works in the cell. In order to really understand the kinetics of these processes, you have to study them at the single-molecule level. For this reason, I believe sptPALM is going to become a very important tool for understanding the "glue" that holds the cell together.

BSJ: We also read about your recent study combining expansion microscopy (ExM) with lattice lightsheet microscopy (LLSM) to image subcellular structures in the mouse and Drosophila brains.[6] What are ExM and LLSM and why did you choose to combine them for imaging brain tissue?

EB: Expansion microscopy came out of Ed Boyden's group at MIT.[7] A tissue is infused with a polymer gel, and fluorophores are chemically linked to the gel. Then you add chemicals that digest all of the biological tissue. You change the osmotic balance of the solution, and the gel expands, giving you a sample that is physically four times bigger than it was before. With super-resolution microscopy, you only focus on one part of one cell—it's like imaging while looking through a straw. You get a really good view of one region, but you have no idea what's happening all around it. Expansion microscopy allows you to see over a much wider field of view. The disadvantage is that as you image sections of tissue, the surrounding fluorophores burn out. Instead of bringing the light from above, lattice light sheet microscopy allows you to bring the light from the side. It also uses ideas derived from Bessel beams, which allow you to make a narrow sheet of light that illuminates only a thin slice of sample at a time, so you don't bleach the fluorescence from regions above and below the layer. The LLSM is very fast because it illuminates an entire plane at once. We can easily acquire

> "You want a microscope that can get down to the molecular resolution if you want to understand how the cell works."

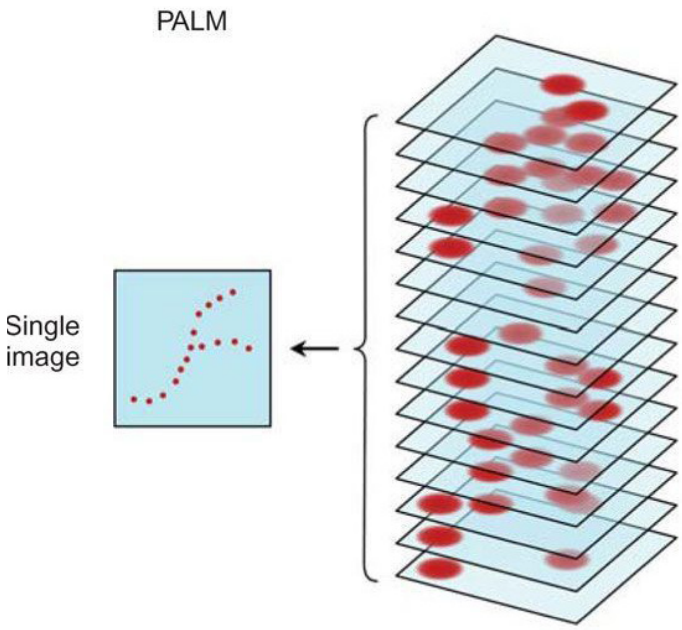


Figure 3: Principle of photoactivated localization microscopy (PALM).[5] *Photoactivatable fluorophores are attached to a protein of interest. Sparse subsets of fluorophores are repeatedly activated, imaged, and bleached. The position of each fluorophore is precisely determined in each frame. Summing together all the frames results in a single super-resolution image of the protein distribution in a sample.*

terabytes of data per day. The normal fly brain is already gigantic on the super-resolution scale, but we were able to image it in just a couple of days. Electron microscopy took about 10 years to accomplish the same task—that's the kind of gain you get with ExLLSM.

BSJ: Why was the *Drosophila* brain an ideal sample for ExLLSM?

EB: First, the study was conducted at Janelia Research Campus, where half of the building studies the fly brain. It is probably one of the most studied systems in neuroscience. Researchers at Janelia have developed fluorescence tools to study the fly brain; they have over ten thousand different genetically perturbed flies. Another nice thing is the scale. While it is certainly a lot bigger than other systems, the fly brain is still tiny enough to fit in our microscope without us having to carve up the brain into many different pieces.

BSJ: In this study, you calculated the brain-wide distribution of synapses in the fly (Fig. 4). How did you validate that the signals accurately represented true synapses and not non-specific background?

EB: That's a good question. In this case, we used fluorescent antibodies, which don't always go to the protein that you want. Electron microscopy (EM) researchers have been working on the fly brain for a long time, and one of the areas of the brain that they have studied exhaustively is the mushroom body, where a lot of learning and memory occurs. By EM, they were able to determine the number and sizes of the synapses. In our study, we developed a pipeline to count the synapses, and we removed any fluorescence signal that was below or above the size of a typical synapse. We then compared our data to the distribution of synapses that the EM researchers found. Ours was a two-color experiment, where we also labeled the dopaminergic neurons across the brain. Some dopaminergic neurons don't have any synapses on them, which has been known from EM. So, you can determine the rate of false positive signals by seeing if there are any synapses on a neuron that shouldn't have synapses. We counted no synapses on the neurons that shouldn't have any and tens of thousands of synapses on the neurons that should. Everything matched up perfectly in all of these controls.

BSJ: How might ExLLSM allow for the correlation of neural structure with neural activity?

EB: There are two ways that people look at neural activity. The gold standard is to use electrodes to determine the electrical signals coming from neurons. That's great, but it's difficult to stick a needle into each neuron. A big area of the last 15 years has been genetically encoded calcium indicators. You can express a calcium-sensitive fluorescent protein in any subset of neurons you want. When an action potential is fired in a neuron, there's an influx of calcium and the neuron lights up for the amount of time that it's firing. This calcium influx is optically read out. At Janelia, researchers have done experiments in which a fly looks at a virtual reality screen and has to follow prey. There are little microscopes that look at the neurons as they fire. You can then kill the fly, take out its brain, and do ExLLSM. This way, you can relate the neural activity to the neural structures and behavior of the fly.

BSJ: You have experience in industry and academia, and you have worked as both an engineer and a scientist. What do you think are the most valuable elements of this skill set?

EB: Several things. First, regardless of what you do, be true to yourself and listen to your internal voice. There's a saying that 90% of everything is BS—that's totally true. You have to be your own toughest critic. There are a lot of things one can do to cut corners or go with the crowd, even if you don't believe what the crowd is saying. But I think there's a larger price to pay in the long term if you do that. Be focused on what you believe is important. The other thing is the value of hard work, which is universal regardless of what you do. I've known many brilliant people in my life, but the people who have succeeded in the end are the ones who are driven, passionate, and work, work, work. If you work 20% harder, that added

"Be true to yourself and work hard—that's true whether you're in academia or industry."

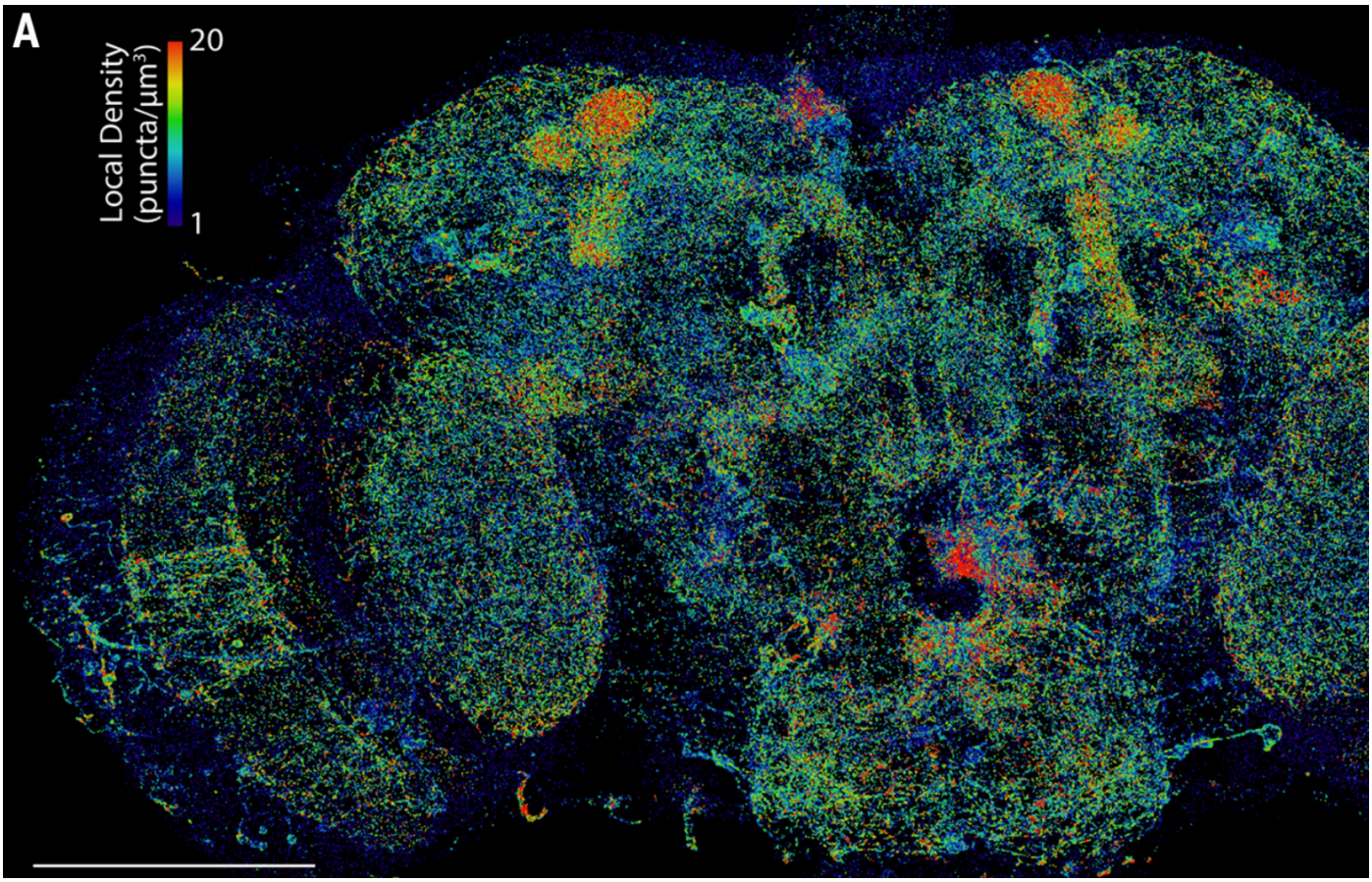


Figure 4: Maximum intensity projection of synaptic proteins. *Synaptic proteins at dopaminergic neurons in the adult Drosophila brain, color-coded by local protein density.*[6]

20% gives you more experience and makes you more productive every year. It really compounds over time until you're two or three times as productive as the next guy. That's not easy, because in life you always have to make compromises of family, work, sanity, and health. But still, the hard work is what will make you stand out and have a real contribution in the end. Be true to yourself and work hard—that's true whether you're in academia or industry.

IMAGE REFERENCES

1. Barande, Jérémy (Photographer). (2015, June 16). Eric Betzig (Photograph). Retrieved from https://www.flickr.com/photos/117994717@N06/18736515949
   Banner image: ZEISS Microscopy (Photographer). (2016, September 9). SK8/18-2 human derived cells, fluorescence microscopy (Photograph). Retrieved from https://www.flickr.com/photos/zeissmicro/29942101073

REFERENCES

2. Jablonski diagram [Online image]. Adapted from https://www.wpiinc.com/blog/post/ca-sup-2-sup-detection-in-muscle-tissue-using-fluorescence-spectroscopy
3. Liu, Z., Lavis, L. D., Betzig, E. (2015). Imaging live-cell dynamics and structure at the single-molecule level. *Mol Cell, 58*(4), 644-659. doi: 10.1016/j.molcel.2015.02.033
4. Betzig, E., Patterson, G. H., Sougrat, R., Lindwasser, O. W., Olenych, S., Bonifacino, J. S. (2006). Imaging intracellular fluorescent proteins at nanometer resolution. *Science, 313*(5793), 1642-1645. doi: 10.1126/science.1127344
5. Gustafsson, M. G. L. (2008). Super-resolution light microscopy goes live. *Nat Methods, 5*(5), 385-387. doi: 10.1038/nmeth0508-385
6. Gao, R., Asano, S. M., Upadhyayula, S., Pisarev, I., Milkie, D. E., Liu, T. (2019). Cortical column and whole-brain imaging with molecular contrast and nanoscale resolution. *Science, 363*(6424), eaau8302. doi: 10.1126/science.aau8302

DRUG USE AND POLICY: A CROSS-DISCIPLINE DIALOGUE

BY SHEVYA AWASTHI, MATTHEW COLBERT, DOYEL DAS, EMILY HARARI, CASSIDY HARDIN, ROSA LEE, MICHELLE LEE, ELETTRA PREOSTI, MELANIE RUSSO, AND SAUMI SHOKRAEE

DR. VERONICA MILLER[2]: *I'm in the School of Public Health, and my perspective is on the regulation of new drug products. I teach a class on the Food and Drug Administration (FDA), drug development, and public health. My research program is concentrated in specific disease areas, in which we facilitate the drug development path.*

DAVID SHOWALTER[1]: *I'm a sixth-year PhD student in Sociology here at UC Berkeley. I use qualitative and ethnographic methods to study drug use and drug policy. In particular, I focus on opioid use and injection drug use. I come from a background of harm reduction work—for the past ten years, I've been involved in syringe exchange and overdose prevention programs. An important underlying principle to my research is helping people who use drugs live healthier and happier lives.*

BREANNA FORD[3]: *I am a fifth-year graduate student in Endocrinology, and most of my research is about molecular toxicology. I primarily look at pesticides and their harmful effects on humans. I also look at endogenous metabolites formed by pharmaceuticals and their direct interactions with the body.*

DR. JOHANNES 'HAN' DE JONG[4]: *I'm a postdoc in the Molecular and Cell Biology department. I study drugs as chemicals and how they affect the brain. Before that, I did my PhD in the Netherlands, where I studied sugar in the context of food addiction. I was also a member of a liberal political party called Democrats 66, which was the first party to legalize drugs in the Netherlands in the 1990s. I have also been involved in several harm reduction programs as a volunteer and an educator.*

Research on both recreational and pharmaceutical drugs spans fields from neuroscience to sociology. This multidisciplinary approach informs regulatory drug policy and shapes the way drugs are perceived in society. *The Berkeley Scientific Journal* sat down with a diverse group of researchers (Fig. 1) to hear their insights into the mechanisms of drug addiction, challenges in drug regulation, and international attitudes toward drug rehabilitation.

BSJ: What are the neurobiological mechanisms underlying addiction?

JDJ: The first phase in the development of addiction is called sensitization. Drugs affect the dopamine system in your brain, which reinforces behaviors that make you feel good. For instance, if you smoke cigarettes, you might not actually enjoy smoking, but the nicotine stimulates your brain to reinforce the behavior. That leads to the second phase, which is the conditioning phase. If you are trying to quit smoking and you see cigarettes on the table, then they are a salient cue for you to start smoking again. Over time, the drugs hijack the reward systems in your brain—both the dopamine system and the system that controls it break down. You cannot tell someone to stop being addicted—the very brain areas that they need to do that are destroyed.

BSJ: There is the historical example of veterans of the Vietnam War who used heroin while in Vietnam, but no longer sought it out when they got back to the United States. Could you elaborate on this phenomenon?

JDJ: This is a famous example, along with the example of Rat Park, which was a series of studies in the 1970s on drug addiction in rats.[5] The rats were exposed to a solution of morphine and would constantly drink it when in an isolated environment. However, when exposed to a more socially enriching environment, the rats would stop drinking the morphine. The same happened to heroin users who came back from Vietnam. Stressful environmental and social factors, combined with the effects of the chemical itself, caused the heroin addiction. After Vietnam, veterans came back to an environment where these social and psychological stressors no longer existed, so many people were suddenly not addicted anymore. However, there was a certain percentage of veterans who continued to be addicted to heroin despite having a family and being happy at home. We might conclude that these people were simply addicted to the chemical itself. Thus the biological, psychological, and social factors lead to addiction, but there's an ongoing debate about which factors are more influential.

DS: These examples are extreme cases, since most people who try heroin don't become addicted to it. There's a big gap between the number of people who have ever used heroin and the number of people who fit the criteria for heroin use disorder. There's so much more about what leads people to destructively use drugs beyond simply consuming the chemical. It's important to underline the difference between using a drug and being addicted to it.

VM: I wanted to ask a question from the perspective of someone who studies the regulation of pain medicine. Suppose someone breaks their leg while skiing. They get a prescription for pain relief medication, and two weeks later they become addicted to the medication. How do pain relief and addiction interact in the brain?

JDJ: The mu opioid receptor is the brain's natural pain-regulating system. That system is under tight control. Every pain medication that works on this system in some way down-regulates the mu opioid receptors. Over time, these interactions can change the system in a way that makes you addicted to the medication. Morphine has one of the strongest interactions with the mu opioid receptor. The trick is that morphine works in your body, but the blood-brain barrier prevents it from entering your brain. This makes morphine a good pain medication but not super addictive. On the other hand, heroin can sneak into the brain and become addictive. The holy grail of pain medication is to make a chemical that will only act in your body and not in your brain.

DS: Pain, whether physical or psychological, is at the root of why people use drugs. In a clinical setting, it's definitely true that widespread availability of prescription opioids is what got a lot of people hooked on the pills. However, about three quarters of people who are dependent on prescription opioids will say they first got them not from a doctor, but from a friend who got them from a doctor. We can't just cut people off from the pills, because the people who are prescribed the pills aren't the ones who are having problems using them. Instead, we need to think about how to prevent strong medications from getting into the hands of people who aren't able to use them in a safe way. The concern about addiction is sometimes misplaced when we just blame doctors or pharmaceutical companies.

BSJ: How does the FDA treat clinical and recreational drugs differently? Are there any instances of hypocrisy in the legislation?

VM: The FDA regulates medicines. These types of drugs include monoclonal antibodies, biologics, and vaccines; they all have a specific medicinal purpose. The drug packaging insert tells you what the drug is supposed to be used for and how it is supposed to be used. The FDA also has a regulatory oversight. They regulate what drugs come into the country, and they can go to the border and inspect shipments. Today, the FDA's efforts have expanded into social networks, as they examine websites that sell illicit drugs. The FDA has three primary roles: the first is to make sure we

have safe medicines to treat pain. The second is to make sure we have medicines to counteract addiction. The third is to oversee the import of drugs. Ultimately, the FDA looks at benefit and risk: what is the benefit that the drug provides, and what is the risk that the drug poses? The FDA cannot regulate clinical practice, because once a drug is approved, doctors have the discretion to prescribe drugs off-label, meaning in a way that is not indicated on the packaging insert. In a way, the FDA has its hands tied in directly interfering with the opioid epidemic besides encouraging the development of new drugs to treat addiction.

BF: I have a follow-up. The question addressed hypocrisy, but I'm not sure if hypocrisy is the most holistic term for what I think of as gaps in FDA oversight. To me, something that potentially falls into that category is that the FDA regulates food and pharmaceutical medications, but it doesn't regulate supplements and natural products that don't have a stated therapeutic value. Is it a problem that the FDA does not regulate these products?

VM: There are other agencies that regulate food, like the USDA. But the minute a supplement is claimed to have a medical benefit, it becomes a drug, and then the FDA can bar it from being sold as a drug. So I think that whole area is a wide open field, and much more could be done by the FDA.

BSJ: What obstacles posed by the FDA interfere with the process of drug research?

JDJ: A major issue is drug scheduling. In terms of addiction research, most addictive drugs are not Schedule I. I can easily do cocaine research because it's a Schedule II drug. It's the same for ketamine, a Schedule III drug that is frequently used for depression research. Meanwhile, marijuana is Schedule I, so nobody in the US can easily study it. For scientists, the bureaucratic process makes it difficult to conduct this research. MDMA, a methamphetamine, is another example. MDMA might have potential for treating PTSD, but it is very difficult to study because it is a Schedule I drug.

BF: Our lab does a lot of drug development research, and all of it is very pre-clinical—it will be years before our research interacts with the FDA. For most drug development research, the FDA does not have any direct interaction with researchers in the academic environment unless they are studying items that are analogs of known Schedule I substances or are part of the synthesis pathways of these substances.

BSJ: Besides drugs, pesticides are other chemicals that can be taken up by the human body. How do pesticides affect human health?

BF: We have this general belief that things that are natural must be good. This is reasonable in many ways. We eat food, which grows from the ground, and therefore we think things that grow from the ground must be okay, whereas things that are synthesized in a lab must be horrible for us. Pesticides are an example of this; we think of them as intentional, synthesized poisons, and therefore they must be bad. But if you think about pesticides as they relate to pharmaceuticals, there's a huge amount of overlap. Antibiotics and antifungals are really just pesticides that we use for a very specific purpose. This is where some of the ethical and psychological issues around what we think of as natural and unnatural come into play, and where the regulatory aspect becomes really important, because all of this is about managing risk and benefit. Is the risk of potential adverse effects of a pesticide, whether used in an agricultural or pharmaceutical way, worth the benefits it affords us both as individuals and as a group?

Figure 1. Cross-discipline panel. *From left to right: Researchers David Showa*
a group of BSJ writers to share their insights on the role of drugs in modern

VM: Without getting too much into the endocrine system, most foreign things introduced to the body get metabolized by the liver. If the liver's enzymes have too much competition with these foreign elements, whether it's aspirin, Tylenol, ibuprofen, or pesticides sprayed on your apple, this interferes with other liver functions and drug metabolisms. It's all part of the same system.

BF: Exactly. The same can be said about the use of non-regulated drugs. They're all undergoing the same metabolism, and cross-interactions between illicit drugs and established pharmaceuticals are going to occur in your body. Understanding these interactions is incredibly important to maintaining human health.

alter, Breanna Ford, Dr. Han de Jong, and Dr. Veronica Miller sat down with society.

VM: A famous case of this is with St. John's wort, [a flowering plant with possible antidepressant activity]. It plays around with those liver enzymes and can seriously diminish concentrations of metabolites that you actually need. This is why a doctor will always ask a patient to list all of the drugs they are taking, even if the drugs are over-the-counter.

BSJ: What are some shortcomings of our current attitudes towards addiction rehabilitation, especially when it comes to illicit substances?

JDJ: What is generally true in the US is that people are high-minded and have strong principles. Whereas in Dutch culture, we are not proud at all.

VM: The Dutch are very pragmatic.

JDJ: Exactly! When I tell people in the US that I am Dutch, they immediately think about parties in Amsterdam. We did not legalize marijuana because we are all hippies. We legalized marijuana because Dutch policymakers, who are literally the most boring people in the world, looked at the data and worked out the best way to approach the problem. In the Netherlands, a country of 20 million people, there are 30 thousand heroin addicts. All of them are in treatment. The average age of the heroin addict increases by one year every year because the population is contained to the people in treatment. Additionally, no market exists for a heroin dealer, since heroin addicts are able to get the drug from clinics for free and use them in a safe space with a clean needle. All it takes is for us to step away from our principles. It might feel wrong that a certain percentage of your income tax supports heroin addicts. After all, they do not work, they receive money from the state, they get a free place to live, and they are allowed to take drugs. But it is important to take a step back and think pragmatically about what the future is going to look like if we implement these policies. That is what happened in the Netherlands. I think this is perhaps the major flaw in US policy: policies are based in principles and not in facts.

VM: It is a science, whether social or medical, of what works and what doesn't.

DS: Another thing is the assumption that if you use drugs, you have to go to treatment. Most people who quit using drugs don't go to treatment to do it. Instead, most people quit using drugs by basically aging out of it. Drugs are predominantly used by young people, those who do not have other things going on in their lives, or those who are seeking relief from something in their lives. But people grow up and get jobs, start families, fall in love, or find things that matter to them. These things fill the gaps that the drugs previously filled. As a result, these people no longer need drugs. There is also a lot of talk about the supply-side factor, such as overprescription. However, opioids were overprescribed in places that already had a large demand for drugs. These are usually places that have had substance abuse problems for decades. If we fix those root causes, there is going to be less of a demand for drugs. Therefore, I believe that the best treatment policy is ensuring that people have a good place to live, have a way to support themselves either through work or the welfare state, can form meaningful relationships, and aren't separated from their children or family members. If we achieve these things, the downstream consequence is that there will be fewer reasons for people to turn to drugs in the first place.

IMAGE REFERENCES

1. David Showalter [Photograph]. Retrieved from https://sociology.berkeley.edu/graduate-student/david-showalter
2. Veronica Miller [Photograph]. Courtesy of Veronica Miller.
3. Breanna Ford [Photograph]. Retrieved from https://nst.berkeley.edu/users/breanna-ford
4. Johannes 'Han' de Jong. [Photograph]. Retrieved from https://www.lammellab.org/people/

REFERENCES

5. Alexander, B. K., Coambs, R. B., & Hadaway, P. F. (1978). The effect of housing and gender on morphine self-administration in rats. *Psychopharmacology, 58*(2), 175-179. doi: 10.1007/BF00426903

BRIDGING SCIENCE AND BUDDHISM: TOWARD AN EXPANDED UNDERSTANDING OF MIND

Interview with Professor David E. Presti

BY SHEVYA AWASTHI, DOYEL DAS, CASSIDY HARDIN, ROSA LEE, MELANIE RUSSO, AND ELENA SLOBODYANYUK

Dr. David E. Presti *teaches neurobiology, psychology, and cognitive science at the University of California, Berkeley. Before coming to Berkeley, Professor Presti worked in the clinical treatment of addiction and post-traumatic stress disorder. He also teaches neuroscience to Buddhist monks and nuns. In 2018, he wrote the book "Mind Beyond Brain: Buddhism, Science, and the Paranormal." In this interview, Professor Presti discusses the nature of the mind, empirical approaches to studying consciousness, and the value of fostering a dialogue between science and Buddhism.*

BSJ: We read that your training is in biophysics and clinical psychology. What led you to integrate spirituality into these fields?

DP: As an undergraduate, I developed a keen interest in quantum physics and Einstein's theories of relativity because they seemed to address what science could reveal about the nature of reality. This was also an era when ideas from Asian spiritual traditions were beginning to penetrate American culture. I read many books related to these traditions, which spoke to a view of mind and nature somewhat different from that of the Western world. After graduating, I came to California to study theoretical physics as a graduate student at Caltech. I joined the research group of Kip Thorne, who recently received a Nobel Prize for his role in the first measurements of gravitational radiation. Stephen Hawking was a visiting professor at Caltech the year I started, and I attended lectures on quantum mechanics by Richard Feynman. I was really steeped in theoretical physics, and at the same time continued to be very interested in the mind. I heard about Max Delbrück, one of the founders of modern molecular biology, who at this point was interested in the evolution of human cognitive capacities. I took his class on biophysics, and he gave me an opportunity to work in his microbial genetics lab for the summer. He also advised me to learn some biology. So, I switched from theoretical physics to experimental molecular biology. After receiving my doctorate, I did postdoctoral work in neurobiology, and then studied cognitive psychology at the University of Oregon because I wanted to learn more about human perception. I ended up getting another PhD in clinical psychology and did a clinical internship at the Veterans Hospital in San Francisco. I got a job there and worked in the treatment of addiction and post-traumatic stress disorder for the next 10 years. While still engaged in that work, I began teaching neurochemistry at UC Berkeley and then got the opportunity to come here full-time. My primary interest throughout has always been expanding the way we think scientifically about the nature of mind and consciousness.

BSJ: Your most recent book, *Mind Beyond Brain: Buddhism, Science, and the Paranormal*, explores what the encounter between science and Buddhism can teach us about consciousness and reality.[2] What motivated you to write this book?

DP: I have been wanting to produce this book for quite a few years. My first book covers the content of my introductory neurobiology class here at Cal.[3] In that book, I make it clear that there is still a great deal of mystery remaining in neurobiology, and at the end I propose a number of ways to broaden the science of consciousness. *Mind Beyond Brain* (Fig. 1) is a sequel to that, expanding upon one particular way forward in the study of mind. It's partly motivated by my close association with researchers who study anomalous phenomena at the University of Virginia School of Medicine. These phenomena include near-death experiences, in which approximately 20% of survivors of clinical death recall extremely vivid experiences after they are revived.[4] Sometimes this includes a vivid perception of the scene of their near-death from an out-of-body perspective. Such phenomena are completely inexplicable in terms of present assumptions about the mind-body relationship. The connection with Buddhism is related to the longstanding interest that the Dalai Lama has had in science. For several decades he has engaged in conversation with scientists on the study of mind and the physical world, and out of these engagements programs have developed to teach science to Tibetan Buddhist monks and nuns. I was fortunate to cross paths with the first of these programs 15 years ago and have on multiple occasions taught neurobiology and dialogued about science with Tibetan monastics in India, Bhutan, and Nepal (Fig. 2). They are deeply interested in questions about mind and world, and yet their tradition draws upon a worldview that is complementary to our own. That's the story of this book.

BSJ: What are your working definitions for the mind and for consciousness? How do these relate to the mind-body problem?

DP: I define mind as our mental experience: our thoughts, feelings, and perceptions. Consciousness is the awareness of this experience, awareness of what it's like to be us. We are not robots that mechanically perform things without any experiential awareness.

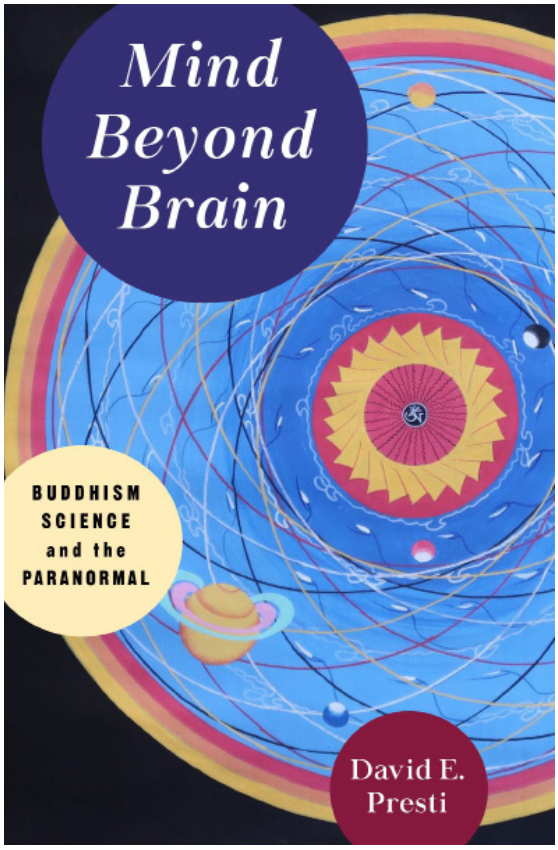


Figure 1: Professor Presti's recent book, "Mind Beyond Brain: Buddhism, Science, and the Paranormal." *This book explores how evidence for anomalous phenomena can productively impact the Buddhism-science conversation.*[2]

We have mental experience, something that is irreducibly subjective and not manifestly physical. The experience of sweetness or saltiness may be related to the presence of sucrose molecules or sodium ions, respectively, but it is not embedded in the molecule or ion. Rather, it depends on the molecule interacting with our nervous system and somehow giving rise to the experience. We may hypothesize that experience emerges in some way from physical processes in our body and brain, but we don't have a description of how that happens. This is the so-called mind-body problem.

BSJ: Why do you believe a paradigm shift for understanding consciousness is forthcoming?

DP: There has only been a handful of major paradigm shifts in the history of modern science. When Earth got displaced from the center of the universe in the Copernican Revolution, that led to hundreds of years of physics and astronomy explaining the organization of the cosmos. In biology, revolution around evolution led to understanding all of life on Earth as interconnected and developing diversity over long periods of time via processes of variation and selection. Einstein's work on relativity indicated that space and time are dynamically interconnected and vary as a function of relative motion and the presence of matter. Finally, one of the biggest revolutions so far has been quantum physics—suggesting that the fundamental structure of the material world is much more fuzzy and interconnected across space and time than we previously imagined. So, we have these four big revolutions in physics and biology, and my guess is that something even bigger will take place when we appreciate an enfolding of our own conscious awareness into the nature of the physical.

BSJ: What are psi phenomena?

DP: Psi phenomena are phenomena that transcend our current capacity to explain by any known physical mechanism. In the late 19th century, a group of British researchers created the Society for Psychical Research. They investigated phenomena that were related to various human experiences—hence the term "psychic," or "psyche," which refers to the mind. The term "psi" came to describe the phenomena. For example, there might be some kind of direct mind information transfer between people, called telepathy. Someone might get information about something happening at a distance, called clairvoyance. Someone might get information about something that hasn't happened yet; that's called precognition. These are phenomena that go beyond our ability to explain via what we presently know about sensory perception and information transfer. There may be straightforward physical explanations that we simply haven't uncovered yet, or they may indicate the need to radically alter the way we think about the relationship of consciousness and the world.

BSJ: What are some of the current psychophysiological methods used to measure psi phenomena? What are some challenges associated with these methods?

DP: There are small numbers of scientists at places like the Institute of Noetic Sciences in Marin County who are conducting laboratory studies of psi phenomena. Such studies are difficult and the effects, though highly significant, are often small. In addition, observations may be prone to perturbations from the environment or the participant's thinking. If there are ways in which mind is impacting the physical world, then all kinds of things might happen in constructed experiments, things that may be very difficult to control. To me, most of the juice is in increasingly sophisticated empirical documentation of spontaneous phenomena, looking for patterns that may suggest hypotheses for further investigation. These spontaneous occurrences are generally experienced in the context of powerful emotionality, often traumatic events: for example, death, near-death, or serious accident or illness.

BSJ: At the end of *Mind Beyond Brain*, you describe a distinction between the "supernatural" and the "super natural." What do these two concepts represent?

DP: Terms like "psi," "paranormal," and "supernatural" are often used synonymously to describe weird and inexplicable phenomena. Especially the latter two are saddled with a great deal of pop-culture baggage. "Super natural" (two words, the space is important)—a phrase coined by my colleague Jeffrey Kripal at Rice

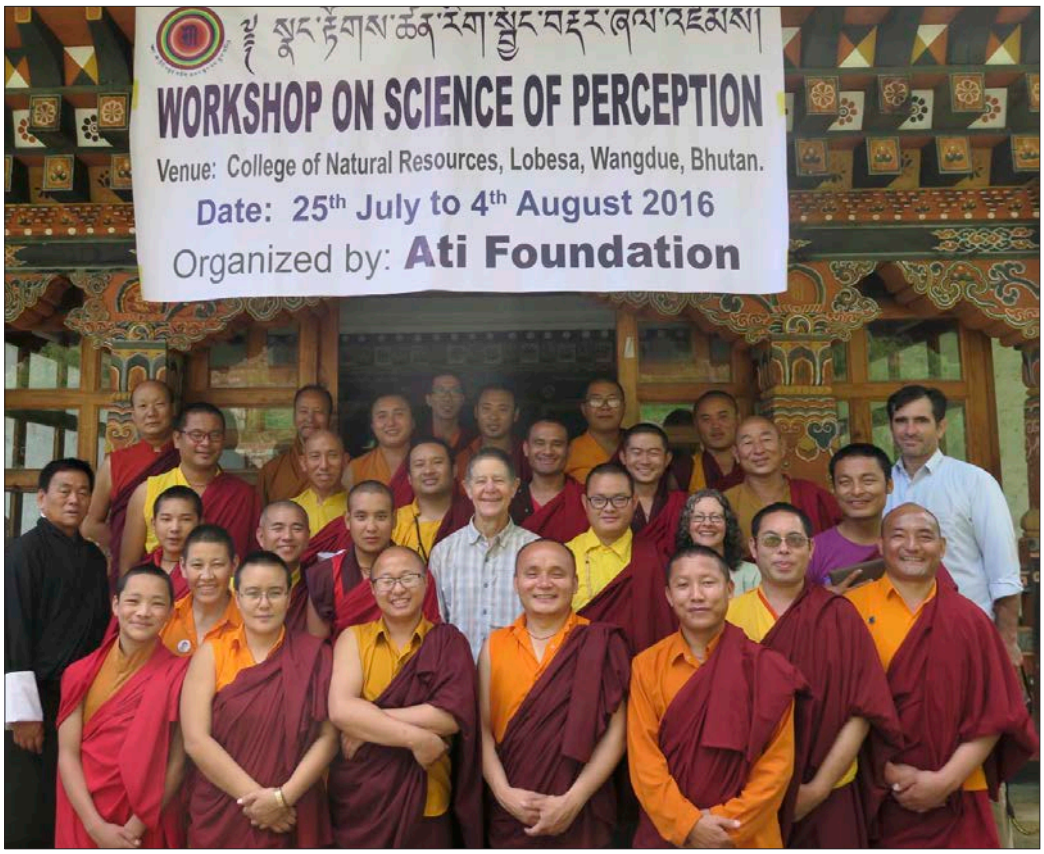


Figure 2: Professor Presti with a group of Buddhist monks and nuns in Bhutan. *Presti has been involved with the Science for Monks and Nuns program for 15 years. Also pictured are Dr. Kristi Panik, psychiatrist at UC Berkeley University Health Services (and Presti's wife), and Dr. Bryce Johnson, UC Berkeley PhD in Environmental Engineering and director of the Science for Monks and Nuns program. Image courtesy of David Presti.*

University[5]—refers to natural phenomena that go beyond ordinary experience in a way that is currently inexplicable: things like out-of-body and other vivid perceptions during near-death experiences; apparitions associated with the death or serious injury of someone with whom one is emotionally connected; precognitive thoughts and dreams; and so forth. This two-word phrase is meant to emphasize that these are natural phenomena, occurring widely, and at the same time kind of super. They are in no way beyond science, and they can be investigated using the methods of science.

BSJ: You touched on near-death experiences. Why do many scientists not recognize these and other anomalous phenomena as meaningful?

DP: That is a hugely interesting question. Many folks who vigorously refuse to be open to the occurrence of these phenomena often don't know much about the empirical data, and moreover there is frequently a reluctance to learn more about it. It can be a very emotional resistance—people can get really upset about this stuff. One guess is that it's threatening to our sense of security in understanding the nature of reality, that we know what's going on. Even though there are mysteries like dark energy, dark matter, and how the brain generates consciousness, we generally feel pretty secure about how biophysical science places us within the material universe. However, these anomalous phenomena force us to step back and say, "Well, perhaps things are way more weird and mysterious than I thought."

BSJ: You also mention a "transmission hypothesis" for explaining anomalous phenomena. Can you tell us more about this framework?

DP: The working assumption in mainstream neuroscience is that whatever our mind is, it's completely generated by our body and brain. That is certainly consistent with a lot of things, but an equally consistent hypothesis is that our awareness comes partly from within our body and partly draws from other places. That's the "mind beyond brain." It's similar to what a lot of spiritual traditions would say—that we're channeling an aspect of something divine or cosmic. Historically, folks have drawn parallels with radio and television, and these days the best analogy would be connectivity to the internet. Using a smartphone, you can connect to an enormous amount of information from all over the planet. If you've never seen such a device before, you would probably assume that all the information is coming from inside the phone. But that's not the case; the phone is receiving what's transmitted through the cellular network. That's essentially the transmission hypothesis: part of what we are able to experience may be coming from something beyond our direct senses, at least insofar as we currently understand them.

BSJ: You write of a refined approach to studying subjective experience that involves careful introspection and analysis. How does Buddhism embody this approach?

Figure 3: Workshop on sensory neurobiology at the Kopan Nunnery in Nepal (2018). *Image courtesy of David Presti.*

DP: In our modern scientific tradition, we try to understand the world as detached observers—the essence of our science is objectivity. We build telescopes and microscopes and all sorts of probes to investigate things outside of our mental space. In Buddhism and a number of Asian philosophical traditions, they have a different worldview. The Dalai Lama and others have described aspects of Buddhist tradition as being a kind of internal science. They use contemplative practices to focus inward and deeply investigate the nature of mind. It's not that one's wrong and one's right—they're both right—but they achieve different things. It's easy to see that all of modern technology came out of Western science—the Tibetan Buddhists didn't invent iPhones. But the Western scientific tradition didn't lead to deeply introspective and analytic meditation practices.

BSJ: How do you see the relationship between Buddhism and science changing in the future?

DP: Buddhism is an ancient spiritual tradition that has long been deeply interested in the nature of mind and reality. Our own scientific tradition has also been around for a while and is interested in the same things. I think the conversation between science and Buddhism is really just getting started. While it has taken off largely due to the Dalai Lama's interest and influence, now all these Buddhist monks and nuns are also learning enough about science to be able to join the conversation (Fig. 3). In addition, scientists are learning about and exploring contemplative traditions and the worldview from which they arise. My guess is that deepening the science of consciousness will benefit enormously from engagement with a tradition that gives mind a far more central role in the world. This relationship is a multi-generational experiment; we'll see how it goes!

BSJ: What are some future directions in consciousness research?

DP: One way is to continue to ever more deeply investigate the structure and function of the brain and body. Biophysical science is telling us that wherever we look, we see more layers of interconnection. The immune system, the endocrine system, the nervous system—they're all constantly talking to each other. Moreover, our beliefs have a huge impact on the physical functioning of our bodies. In medicine, we call this the placebo effect. The placebo effect often gets dismissed, but it's truly the most amazing thing! Simply believing in something has an impact on its physical efficacy. Investigating these phenomena will give us more insight into mind-body connectivity. Another way is to expand our capacity to empirically explore subjective experience, learning some things from contemplative traditions such as Buddhism. And in addition we can pay attention to and further investigate anomalous psychological phenomena, the things that do not fit into our current framework of explanation. Ultimately this is about deepening our understanding of who we are as conscious living beings and how we fit into what we call the physical world. Are we completely explicable in terms of configurations of atoms that after billions of years of physical and biological evolution somehow bubble up consciousness? Or is something weirder going on, something where mind plays a more central role? How we choose to answer these questions has enormous social repercussions because it informs how we see who we are and what our place is in the world. That is likely to impact how we treat one another, the environment, and future generations. That's the deep reason I'm interested in this subject.

REFERENCES

1. Presti, D. E. (2018). *Mind Beyond Brain: Buddhism, Science, and the Paranormal*. New York, NY: Columbia University Press.
2. Presti, D. E. (2016). *Foundational Concepts in Neuroscience: A Brain-Mind Odyssey*. New York, NY: W. W. Norton.
3. Greyson, B. (1998). The incidence of near-death experiences. *Med Psychiatr, 1*, 92-99. doi: 10.1080/08998280.2018
4. Strieber, W., & Kripal, J. J. (2017). *The Super Natural: A New Vision of the Unexplained*. New York, NY: TarcherPerigee.

BRINGING A NEW PERSPECTIVE TO GENE REGULATION

Interview with Professor Elçin Ünal

BY MATT COLBERT, EMILY HARARI, MICHELLE LEE, ELETTRA PREOSTI, SAUMI SHOKRAEE, AND NIKHIL CHARI

Dr. Elçin Ünal *is an Assistant Professor of Genetics, Genomics, and Development at UC Berkeley. After growing up in Turkey, Ünal came to the United States for graduate school where she began her research career. Her lab currently studies meiosis in the context of gene regulation and cellular quality control. We spoke to Professor Ünal about her work on the processes of kinetochore inactivation prior to meiosis and mitochondrial segregation during cell differentiation.*

BSJ: How did your experiences growing up in Turkey make you interested in biology, and what led you to continue that type of research in the United States?

EU: I had a very engaging biology teacher in high school. Normally, you think of biology as memorization-based, but it wasn't taught that way in his class. In Turkey, the college admissions process is a little bit different than in the U.S.; you need to take a national exam, which is basically the only determinant of whether or not you make it into college. It's very competitive, and you announce your major before you actually get there. My teacher informed me about a new department, Molecular Biology and Genetics, that just opened a few years ago. The year I applied to college was the year Dolly the sheep was cloned, so there was a lot of excitement around molecular biology. Every year, my department would only get about 15 students. The training was great, but doing research was difficult for economic reasons. So I followed the footsteps of the previous years' classes, and went for an internship abroad. I actually went to graduate school at the same place abroad the next year.

BSJ: Where was that?

EU: Johns Hopkins.

BSJ: Much of your current research is in kinetochore inactivation. Could you elaborate on what a kinetochore is and why the activation and inactivation of kinetochores is important for meiosis?[1]

EU: A kinetochore is a large macromolecular structure that connects the chromosomes, which contain a cell's DNA, to the division machinery, namely the microtubules and the cytoskeleton. Kinetochores are essential for genome segregation during both mitosis and meiosis. What's interesting about meiotic cell divisions is that, prior to the divisions themselves, there are a lot of changes that occur to the chromosomes. The DNA is first shattered into pieces by programmed double-strand breaks, which then allow chromosomes to undergo recombination. Meiosis is unique because the genome size is reduced by half; chromosomes are replicated, and then segregated twice. To accommodate that, the kinetochore also undergoes changes, and this is observed in any model that has been studied so far, from yeast to mice. It has been found that the kinetochore loses its ability to interact with microtubules during prophase, while chromosomes are being repaired and undergoing recombination. Then prior to chromosome divisions it gains back that activity.

In my postdoctoral work, we asked why it matters for the cell to go through this sort of inactivation and re-activation, and we found that of more than 40 kinetochore subunits, there was one protein called Ndc80 that dramatically changed in abundance between the two stages of kinetochore activity. We overexpressed this protein in a cell during a time when it's normally not present, and we observed the cells revert from a meiotic to mitotic phase. That told us that kinetochore inactivation is probably important for establishing a meiosis-specific chromosome architecture during prophase. Since one lynchpin subunit dictated when this entire complex was functional during meiosis, we really wanted to understand how this gene is regulated at all steps, and that's how we got into this LUTI mRNA business.

BSJ: In addition to the main open reading frame (ORF) which codes for a specific protein, long undecoded transcript isoform mRNAs, or LUTI mRNAs, contain several upstream open reading frames (uORFs). What's the difference between the ORF that codes for the Ndc80 protein and the uORFs we find in LUTI mRNAs?[2]

EU: The LUTI mRNA is basically a longer version of the canonical mRNA. It has an entire ORF for a given gene—in this case, the Ndc80 ORF. In addition to this, due to its extended 5' region, it has uORFs. These are short translation units that have their own start and stop codons. How the translational machinery normally works is that the 40S ribosome subunit engages with the mRNA and starts scanning from the 5' end; when it finds an AUG start codon the ribosome is fully assembled and starts translating. Then, when the ribosome encounters a stop codon, it is released from the mRNA. In LUTIs, the uORFs compete with the main ORF, so the ribosome abortively engages with the short translation units but never gets to properly translate the actual ORF (Fig. 1). This molecule looks like an mRNA that would normally be decoded into protein, but in the case of LUTI, it does the opposite. The message itself is not translated, but when the LUTI is transcribed, the production of the canonical mRNA is repressed. In this way, by coupling both transcriptional and translational repression, mRNAs can effectively shut down protein synthesis. Basically, the cell can use the same components that it employs to activate gene expression to repress gene expression.

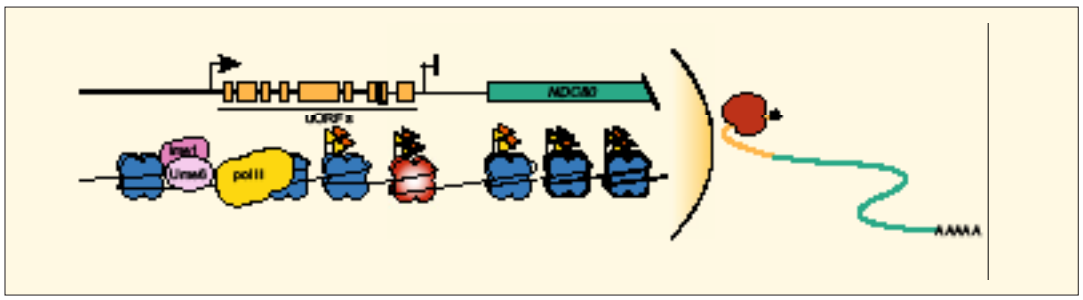


Figure 1: Translation diagram. *Upstream ORF prevents the ribosome from translating the downstream NDC80 ORF.*[2]

BSJ: Why can we call LUTI a mRNA if it doesn't encode a protein?

EU: It has all the information we normally find in mRNAs. By all criteria, an mRNA should have a cap, a poly-A tail, and an open reading frame, and it should be transcribed by RNA polymerase II. LUTIs fit all that criteria. In an mRNA-seq experiment, these non-coding LUTIs will be clustered together with all the other mRNAs because they contain all the other necessary signatures. They still have a coding potential, but because of the uORFs they become non-coding. We classify it as a LUTI because even though all the essential components of mRNA are there, it's not making protein.

BSJ: Could you elaborate on the transcriptional and translational mechanisms affected by LUTI-mRNA gene-regulation, specifically in kinetochore inactivation?[2]

EU: For meiosis in yeast, there are two major transcription factors that drive meiotic progression. One turns on early in response to both intrinsic and extrinsic cues. This first transcription factor activates genes involved in double-strand break formation, recombination, and DNA replication, but it also turns on about a hundred of these LUTIs, one of which is this NDC80LUTI. The transcription factor binds approximately 600 base pairs upstream of the ORF. When that happens, a message is transcribed that the mRNA is starting from an alternative upstream promoter. That message's transcription actually inhibits transcription from the canonical promoter. That's how the transcriptional repression works. Then you make a longer version of an mRNA—a LUTI mRNA—which is different from the canonical mRNA due to an approximately 500-nucleotide upstream extension. That extension contains uORFs which repress translation.

The second key transcription factor is activated just prior to the meiotic divisions, as the first one's activity goes down. It has a binding site in the canonical promoter, which codes for the regular NDC80 mRNA. This leads to a switch in production between two different NDC80 isoforms, leading to re-activation of the kinetochores. So, a cell can turn on either an ORF-directed or uORF-directed transcription factor, depending on whether it wants to activate or inactivate kinetochores.

BSJ: Besides kinetochore inactivation, are there any other events in meiosis or elsewhere that may be regulated by LUTI expression?

EU: I have a team lab with another professor, Gloria Brar. We're collaborators and basically best friends. Her group does a lot of global measurements of mRNA translation and quantitative protein measurements. When you look at meiotic prophase, NDC80 mRNA is expressed very highly, but because it's a LUTI it actually has a negative impact on protein production. Following our lab's work on NDC80, she used these global datasets taken over time during meiotic progression to see if there are other mRNAs that anti-correlate with their protein levels. Her lab found that 389 genes, about eight percent of the yeast genome, are expressed at different times through these luti-mRNAs, recognized by their signature uORFs. We're specifically looking at the genes regulated by this first meiotic transcription factor to understand the genome-wide rules for both transcriptional and translational repression. Beyond that, we looked at individual transcripts in human and fly cells, to see whether this kind of combinatorial integrated mechanism exists in other species. It seems to be conserved, and any kind of transcription factor-dependent gene expression program (which is pretty much every process in the cell) can, in theory, be regulated this way. It's just a matter of looking with a different lens and not making the assumption that mRNAs needs to make proteins. If you drop that simple assumption, then you start finding these mechanisms.

> "By coupling both transcriptional and translational repression, mRNAs can effectively shut down protein synthesis."

BSJ: How do you think the LUTI mRNA mechanism is evolutionarily advantageous?

EU: I think there is a nice parsimony in the system, in that the same trans-acting machinery is used to coordinate up-regulation and down-regulation of genes at the same time. Normally, when we think about anything in the cell that is dependent upon transcription factor regulation, like differentiation or sensing environmental stresses, we mostly think about how the cell activates a group of genes. But there are also genes that need to be inactivated. Cells can evolve a transcription factor and repressor separately to turn on and off genes at the same time. In this case, however, you evolve a single transcription factor, and the cis binding sites on the DNA sequence are what determine gene expression. If the transcription factor binds an upstream promoter, a LUTI mRNA can be produced; if it activates a downstream promoter, a normal protein-producing mRNA will be produced.

The other trans-acting factor is the ribosome. Ordinarily, we assume the ribosome always makes proteins. However, in this case, the ribosome associates with cis elements within the mRNA, which then prevent productive translation. Thus, the body is relying on the same toolkit to do both jobs, which is evolutionary advantageous.

BSJ: Like a transcriptional Swiss Army knife?

EU: Exactly!

BSJ: The meiotic mechanisms you study are critical to ensuring healthy cell progeny in yeast. Kinetochores are not the only cellular structures subject to this regulation, and your research also focuses on mitochondrial inheritance. Cells allocate half of their mitochondria to pass down to their daughter cells. Where are mitochondria normally found in the cell?[3]

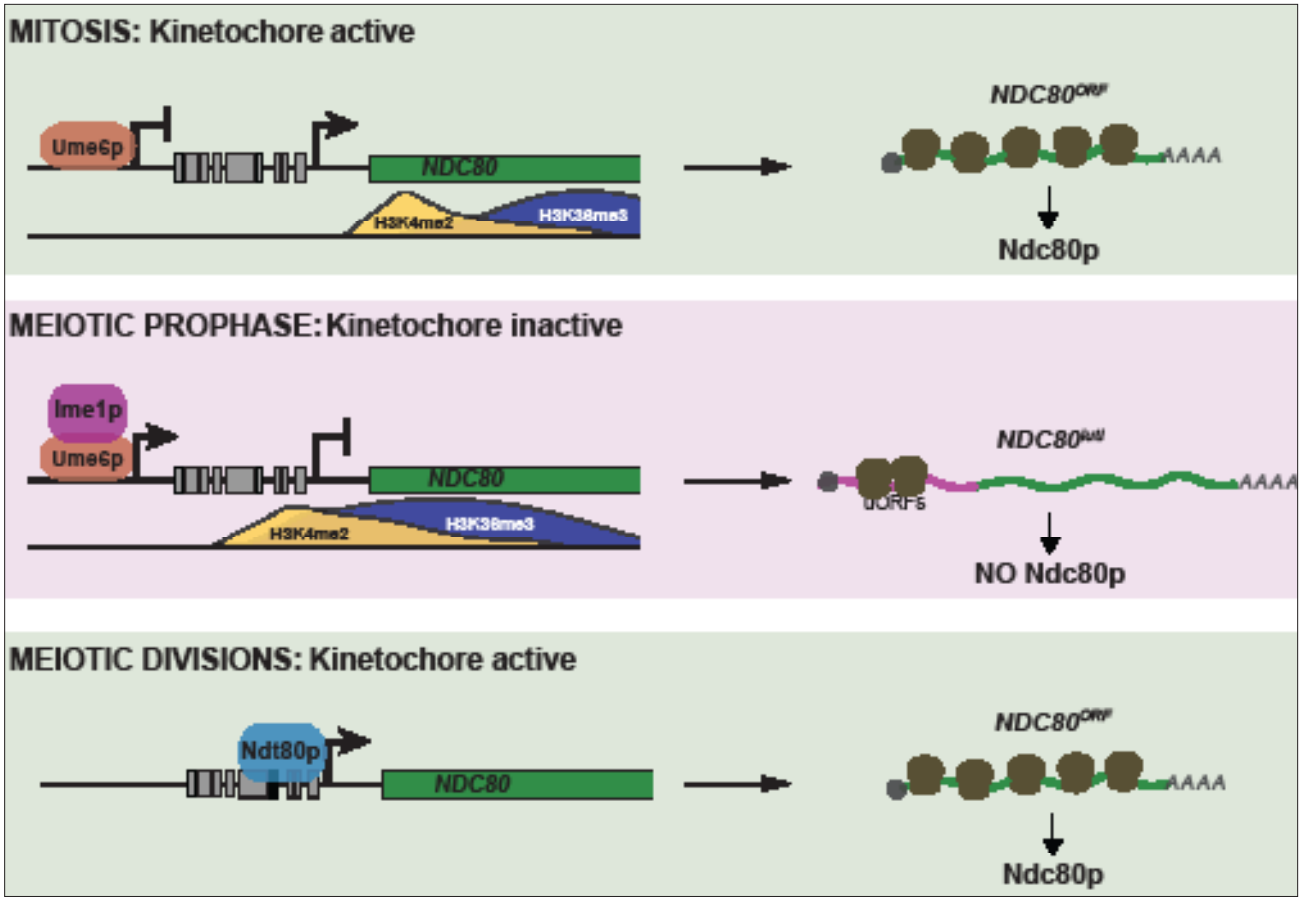


Figure 2: Formation of luti-mRNA during meiotic prophase. *It inactivates kinetochores. During meiosis, Ndt80 transcription factor prevents transcription of uORFs.*[1]

EU: Mitochondria normally reside in the cytoplasm of cells, but unlike the textbook picture of individual organelles floating around in the cell, the mitochondria form an interconnected network. When the cell undergoes division, it needs to inherit mitochondria. Besides being the major ATP production centers in the cell, mitochondria are also involved in many other metabolic regulation processes. The mitochondrial genome co-evolves with the nuclear genome, which contains most of the mitochondrial proteins that are imported into the organelle. So there is a symbiotic-type relationship. In organisms that have sex-specific gametes, the oocyte brings all of its mitochondria into the embryo at the end of fertilization, whereas single-celled organisms like yeast act like oocytes by inheriting a group of mitochondria, but also act like sperm by eliminating 50% of their genome, which won't be inherited. Ultimately, we are interested in whether there are differences between what is segregated into the gametes and what is left behind. We wanted to start by defining distinct stages where there are stereotypical morphological changes in mitochondria towards the segregation.

One of my students discovered that the first step of regulated mitochondrial detachment occurs at the molecular level. Mitochondria are connected to the cell cortex in the plasma membrane by a molecular organelle tether called the mitochondria-ER-cortex anchor (MECA) that binds to the mitochondrial outer membrane on one hand and the plasma membrane on the other. We knew that these contact sites are destroyed during meiosis because we see a very timed change in mitochondrial morphology when the cell is in transition during meiosis. After this, the mitochondria start associating with the nuclear membrane. The nuclear genome needs to get inherited somehow, so the mitochondrial genome "hitchhikes" with the nuclear genome to make it into the gametes. For this reason, we believe that the subset of the mitochondria that interacts with the nucleus is the part that makes it into the gametes, and the subset that does not interact with the nucleus is left behind and degraded.

BSJ: How does phosphorylation of protein kinase Ime2 promote the degradation of MECA and allow proper segregation of mitochondria during meiosis?[3]

EU: In mitotic cells, the MECA complex is very stable—it locks the mitochondria to the plasma membrane without dynamically changing. However, during meiosis, mitochondrial detachment occurs by modulating the activity of this tether. The kinase Ime2 phosphorylates the MECA subunits, changing the structure of these tethers and making them more amenable to proteasome degradation. Because phosphorylation is required for the degradation, we know that degradation occurs at the post-translational level. One

reason this probably happens is because a functional MECA is needed prior to meiotic division. Because it is a stable structure, shutting down its synthesis would not be enough for mitochondrial morphogenesis to be activated.

BSJ: Mitochondrial functions decline with age but are renewed during gametogenesis. How might this be possible?[3]

EU: Even in old cells undergoing meiosis, we see the cellular rejuvenation that occurs as part of meiosis. These progenitor cells are "replicatively aged" since they have undergone prior cell divisions to produce daughter cells through mitosis. They display some defects at the cellular level, but they are still partially functional. This is because aging is a progressive phenomenon.

Before meiosis, we anticipate that some mitochondria are more functional than others. Our question right now is whether the more functional ones are somehow selected over the others at the level of meiotic segregation. We can look at the quality control aspect of this by making basic fusions between young and aged cells or by damaging a subset of mitochondria. We want to find out whether there is selection in these mixed populations. Probably there is, but we don't know for sure.

BSJ: Many of your studies use a budding yeast model. What characteristics of budding yeast that make it an ideal organism for studying meiosis?[1,2,3]

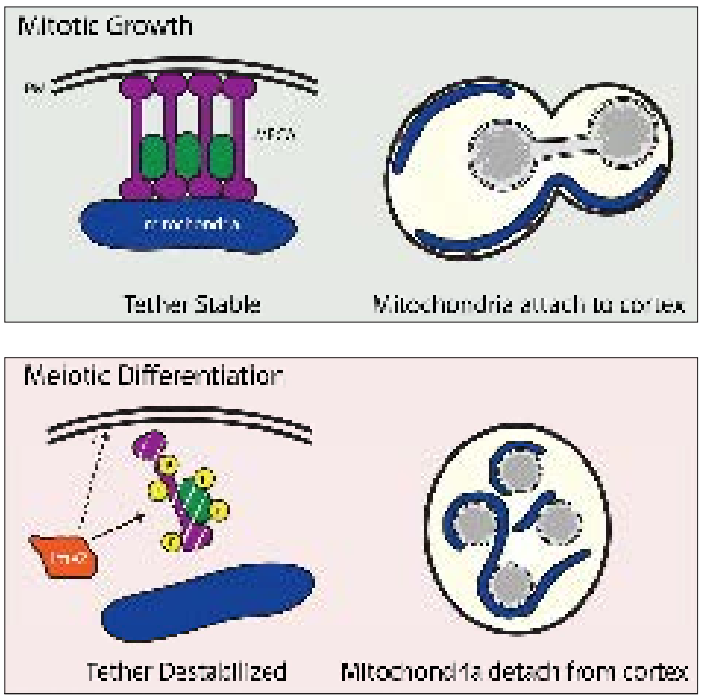


***Figure 3: Mitotic growth and meiotic differentiation**. MECA tether is destabilized with phosphorylation by Ime2 during meiosis, untethering mitochondria from cell cortex.[3]*

EU: When it comes to meiosis, most of what we know comes from a very chromosome-centric perspective. But meiosis is a full cell differentiation program. Meiosis is an evolutionarily conserved process, which involves a lot of changes happening to the cell. If you want to understand those changes, you want to look at the most genetically tractable organism. When it comes to yeast, one big advantage is that we can induce meiosis. That's very important for looking at population-based studies of gene expression. If cells are performing a differentiation or some other change synchronously, the population behavior is more reflective of individual cell behavior. We can also live image meiosis in budding yeast cells from the very beginning to the very end of meiosis—it takes about 24 hours. Whereas in other cells like oocytes, we can't observe that refined framework or synchronous manner. Because there are so many unanswered questions, we want to start with the most simple and tractable organism—from there, we can begin to look for conservation and divergence in other systems.

BSJ: What motivates you to challenge canonical perspectives of gene regulation and cell differentiation in your research?

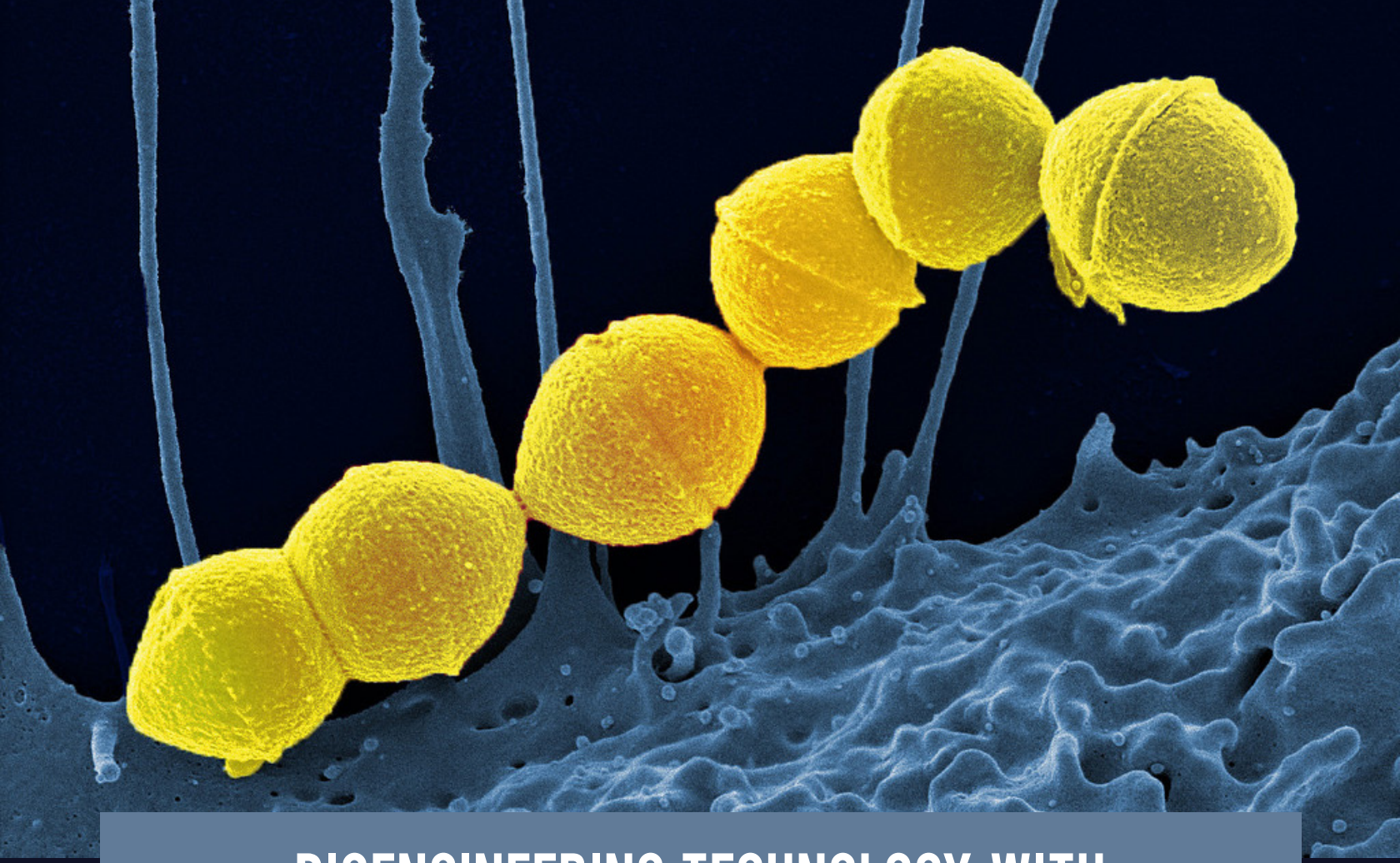
EU: In my mind, it's more a sense of curiosity than a desire to challenge. If I make an odd observation, I think, "How does this happen?" In our lab, we have the ability to observe cellular process in great detail, and I also have incredible graduate students. It all comes together because of their hard work, dedication, curiosity, and ability to make unique observations. When you do research, it mostly doesn't work. But when a discovery comes, for example, you check something in the scope and you're the first person to ever see it, you go back to childhood excitement. When that happens, you get the science high, which is better than any drug.

REFERENCES

1. Chen, J., Tresenrider, A., Chia, M., Mcswiggen, D. T., Spedale, G., Jorgensen, V., . . . Ünal, E. (2017). Kinetochore inactivation by expression of a repressive mRNA. *ELife, 6*. doi: 10.7554/elife.27417
2. Tresenrider, A., & Ünal, E. (2017). One-two punch mechanism of gene repression: A fresh perspective on gene regulation. *Current Genetics, 64*(3), 581-588. doi:10.1007/s00294-017-0793-5
3. Sawyer, E. M., Joshi, P. R., Jorgensen, V., Yunus, J., Berchowitz, L. E., & Ünal, E. (2018). Developmental regulation of an organelle tether coordinates mitochondrial remodeling in meiosis. *The Journal of Cell Biology, 218*(2), 559-579. doi:10.1083/jcb.201807097

IMAGE REFERENCES

4. Newton, I. P., & Appleton, P. L. (Photographers). (2014). Madin-Darby Canine Kidney epithelial cells [digital image]. Retrieved from https://commons.wikimedia.org/wiki/File:Mitotic_MDCKs.jpg



BIOENGINEERING TECHNOLOGY WITH A SOCIAL RESPONSIBILITY

Interview with Professor Luke P. Lee

BY MATT COLBERT, CASSIDY HARDIN, MICHELLE LEE, ROSA LEE, MELANIE RUSSO, AND NIKHIL CHARI

Dr. Luke P. Lee *is a Professor of Bioengineering, Electrical Engineering and Computer Science at UC Berkeley. He is a pioneer of technologies such as rapid microfluidic PCR and optofluidic spectroscopy. Dr. Lee is also a strong believer in tackling problems with broad-scale social implications. In 2016, he founded the Biomedical Institute for Global Health Research and Technology (BIGHEART) at the National University of Singapore. We spoke with Dr. Lee about his mission at BIGHEART, his work in establishing microfluidic platforms for rapid polymerase chain reaction (PCR) and waterborne pathogen detection, and the importance of bringing socially responsible technologies to market.*

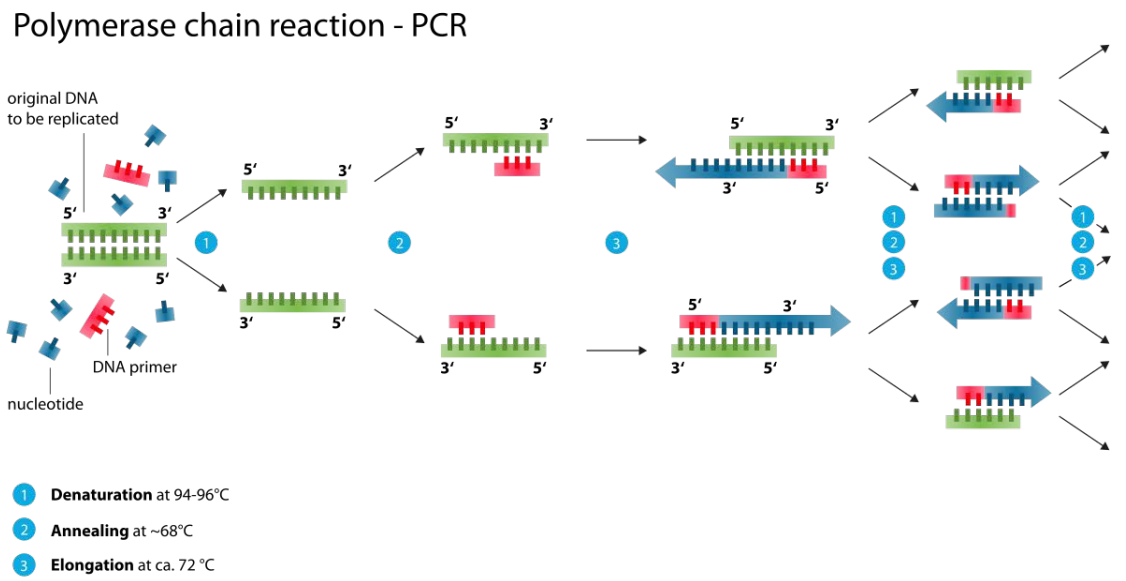


Figure 1: Steps of PCR cycle.[5] *Denaturation of double stranded DNA at high temperature, followed by annealing of polymerase at lower temperature, and elongation of new DNA strands.*

BSJ: We'd like to start off by talking about BIGHEART. Can you briefly explain the purpose of BIGHEART and how BIGHEART uses multidisciplinary approaches to address modern humanitarian and medical challenges?

LL: I originally started my career here in Berkeley, but back then, I didn't have much funding, physical space, or manpower to set up this institution. I wanted to create an environment where people from different disciplines could work together to solve one problem—global health. Of course, there are so many different issues within global health that need attention, but I wanted to focus on early infectious disease diagnostics and precision medicine using organoid chips—tissue cultures that can replicate the behavior and complexity of a real organ. Everyone's metabolic activity is different, but even if we assume there is a universal recipe for treating a patient for a certain disease, certain people may still not be able to handle that treatment. In the case of cancer in Third World countries, if the patients are malnourished, they might not be able to handle the same toxic compounds that are found in cancer medications. We were thinking about how to find the best personalized medicine for organoid-on-a-chip technologies. But molecular diagnostics was our first priority. That's why I spent a lot of time developing more efficient technologies to detect DNA and protein biomarkers. That's one of the reasons we're working on PCR-on-a-chip. If possible, we want to detect the pathogen directly without any labeling.

BIGHEART's goal is to bring scientists together—whether they're engineers, biologists, physicists, even clinicians—to communicate and solve one project, such as malaria, as a group. But all those scientists are busy with their own work, so it's not always easy for them to work on a specific medical issue. We wanted to bring these people into one place and allow them to work together without any physical or mental barriers.

BSJ: You spoke briefly about your work in PCR-on-a-chip technology. PCR is a technique that is commonly used in biology to amplify a certain section of DNA. What are the current applications of PCR?

LL: There are many, many applications. One is rapid and accurate measurement of different diseases. In our microfluidic PCR device, each circle is a reaction chamber. You can do simultaneous tests of different biomarkers for different cancers or infectious diseases in each one. The PCR solution is automatically pumped into the chambers using a vacuum, and each chamber has an individual PCR reaction.

BSJ: Could you briefly highlight the issues and potential inefficiencies with current PCR systems?

LL: Current PCR takes hours. You have to use a heater to change the temperature of the reaction several times to denature the DNA and then allow the polymerase to bind (Fig. 1). Many companies build heater blocks, so you are heating up for denaturing and cooling down for annealing, and then you go back up a little bit for extension. You have to make a cycle, which takes a lot of power and time. We created a technique called ultra-fast photonic PCR using LEDs and so-called "plasmonic concept antenna" to heat up and cool down the reaction really fast. Instead of one hour for 30 cycles, you can do it in two to three minutes. The idea is to use a specific wavelength to resonate this plasmonic structure, like gold. The other issue with microfluidic PCR is when you heat up the PCR fluid, you generate bubbles. We are trying to remove these bubbles by pumping them out of the system.

This is not only important for infectious disease detection or cancer diagnostics. Everyone in the life sciences needs to use PCR to quantify gene information; it's the only way to amplify DNA. That is why PCR is so important and there are so many people competing with each other to claim that they made ultra-fast and accurate PCRs. Some labs might not mind waiting one hour, but if you want to screen a lot of samples or build a library and you have to wait one

hour for one experiment, it can take a lot of time. With microfluidic PCR, we can go from one experiment per hour to 100,000 experiments in five minutes, because we also have many wells or chambers in each PCR chip (Fig. 2). 15 years of work can be finished in 15 days. You can speed up an enormous amount of life science automations and capture more reliable biomarker discovery and detection.

A lot of people don't think we need fast PCR, but it's not so much about speed as it is about collecting massive amounts of data. If you really want to make a precision personalized medicine, you need to collect all the information about how cellular behaviors change with various types of cancers or diseases. We need to build reliable, high-density information about these biomarkers. Not only for disease, but also for our food and our environment. We can prevent disease by correlating it with the food we eat. For example, say I got cancer and the doctor hypothesized it was because I was eating a lot of junk food. If we are able to build these biomarker databases, we could trace my eating habits and my DNA information to the source and verify the doctor's hypothesis. In other places, water is very important. The DNA of the pathogens that introduce infectious diseases is hidden somewhere—in the water, the environment, the food—we need to find it.

BSJ: Going back to microfluidic PCR, we wanted to ask: how does the generation of bubbles hamper the efficacy of microfluidic PCR?[1]

LL: If there is air trapped in the reaction chamber, when you introduce the PCR solution into the chamber, bubbles will dominate the space and you will not have any reactions. This is because when you heat the chamber, the bubble will expand and push all of the DNA and PCR solution out of the chamber. You need to pump out the bubble before you begin. We built this ring (Fig. 3) to be intentionally gas-permeable so we can remove the bubble before the reaction begins, and we developed a degassing method that uses a vacuum battery to pump out all of the residual gas trapped inside the chamber.

BSJ: To aid in the prevention of bubbles forming in the PCR microfluidic machine, you used a layer of polyethylene (PE) and two layers of polydimethylsiloxane (PDMS) (Fig. 3). Can you explain how these two substances prevent the formation of bubbles?

LL: The gas permittivity of PE is much lower compared to PDMS. We made the top surface of the reaction chamber PE to ensure that all the gas would be pumped out laterally through the PDMS around the sides of the chamber.

BSJ: Could you briefly describe the concept of degas-driven flow?[1]

LL: The PDMS is very flexible and contains lots of nanopores. After we fabricate a PCR chip, we pump out all of the gas inside by putting it in a desiccator. You can remove even the small amounts of gas trapped in the nanopores. You cannot see them with the naked eye because they are so small—less than a nanometer. If

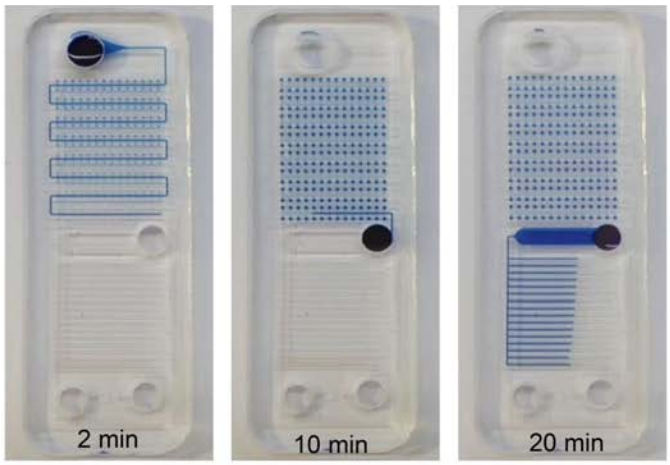


Figure 2: PCR reaction chambers in one of Dr. Lee's microfluidic PCR chips.[3] *The chambers are filling up over time.*

you pump out all of the gas from the PDMS, you can build a negative pressure environment, lower than the atmosphere. Then you package it. When you are ready to use it, you can open the package just like a potato chip bag—of course, you cannot store in in this negative pressure for a long time. Just like potato chips—if you eat the potato chips after a certain amount of time they're not crispy anymore, right?

Anyway, you pump it down, and then you package it. When you are ready, you drop in the PCR solution, and the negative pressure draws the solution into the chip. Building that negative pressure in the first place is what we call degassing. But our design also adds a vacuum battery. If we rely only on the properties of the material to degas, the operator of the chip only has about two minutes to drop in the PCR solution before the pressure returns to atmospheric. But with the vacuum battery, we increased the volume of space you can pump air out of, so it's much more than just the small nanopores in the PDMS. Now, we can wait for 15 minutes instead of two. This accounts for less efficient human operation of the chip, and the operator can take more time to drop in the solution.

BSJ: Could you describe what the function of the circumferential degas pump surrounding each chamber is?[1]

LL: The circumferential degas pump helps to pump out any gas within this chamber. As I mentioned, PDMS can pump out gas because there are small, invisible pores which you have already vacuumed. It works, but it takes time. Here, we intentionally made a space, the circumferential degas pump, where you can have more vacuum volume (Fig. 3). This makes it faster to pump out residual gas. Also, it's easier to pump laterally. If you have this kind of design,

"With microfluidic PCR, we can go from one experiment per hour to 100,000 experiments in five minutes."

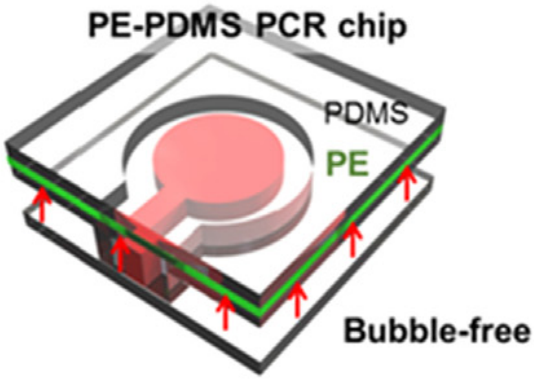


Figure 3: The design of the microfluidic PCR chip features a thin layer of PE sandwiched between two thicker layers of PDMS.[1] *The PE layer prevents PCR fluid (red arrows) from flowing out of the cell vertically, whereas gas flows out laterally through PDMS.*

all the gas bubbles are pumped to the outside.

BSJ: Earlier, you talked briefly about methods of detection for waterborne pathogens. Could you elaborate on what some of the limitations of the current methods of waterborne pathogen detection are?

LL: All pathogen detection nowadays is culture-based. It takes time. For example, let's say your kid has an infection. You don't know whether it's from the water or if it's an airborne pathogen. It happened to my daughter. She had an infection, but we didn't know what happened, so we just drove down to the best children's hospital—at the time, I thought that Stanford's Children's Hospital was better. Doctors just kept injecting her with different antibiotics without knowing what the real problem was. It was really disastrous; she was hospitalized for many months. Thankfully, she survived, but some people can die. If you just keep on adding antibiotics without knowing what the infection is, it's actually damaging the patient. It's better to treat it with the proper drug that will kill that particular pathogen. I had food poisoning the last four days, I couldn't even eat. I don't know what caused it, but it would've been nice. It is critical for certain diseases to detect pathogens precisely.

How can we identify pathogens right now? Either ELISA, which is protein-based detection, or PCR, which is DNA-based. We are trying to quickly detect the specific fingerprint of the pathogen from its outer surface. It's challenging, but we're trying to do our best.

BSJ: Could you briefly describe what plasmonic bacteria are? How did you modify water-borne pathogens to create plasmonic bacteria?[2]

LL: This kind of solution we use is nothing but gold nano-plasmonic particles. Actually, it's not very expensive; it's a salt solution—it costs less than a few cents! We wanted to be able to detect pathogens without lysing them, which you have to do to detect the DNA inside. Our idea was to detect a chemical signature on the surface of the pathogen. Just like how your faces all exhibit different phenotypes, the surfaces of pathogens exhibit unique chemical structures. Unfortunately, the surfaces of all pathogens look quite similar. There are a few vibrational spectra that are different, but they're difficult to differentiate because the signal is very weak. Wrapping the bacteria in gold nanoparticles helps to amplify the chemical signal (Fig. 5).

All chemicals have a vibrational spectrum, and we're using vibrational spectroscopy called surface-enhanced Raman scattering (SERS) to detect these different spectra. Since these vibrations are normally so weak, we have to use the plasmonic bacteria to amplify the vibrational scattering signal. We also measure the signal from the

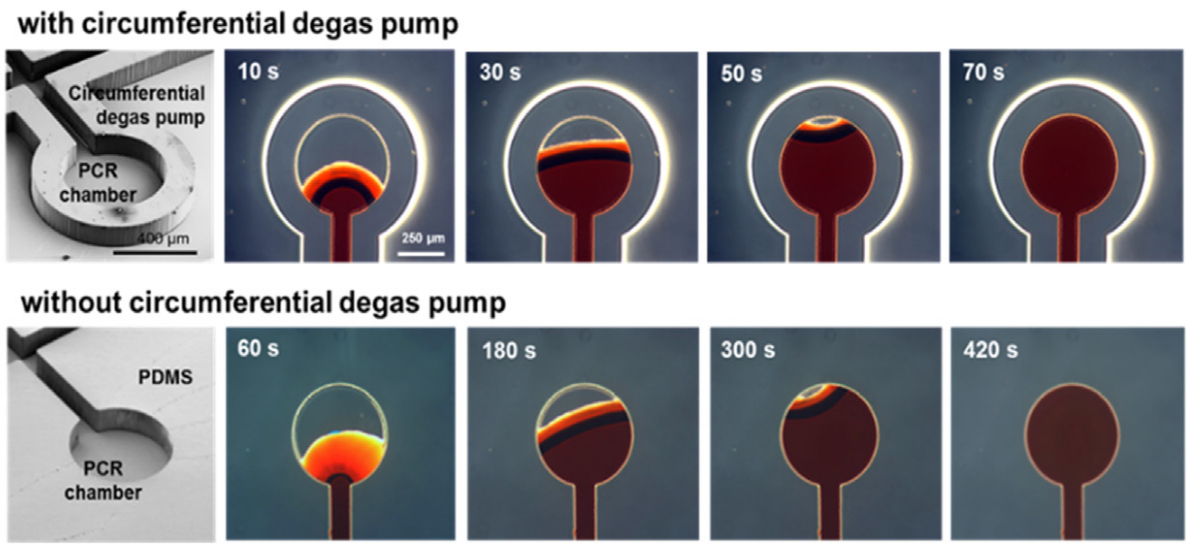


Figure 4: The role of the circumferential degas pump.[1] *It ensures uniform flow of the PCR fluid into the reaction chamber.*

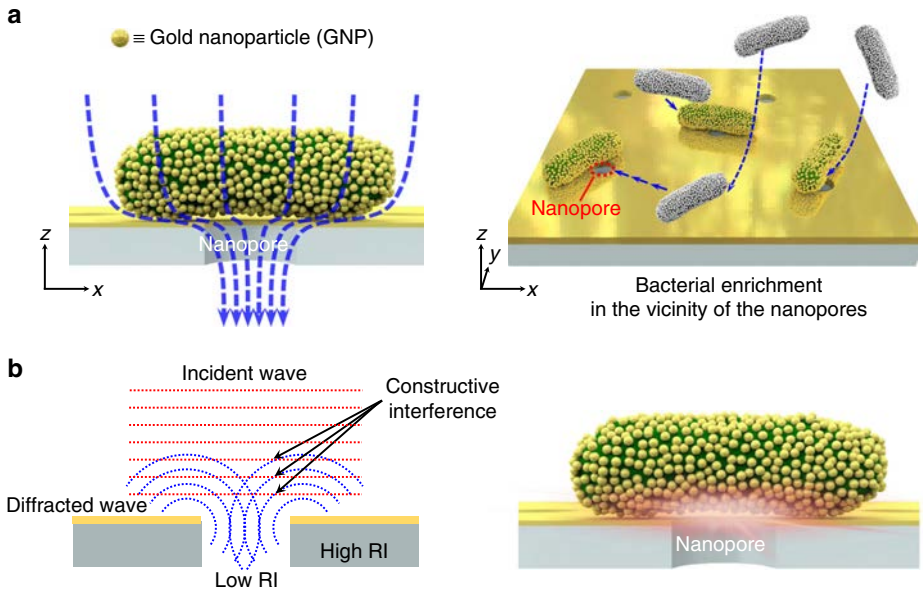


Figure 5: Lee's optofluidic platform.[2] *It aims to enhance surface signals from pathogenic bacteria using GNP coatings and constructive interference with nanopores.*

plasmonic bacteria on the surface of a gold mirror, because we're trying to amplify as much as possible. We're doing all kinds of different things to increase the vibrational peak of surface proteins on these pathogens. It's not an easy job, but we want to figure out whether we can accumulate enough data to distinguish from pathogen to pathogen. But without the gold we cannot get any signal.

BSJ: You explained that you measure the plasmonic bacteria signal on a thin gold mirror dotted with nanopores. How do these nanopores contribute to trapping plasmonic bacteria?[2]

LL: In our system, polycarbonate plastic is cased with a thin gold layer, which is dotted with nanopores. Around the edges of the nanopores, we see what are called "hot spots," which are generated by more electron oscillation (Fig. 4). More electron oscillation means more radiation. It's like an antenna—which is nothing more than a simple rod structure with oscillating electrons. What happens if you oscillate electrons in the surrounding area? An electromagnetic field is generated, which we can detect. But we want to get as much information as we can out of that field. Many people have tried to make nanopillars or nanostructures to enhance this signal - something with a sharp tip. But here, instead of using a sharp tip, we use nanopores because we want to trap more of the pathogen. If we weren't trying to filter out bacteria, we wouldn't use nanopores —we would use a sharp tip structure.

BSJ: How may rapid detection of waterborne pathogens enabled by your optofluidic mechanism contribute to our understanding of the pathogenic genome?

LL: We need to mass-produce these technologies to disseminate to the Third World. That was my dream. It's a "chicken and egg" problem because many venture capitalists want to make money right away, and this kind of thing doesn't make money right away. They value a winner's market. Waterborne pathogen detection is not really a big market. That means we have to invest in technologies with a social responsibility. That's why I started BIGHEART. After all, we are in Berkeley and we shouldn't give up. Social responsibility is important, because these pathogens are connected with real health problems. You may think that the health problems in the Third World will not affect us, but actually, these pathogens circulate through the air, or traffic, or even the food that we import. We have to think about how to use our resources wisely to prevent and mitigate these global health problems.

REFERENCES

1. Lee, S. H., Song, J., Cho, B., Hong, S., Hoxha, O., Kang, T., . . . Lee, L. P. (2019). Bubble-free rapid microfluidic PCR. *Biosensors and Bioelectronics, 126*, 725-733. doi:10.1016/j.bios.2018.10.005
2. Whang, K., Lee, J., Shin, Y., Lee, W., Kim, Y. W., Kim, D., . . . Kang, T. (2018). Plasmonic bacteria on a nanoporous mirror via hydrodynamic trapping for rapid identification of waterborne pathogens. *Light: Science & Applications, 7*(1). doi:10.1038/s41377-018-0071-4
3. Yeh, E., Fu, C., Hu, L., Thakur, R., Feng, J., & Lee, L. P. (2017). Self-powered integrated microfluidic point-of-care low-cost enabling (SIMPLE) chip. *Science Advances, 3*(3). doi:10.1126/sciadv.1501645

IMAGE REFERENCES

4. Banner image: National Institute of Allergy and Infectious Diseases. (Photographer). (n.d.). *Streptococcus* [digital image]. Retrieved https://ysnews.com/news/2019/03/calamity-day-the-halls-are-alive-with-the-sound-of-coughing
5. Microbeonline. (Photographer). (n.d.). Steps of PCR [digital image]. Retrieved https://microbeonline.com/polymerase-chain-reaction-pcr-steps-types-applications/

Bird Health in California's Central Coast: Interactions Between Agricultural Land Use and Avian Life History

By: Victoria Marie Glynn; Research Sponsor (PI): Dr. Claire Kremen

ABSTRACT

The Central Coast of California has implemented bare-ground buffers to deter the presence of food-borne pathogens in produce. The destruction of natural habitats surrounding farms may place avian communities at risk. To ascertain bird health in this rapidly-changing landscape, we sampled passerine and near passerine birds on organic strawberry farms in Monterey and Santa Cruz counties. The ratio of two white blood cell types, heterophils and lymphocytes (H:L ratio), served as a proxy for bird health. Mixed-effects models revealed that song sparrow health slightly increased on farms with high proportions of agriculture ($p = 0.08$). High levels of reproductive readiness were also linked to improved song sparrow health ($p = 0.007$). The study's findings suggest that foraging and habitat resources created by agriculturalists, as well as fledging survivorship may be impacting bird health in the Central Coast. There is a need to re-evaluate human-wildlife relationships as agricultural spaces may be safeguarding avian communities.

Major, Year, and Department: Environmental Sciences, 4th year, Department of Environmental Science, Policy, and Management.

Keywords: Passerine and near passerine birds, organic strawberry farms, H:L ratio, bird health, mixed-effects models, ArcGIS.

INTRODUCTION

Avian communities near agricultural fields impact both human health and surrounding ecological communities. Birds provide critical ecosystem services to farmers by predating on crop pests in a variety of agroecological systems including coffee, cacao, and palm oil farms.[1,2,3] However, birds also pose challenges to agricultural production because they eat crops, and their feces can be found in adjacent waterways and on produce.[4,5,6,7] Birds are vectors for food-borne pathogens such as *E. coli*, *Salmonella* spp. and *Campylobacter* spp., and, as a result, destroying natural bird habitats near farms potentially deters the spread of infectious agents.[8,9,4,10,11] Bare-ground buffers, swaths of unvegetated land adjacent to farmland, destroy critical resources for wildlife cohabiting with agroecological systems and disrupt ecosystem services. These landscape changes are linked to decreases in bird biodiversity in the surrounding landscape.[12,13] It is evident that the study of birds in agricultural areas has been centered on human concerns; nonetheless anthropogenic actions may likewise affect avian communities.

To shift the focus from humans to wildlife in agricultural systems, the landscape matrix approach allows for discerning at multiple scales the complex interplay between an organism's health and its surroundings. This approach envisions a collection of natural habitat patches coalescing to form the landscape at large, referred to as the matrix.[14,15] Carving out portions of the natural environment for agricultural purposes potentially disturbs the overall matrix.[16] Thus, areas with more connected natural habitat patches are preferred. Adopting the established landscape matrix approach provides a more robust theoretical base for the idea of a "quality" landscape. Wildlife health studies have widely implemented this framework to discuss how landscape changes impact wildlife health in terms of: species composition, abundance and richness, gene flow, and parasitism.[17,18,19,20] Yet few molecular and cellular tools have been implemented at the landscape scale to discuss community well-being.

Immunology can assess how wildlife health is being impacted by changing landscape matrix compositions. The ratio of two white blood cell types, heterophils and lymphocytes (H:L ratio), has been used to infer a bird's future and present state of health in a variety of contexts, from confined feeding operations to national reserves.[21,22,23,24] Although the H:L ratio is a high fidelity marker of bird health, it has not been used to quantify how bird health is impacted by changing landscapes. By linking the H:L ratio to landscape quality, specific land use changes can be discerned as detrimental to birds in agroecological systems.

Using the H:L ratio as a measure of bird health within a landscape matrix framework can determine how bird health is impacted by landscape composition in agricultural areas. This study asks whether certain agricultural land use types are more critical to avian health in comparison to others. We also consider if particular species are more vulnerable to certain land use types. In addition, birds of different reproductive states are examined to determine whether this factor impacts overall health in the face of variable environmental conditions. By understanding how changing agricultural landscapes are impacting avian communities, farmers and food regulators will be able to balance human and wildlife concerns more equitably.

MATERIALS AND METHODS

I. Study Sites

During July and August of 2018, as part of the Kremen Lab at UC Berkeley and the Karp Lab at UC Davis, I mist-netted for passerine and near passerine (tree perching and dwelling) birds on 20 organic strawberry farms in Monterey and Santa Cruz Counties (Fig. 1). These 20 farms were selected to capture a spectrum of agricultural and landscape conditions. Farm sizes ranged from 0.04 to 9 km^2, with production models greatly differing. Some sites were monocultures, while others contained over 60 crops. These farms are also located along a land-use gradient, where the density of natural landscapes was 53% in some areas while in others 87% of the area corresponded to agriculture.

II. Landscape Diversity

To determine each site's landscape diversity, I digitized land use types on and surrounding each farm. I downloaded National Agricultural Imagery Project (NAIP) photographs corresponding to Santa Cruz and Monterey Counties and imported them into ArcGIS.[25,26] I overlaid the farms' GPS waypoints onto NAIP imagery to locate the sites within the larger landscape matrix. A one kilometer buffer circle was drawn around each farm and all land uses within this region were digitized. Using the Gonthier Lab's landscape digitization protocol (K. Garcia, personal communication), land uses were categorized into the following types: forest and woodlands; shrublands; herbaceous vegetation; low to no vegetative cover; agriculture; urban or built up; exurban; suburban; or water features. Google Maps was also used to confirm land use categorizations.[27] Based on the focal bird species' life histories, I used the proportion of agriculture and natural habitats (oak woodlands and shrublands combined) in the final mixed-effects model (Table 1). For instance, when considering breeding habitat (Table 1), the focal species require either forested areas, brushland, or thickets, which roughly corresponded to the oak woodlands and shrubland vegetation types I digitized.

III. Study Organisms

Although California's Central Coast has diverse flora and fauna, I focused on passerine and near passerine birds as these taxa often serve as indicators of environmental health, and more specifically landscape changes in agricultural areas.[32] Of the captured birds, I selected a subset representing the 4 most common agricultural species: song sparrows (*Melospiza melodia*), house finches (*Haemorhous mexicanus*), Oregon juncos (*Junco hyemalis*), and spotted towhees (*Pipilo maculatus*). Each bird species' banding alpha code is: house finches (HOFI), Oregon juncos (ORJU), song sparrows (SOSP), and spotted towhees (SPTO). The subset selected amounted to 200 birds, approximately 15% of the entire mist-netted sample of 1303 birds. These 200 birds represent 13 of the 20 sampled farms. Each bird species has different foraging, nesting, and breeding habits (Table 1). The differences in biological functional traits are expected to be predictive of certain bird species being more vulnerable to changes in the natural landscape.

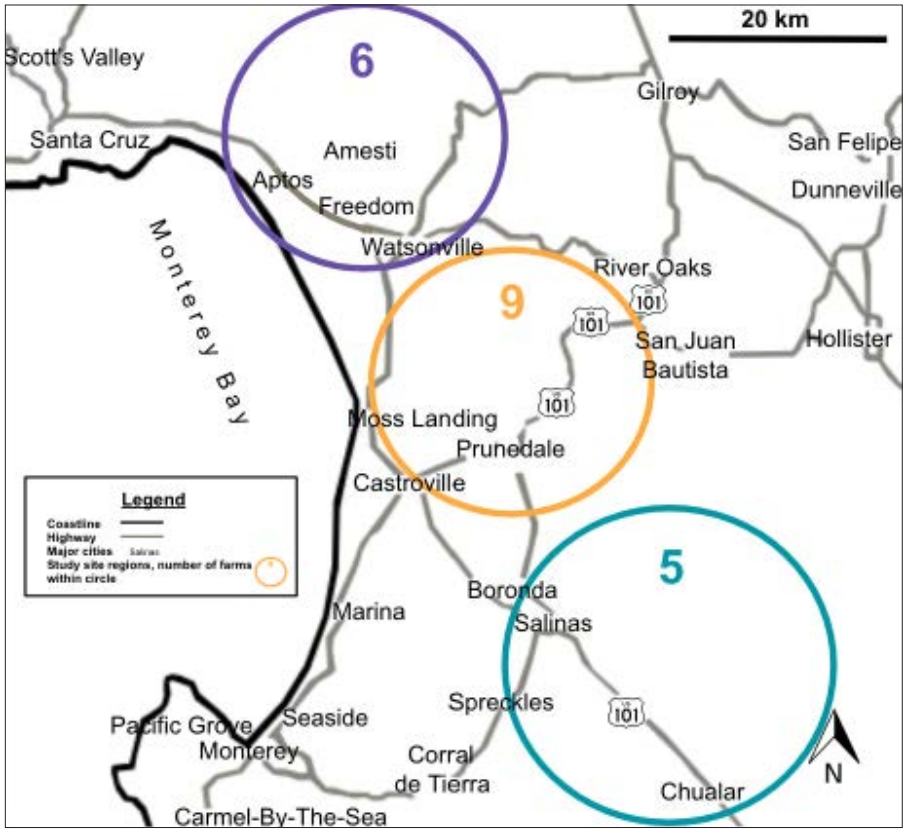


Figure 1. Spatial distribution of the 20 sampled farms in Monterey and Santa Cruz Counties. *Each circle encloses a region of sampled farms. The number of farms found within each circle is denoted at the top of the region. The base map was created using National Agricultural Imagery Project (NAIP) photographs.*[25]

Species	Food Guild	Cover	Nestling	Clutch Size	Start of Breeding Season	Breeding Habitat
House finch (*Haemorhous mexicanus*) HOFI	Fructivore and granivore	Trees, tall shrubs, and buildings	Trees, abandoned nests, and human-made structures	Lay 2-6 eggs, 2-3 broods per year	March or April	N/A
Oregon junco (*Junco hyemalis*) ORJU	Insectivore and granivore	Trees, shrubs, and ground herbage	On ground, small tree or shrub near water	Lay 3-5 eggs, 2-3 broods per year	April into August, peaking in May and June	Forests, woodlands, and forest edges
Song sparrow (*Melospiza melodia*) SOSP	Omnivore, eats litter	Low dense vegetation near moist areas	On ground, small trees and shrubs	Lay 3-6 eggs, 2-3 broods per year	April	Dense riparian thickets and wetlands
Spotted towhees (*Pipilo maculatus*) SPTO	Omnivore, eats litter	Shrubs, ground herbage, and thickets with abundante leaf litter	On ground, in slash pile, dense shrub, or vine tangle.	Lay 2-6 eggs, 2 broods per year	Late April to late August, peaking in May and June	Dense brush or thickets with substantial accumulation of litter

Table 1. Summary of house finches, Oregon juncos, song sparrows, and spotted towhees' life histories. *All taxa are from the family passerellidae, order* passeriformes.[28,29,30,31] *Each bird species' banding alpha code is included (i.e. house finches as HOFI).*

IV. Mist-netting

To representatively sample avian communities surrounding farms, we implemented the standard mist-netting protocol. Mist-netting uses nets to capture and sample avian communities in a given area. We set up 10 mist nets per site along field edges, bordering strawberry fields, other crops, and natural areas alike. A diversity of mist-net locations ensured that we captured birds that were using various land use types. We recorded GPS waypoints for each net to later locate them on satellite imagery. Following standard protocol, all nets were opened at sunrise (around 5 AM) and left open for 5-6 hours.[33] Nets were checked at 20 minute intervals and all birds caught were brought back to the banding station for data collection. We worked on each farm for three continuous days to reach sample saturation. Doing so ensured that most, if not all, birds surrounding the farms were sampled.

V. Sample Collection

To collect data on each captured bird, we transported specimens from the nets to the on-site station for banding and morphometric calculations. Each bird was banded with a metal ring, imprinted with a unique serial number provided by the United States Geological Service (USGS) to prevent a bird from being counted as a unique observation after the initial collection. Each captured bird was sexed based on its plumage and/or visible reproductive organs. Birds were aged via the level of skull ossification and/or plumage.[34] An individual's beak length, beak width, tail length, and tarsus length were also measured.[33] Lastly, we noted the presence of strawberry residue on a bird's beak, and evidence of ectoparasites such as wing lice and head ticks.

To determine a bird's state of reproduction, we calculated a "reproductive readiness" index. Reproductive readiness was determined by examining a bird's cloacal protuberance or brood patch, for males and females respectively. Cloacal protuberance and brood patches' size, color, and texture indicate a bird's state of sexual maturity.[34] These organs were assigned a score ranging from 0-4, where larger numbers indicate a bird is more prepared for reproduction. Using cloacal protuberance and brood patch scores as proxies for "reproductive readiness", I calculated z-scores for both male and female breeding parameters and combined them into a single metric called "reproductive readiness." Standardizing the scores via a z-score calculation allowed for models that included a single term to describe a bird's current or potential sexual activity.

VI. White Blood Cell Differential

To determine avian community health, we collected a blood sample from each captured individual to create blood smears. Using a 27-gauge needle, we extracted approximately 50 μL of blood from a bird's brachial vein.[35,36] The blood was then placed in a heparinized tube to prevent coagulation. With the heparinized blood, we made a blood smear for each sampled bird.[37] Blood smears were then placed in a slide box to dry and relocate to the laboratory.

To determine the white blood cell composition of each bird, I stained the blood smears with Giemsa-Wright stain and observed the slides under a microscope. Giemsa-Wright staining was selected because it causes different blood elements to acquire characteristic colors and patterns, resulting in the precise quantification of different white blood cell types.[37,38] I calculated a white blood cell differential for each smear by studying the slides under a microscope with oil immersion fluid,[39] following a snaking pattern from head to tail of the smear to avoid double counting.[40,41] White blood cell types were categorized into one of the following types: lymphocytes, heterophils, basophils, monocytes, and eosinophils. Separately, I made note of any parasites identified, particularly Haemoproteus spp. and microfilariae given they greatly place bird health at risk.[42,43] Once a

Farm	Proportion of Land as Agriculture	Proportion of Land as Shrubland	Proportion of Land as Oak Woodlands
1	**0.73**	0	0.02
2	0.29	0.09	0.19
3	0.36	0.08	0.23
4	0.34	0.01	0.27
5	0.24	0.003	0.09
6	0.02	0.16	0.37
7	0.15	0.18	0.002
8	**0.68**	0.002	0.14
9	0.14	0.1	0.21
10	**0.82**	0	0
11	**0.87**	0.00007	0.03
12	0.32	0.09	0.15
13	0.32	0.02	0.30

Table 2. Proportion of agriculture, shrublands, and oak woodlands on each farm. *Dominant land use types (proportions above 0.5) are bolded. Note that the proportions do not add up to 1 as there were additional land use types not considered in the study.*

smear had been fully analyzed, I calculated its H:L ratio by dividing a sample's heterophil count by its lymphocyte count. High H:L ratios are associated with birds in poor health.[21,22,24]

VII. Mixed-effects Model

To distill the relationship between landscape composition and bird health, I created and ran linear mixed-effects models (LMEs). I used the statistical program R version 3.6.1 (44) with the lme4,[45] lmtest,[46] and stargazer packages.[47] Visualizations were created using the ggplot2 package (48). As the H:L ratio was not normally distributed based on the QQ-plot, I first log-transformed the H:L ratio so it could be used in parametric tests. The model's syntax was determined based on the experimental design and hypotheses as:

Bird Health ~ Reproductive Readiness * Species + Natural Habitat * Species + Agriculture * Species + (1 | Farm)

The "Bird Health" variable is the log transformed H:L ratio. Farm is the random effect, and there are three interaction effects with species: the proportion of agriculture, proportion of natural habitat, and reproductive readiness standardized score. As we are concerned with how bird health is being modulated by changes in the landscape, "Bird Health" is the predictor variable. The random effect of farm assumes that birds sampled from the same location, regardless of their intrinsic characteristics, will have similar H:L ratios given the shared context. Most importantly, by setting each variable (reproductive readiness, natural habitat, agriculture) in an interaction effect with bird species, the model may reveal that some bird species are inherently more sensitive to certain land use types and how different levels of reproductive readiness are impacting bird health.

RESULTS

I. Farm Landscape Characteristics

The 13 farms in the study varied greatly in terms of the proportion of agricultural fields, shrublands, and oak woodlands present

Figure 2. Bird species counts by farm. *Each bird species is referred to by its bird banding alpha code, where each species is shown as its own bar and color. Bars are grouped by farms.*

Factor	Effect Size	Standard Error	*T*-Value	*P*-value
Null	-1.90	1.04	-1.83	**0.07**
ORJU	1.40	2.02	0.70	0.49
SOSP	1.30	1.32	0.98	0.33
SPTO	2.82	2.27	1.24	0.22
Natural Habitat	0.5	2.91	0.17	0.86
Agriculture	1.53	2.08	0.74	0.46
Reproductive Readiness	-2.18	1.77	-1.23	0.22
Natural Habitat*ORJU	-6.68	7.28	-0.92	0.36
Natural Habitat*SOSP	4.84	3.70	1.31	0.19
Natural Habitat*SPTO	4.32	7.10	0.68	0.54
Agriculture*ORJU	-8.21	8.87	-0.93	0.36
Agriculture*SOSP	5.46	3.12	1.75	**0.08**
Agriculture*SPTO	2.12	8.72	0.24	0.8
Reproductive Readiness*ORJU	-2.89	3.18	-0.91	0.37
Reproductive Readiness*SOSP	6.13	2.24	2.74	0.007*
Reproductive Readiness*SPTO	-3.07	3.95	-0.78	0.44

Table 3. Relationship between bird health, proportion of natural habitat and agriculture on farms, and reproductive readiness. *The final model was as follows: Bird Health ~ Reproductive Readiness * Species + Natural Habitat * Species + Agriculture * Species + (1 | Farm). In the output table, one asterisk denote* p < *0.01 and bolded terms denote* p < *0.1. House finches were the reference group. The degrees of freedom associated with each factor was 98. The effect size reported is a standardized effect size, where each predictor variable was subtracted by its mean and divided by two standard deviations. Each bird species is referred to by its bird banding alpha code.*

(Table 2). Farm 11 has the largest proportion of land dedicated to agriculture, followed by farm 1, with values of 0.87 and 0.73, respectively. Farm 6 has the lowest proportion of agriculture at 0.02. Shrublands were the land use type least represented in the sample. The highest proportion of land corresponding to shrubland is 0.18, corresponding to farm 7; farms 1 and 10 do not have shrublands represented (Table 2). Conversely, the proportion of oak woodlands varies greatly between farms. Farm 10 does not have any oak woodlands, while farm 6 has over a third of its area comprised of this land use type (Table 2).

II. Bird Community Composition

Each farm had a distinct bird community composition in terms of the richness and abundance of the four focal species (Fig. 2). The highest bird count occurred in farm 6, while the lowest count occurred in farm 10. On each of these farms, 27 versus 4 birds were sampled, respectively. Only farm 6 had all 4 species of interest present; most farms had only 3 out of the 4 study species present. Within the entire sample, song sparrows were the most represented (69 birds, 35% of sample), while spotted towhees were the least sampled (31 birds, 16% of the sample) (Fig. 2).

III. Modeling Bird Health and Landscape Quality

In the mixed-effects model, only two terms emerged as significant: the interaction effect between song sparrows and the proportion of agriculture on farms, and the interaction effect between song sparrows and reproductive readiness (Table 3). The interaction effect involving agriculture was marginally significant at p = 0.08, while the

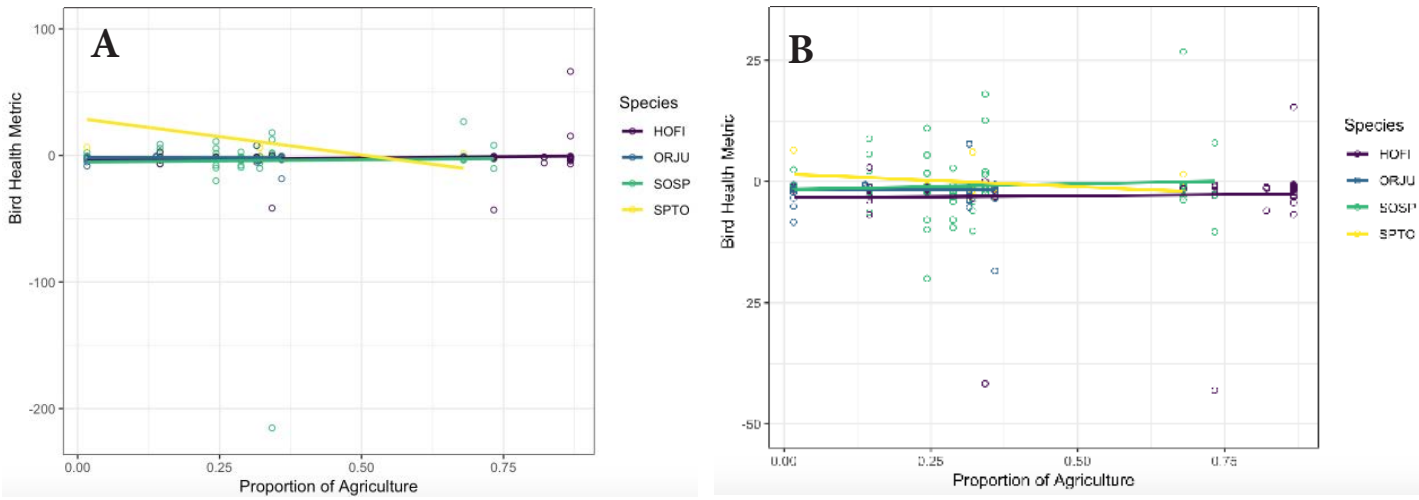


Figure 3. Relationship between the proportion of land devoted to agriculture and bird health. *The "bird health metric" is the log-transformed H:L ratio that has been inverted, such that higher values indicate better health. Each bird species is referred to by its bird banding alpha code, shown in its own color on the plot. (A) The scatterplot depicting the trends between bird health and the proportion of agriculture on farms, with all points in the data set included. (B) The same scatterplot as (a) but zoomed in to better visualize individual species' trends.*

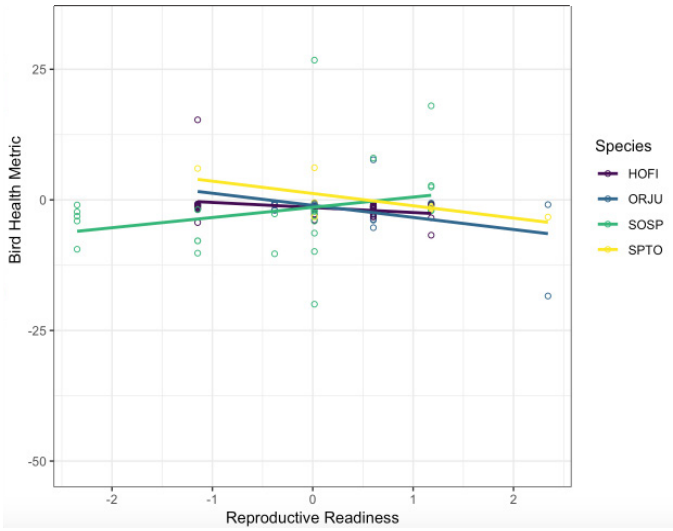


Figure 4. Relationship between reproductive readiness and bird health. *The "bird health metric" is the log-transformed H:L ratio that has been inverted, such that higher values indicate better health. "Reproductive readiness" is the standardized cloacal protuberance and brood patch scores for male and female birds, respectively. Each bird species is referred to by its bird banding alpha code, shown in its own color on the plot.*

interaction effect with reproductive readiness was highly statistically significant at p = 0.007 (Table 3). When comparing the effect sizes of the two significant effects, the interaction effect with reproductive readiness emerged as greater than the interaction effect with agriculture, 6.13 and 5.46 respectively (Table 3).

The positive effect sizes for the significant and marginally significant interaction effects indicate that, compared to house finches (the reference group), song sparrows experience a steeper increase in health with increasing proportions of agriculture on farms and level of reproductive readiness (Fig. 3; Fig. 4). As there is a less statistically significant relationship between song sparrow health and proportion of agriculture, this relationship is less marked on the scatterplot as compared to the relationship between reproductive readiness and bird health (Fig. 3; Fig. 4).

DISCUSSION

The proportion of agriculture on farms and birds' reproductive readiness were the two factors that most influenced song sparrow health (Table 3). Compared to house finches, song sparrows were in marginally improved health on farms with higher proportions of agriculture (p = 0.08). Similarly, song sparrows were the healthiest at higher levels of reproductive readiness as compared to the reference species (p = 0.007). In considering food guilds and the resources present on anthropogenic landscapes, our findings imply that birds may obtain critical resources from agricultural spaces in the form of habitat and forage. Trends in reproductive readiness can be interpreted through the lens of survivorship: birds that survived the breeding season had more robust immune systems. By means of the H:L ratio, we can begin to uncover how changes in the landscape matrix are impacting avian community health in the Central Coast of California.

I. Resources on Farmlands and Bird Health

During statistical modeling, song sparrows were in marginally better health as compared to the reference species as the proportion of agriculture on farms increased (Fig. 3). This trend can initially appear counterintuitive, as human intervention in the landscape has historically negatively impacted wildlife.[12,13,20] Thus the marginally significant p-value could be attributed to the small sample size of 200 birds or could indicate that the H:L ratio is not as strong of an indicator for bird health as previously assumed. Irrespective of the statistical significance of the modeling results, historical land use trends purport that agricultural land may not be the paramount stressor to avian communities. The proportion of agricultural lands in California has remained relatively stable, with the state losing only 1% of agricultural land between 1973 and 2000.[49] There has been a greater landscape pressure from suburban and exurban development. Agricultural intensification may be a more critical factor when discussing wildlife health.[50,51,52] Even if the amount of agricultural land has remained relatively stable, active farmlands favor passerine species richness in Mediterranean climates prior to the breeding season.[53,54] Specifically, this trend is due to the fallow fields, cereal crops, and soil-living invertebrates on Mediterranean farmlands as these provide critical foraging and habitat resources to wildlife.[53,54] It is also important to note that for the other three species, there was no significant trend between health and the proportion of agriculture, even if the scatterplot indicated any directionality (Fig. 3). The lack of statistically significant relationships for the other focal species probes us to consider additional factors that may be impacting house finches, Oregon juncos, and spotted towhees within the study system, such as suburban and urban development. Nonetheless, in this Central Coast study system the marginal trend between agriculture and song sparrow bird health points to the potential benefits associated with agricultural landscapes. However, study limitations require caution in ascribing anthropogenic land uses as positive to wildlife health.

II. Food Guilds in Rapidly Changing Agricultural Landscapes

For song sparrows, food guilds may elucidate why there was a positive association between health and the proportion of agricultural land. Food guilds refer to the food resources a bird exploits for subsistence. Disturbances within the landscape promote the presence of specialist species, while heterogeneous landscapes promote generalists.[54] Song sparrows are generalists as they are omnivorous birds (Table 1) so they are able to utilize various food resources within the landscape. Furthermore, these birds reproduce and survive well on or outside forest reserves.[30,55] These findings imply that song sparrows are a resilient species, as they can persist in changing land use configurations and efficiently use the resources present. Song sparrows' generalist nature may contribute to the modelling results that indicate this species was less severely impacted by increases in agriculture.[55] Considering food guilds provides a more holistic and species-centered perspective on how birds interact in complex landscape mosaics, and yet a bird's reproductive readiness presents an added stressor that may compound the impacts of changing landscapes.

III. Reproductive Readiness and Bird Health

Reproductive readiness was the variable that most strongly modulated a song sparrow's state of health (Table 3; Fig. 4). The relationship between the variables was positive, such that higher levels of reproductive readiness resulted in improved health. A bird's sexual maturity and state of reproduction is known to impact its health, particularly in terms of immunological stress. Although bird species have differing baseline levels of stress associated with reproduction,[56,57] higher levels of reproductive stress are strongly correlated with lower reproductive success and fledgling survival rates.[23,24,21,22] In taking a survivorship perspective, our findings would suggest that only the fittest song sparrows survived the breeding season.[58,59,60] This is because parents often bear the costs of anthropogenic landscape changes in order to supply for their young.[61]

Breeding seasons are loosely defined, so even when attempting to standardize and account for different levels of reproduction, the z-score calculation with cloacal protuberance and brood patch scores may not holistically capture the different levels of sexual activity between bird species. It may be that song sparrows were the only species that had finished their reproductive cycle, while spotted towhees, Oregon juncos, and house finches had just begun brooding when the sampling period was conducted. The index may not have a high enough granularity to distinguish these two life stages.

IV. Further Work

The limited sample size and spatial replication, alongside the species represented, limit the level of generalizability of the study. We can only discuss trends for the four study species, restricted to the study region in the Central Coast of California. Bird-landscape studies typically tend to implement larger spatiotemporal scales.[21,52,62] Our sample represented 13 farms sampled over a two-month period. Further site sampling within the Central Coast would allow for more cogent discussions on large-scale trends of bird health on agricultural lands in the region. It may also be useful to sample over larger spans of time, both during and outside the summer growing season. This wider sampling period would not only allow for more data to be modeled, but could also more rigorously account for differences in the reproductive cycles of species. Limitations in spatiotemporal reproducibility and our sample's characteristics require us to be cautious when discussing larger agricultural trends but also foment further research on bird-agriculture interactions.

CONCLUSION

The Central Coast of California is facing rapid agricultural change that needs to be quantified in order to assess its impacts on wildlife.[11] Within this study, bird health was impacted by landscape quality in an unexpected fashion. Higher proportions of agriculture resulted in better "quality" landscapes in terms of improved bird health for one species, song sparrows. Conversely, reproductive readiness most strongly drove bird health, where higher levels of readiness were associated with improved health for song sparrows. Our preliminary findings suggest that agriculturalists may be providing foraging and habitat resources to song sparrow communities, linked to marginally improved health for this species.[52,53] This finding is not to ignore the deleterious impacts humans have had on their surroundings, such as rapid deforestation and pollution of the biosphere. Nonetheless, our work encourages us to more critically assess and describe humans' impacts on their surroundings. Bird health was either slightly improved or not significantly impacted by landscape composition changes. We need to provide farmers with incentives to foment multi-use spaces that both provide critical resources for birds and maintain productivity. Thresholds for particular crops or land cover types may be a tactic to foment multi-use landscapes.[51] Similarly, it may be necessary to provide additional safeguards for birds at the peak of their reproductive cycle, such as by moving mowing and pesticide application dates, but further research is required to substantiate such policy prescriptions.[63]. A larger sample size, other bird species, and additional metrics of bird health should be considered to ascertain at a larger spatiotemporal scale how agricultural land use changes in the Central Coast are impacting avian community health. Farmers should be supported in becoming active ecosystem managers that improve landscape heterogeneity and avian health.

ACKNOWLEDGEMENTS

Dr. Claire Kremen for her mentorship. Dr. Daniel Karp and Dr. Elissa Olimpi's (UC Davis) for their support during summer 2018. Dr. Ruta Bandivadeka (UC Davis) for Giemsa-Wright staining training. PhD candidate Chris Hoover for modeling guidance. Sponsored Projects for Undergraduate Research (SPUR) and CNR travel grants for the funding.

REFERENCES

1. Railsback, S.D., & Johnson, M.D. (2014). Effects of land use on bird populations and pest control services on coffee farms. *Proceedings Of The National Academy Of Sciences, 111*(16), 6109-6114. doi: 10.1073/pnas.1320957111
2. Maas, B., Clough, Y., & Tscharntke, T. (2013). Bats and birds increase crop yield in tropical agroforestry landscapes. *Ecology Letters, 16*(12), 1480-1487. doi: 10.1111/ele.12194
3. Koh, L.P. (2008). Birds defend oil palms from herbivorous insects. *Ecological Applications, 18*(4), 821-825. doi: 10.1890/07-1650.1
4. Karp, D.S., Baur, P., Atwill, E.R., De Master, K., Gennet, S., & Iles, A. et al. (2015). The unintended ecological and social impacts of food safety regulations in California's Central Coast region. *Bioscience, 65*(12), 1173-1183. doi: 10.1093/biosci/biv152
5. Westerlund, F., Gubler, D., Duniway, J., Fennimore, S, Zalom, F., Westerdahl, B., & Larson, K. (2000). *Pest management evaluation for strawberries in California.* Pest Management Evaluation PMA Grant No. 99-0195 California Department of Pesticide Regulation.
6. Bihn, E., & Gravani, R. (2006). *Microbiology of Fresh Produce* (pp. 21-53). Washington, DC: ASM Press. doi: 10.1128/9781555817527.ch2
7. Clark, L., & Hall, J. (2006). Avian influenza in wild birds: Status as reservoirs, and risks to humans and agriculture. *Ornithologi-*

cal Monographs, (60), 3-29. doi: 10.2307/40166825
8. Wetzel, A.N., & LeJeune, J.T. (2006). Clonal dissemination of *Escherichia coli* O157:H7 subtypes among dairy farms in Northeast Ohio. *Applied And Environmental Microbiology, 72*(4), 2621-2626. doi: 10.1128/aem.72.4.2621-2626.2006
9. Park, S., Navratil, S., Gregory, A., Bauer, A., Srinath, I., & Jun, M. et al. (2013). Generic *Escherichia coli* contamination of spinach at the preharvest stage: Effects of farm management and environmental factors. *Applied And Environmental Microbiology, 79*(14), 4347-4358. doi: 10.1128/aem.00474-13
10. Karp, D.S., Gennet, S., Kilonzo, C., Partyka, M., Chaumont, N., Atwill, E.R., & Kremen, C. (2015). Comanaging fresh produce for nature conservation and food safety. *Proceedings Of The National Academy Of Sciences, 112*(35), 11126-11131. doi: 10.1073/pnas.1508435112
11. Baumgartner, J., & Lowell, K. (2016). *Co-managing farm stewardship with food safety GAPs and conservation practices: A grower's and conservationist's handbook* (pp. 1-5). Watsonville: Wild Farm Alliance.
12. Hallmann, C.A., Sorg, M., Jongejans, E., Siepel, H., Hofland, N., & Schwan, H. et al. (2017). More than 75 percent decline over 27 years in total flying insect biomass in protected areas. *PLoS ONE, 12*(10), e0185809. doi: 10.1371/journal.pone.0185809
13. Inger, R., Gregory, R., Duffy, J.P., Stott, I., Voříšek, P., & Gaston, K.J. (2014). Common European birds are declining rapidly while less abundant species' numbers are rising. *Ecology Letters, 18*(1), 28-36. doi: 10.1111/ele.12387
14. Fahrig, L. (2001). How much habitat is enough? *Biological Conservation, 100*(1), 65-74. doi: 10.1016/s0006-3207(00)00208-1
15. Perfecto, I., & Vandermeer, J. (2010). The agroecological matrix as alternative to the land-sparing/agriculture intensification model. *Proceedings Of The National Academy Of Sciences, 107*(13), 5786-5791. doi: 10.1073/pnas.0905455107
16. Lindenmayer, D., Hobbs, R.J., Montague-Drake, R., Alexandra, J., Bennett, A., & Burgman, M. et al. (2003). A checklist for ecological management of landscapes for conservation. *Ecology Letters, 11*(1), 78-91. doi: 10.1111/j.1461-0248.2007.01114.x
17. Buskirk, J. V. (2012). Permeability of the landscape matrix between amphibian breeding sites. *Ecology and Evolution, 2*(12), 3160–3167. https://doi.org/10.1002/ece3.424
18. Brady, M., Mcalpine, C., Possingham, H., J. Miller, C., & Baxter, G. (2011). Matrix is important for mammals in landscapes with small amounts of native forest habitat. *Landscape Ecology, 26*, 617–628. https://doi.org/10.1007/s10980-011-9602-6
19. Häkkilä, M., Tortorec, E.L., Brotons, L., Rajasärkkä, A., Tornberg, R., & Mönkkönen, M. (2017). Degradation in landscape matrix has diverse impacts on diversity in protected areas. *PLoS ONE, 12*(9), e0184792. https://doi.org/10.1371/journal.pone.0184792
20. Laurance, S.G., Jones, D., Westcott, D., Mckeown, A., Harrington, G., & Hilbert, D.W. (2013). Habitat fragmentation and ecological traits influence the prevalence of avian blood parasites in a tropical rainforest landscape. *PLoS ONE, 8*(10), e76227. doi: 10.1371/journal.pone.0076227
21. Kilgas, P., Tilgar, V., & Mänd, R. (2006). Hematological health state indices predict local survival in a small passerine bird, the great tit (*Parus major*). *Physiological And Biochemical Zoology, 79*(3), 565-572. doi: 10.1086/502817
22. Al-Murrani, W.K., Al-Rawi, I.K., & Raof, N.M. (2002). Genetic resistance to *Salmonella typhimurium* in two lines of chickens selected as resistant and sensitive on the basis of heterophil/lymphocyte ratio. *British Poultry Science, 43*(4), 501-507. doi: 10.1080/0007166022000004408
23. Bienzle, D., Pare, J.A., & Smith, D.A. (1997). Leukocyte changes in diseased non-domestic birds. *Veterinary Clinical Pathology, 26*(2), 76-84. doi: 10.1111/j.1939-165x.1997.tb00715.x
24. Lobato, E., Moreno, J., Merino, S., Sanz, J.J., & Arriero, E. (2005). Haematological variables are good predictors of recruitment in nestling pied flycatchers (*Ficedula hypoleuca*). *Écoscience, 12*(1), 27-34. doi: 10.2980/i1195-6860-12-1-27.1
25. United States Department of Agriculture Farm Service Agency. (2018). NAIP Imagery. Retrieved from https://www.fsa.usda.gov/programs-and-services/aerial-photography/imagery-programs
26. Environmental Systems Research Institute. (2018). ArcGIS Desktop and Spatial Analyst Extension (Version 10.6) [Computer software]. Retrieved May 30, 2018 from https://www.arcgis.com/index.html
27. Google. (2018). Google Maps [Computer software]. Retrieved November 22, 2018 from https://www.google.com/maps
28. Dobkin, D. (1990). SPOTTED TOWHEE (*Pipilo maculatus*). In D.C. Zeiner, W. F. Laudenslayer Jr., K.E. Mayer, & M. White (Eds.), *California's Wildlife* (p. B483). Sacramento, CA: California Department of Fish and Wildlife.
29. Granholm, S. (1990). HOUSE FINCH (*Haemorhous mexicanus*). In D.C. Zeiner, W. F. Laudenslayer Jr., K.E. Mayer, & M. White (Eds.), *California's Wildlife* (p. B538). Sacramento, CA: California Department of Fish and Wildlife.
30. Granholm, S. (1990). SONG SPARROW (*Melospiza melodia*). In D.C. Zeiner, W. F. Laudenslayer Jr., K.E. Mayer, & M. White (Eds.), *California's Wildlife* (p. B505). Sacramento, CA: California Department of Fish and Wildlife.
31. Green, M. (1990). DARK-EYED JUNCO (*Junco hyemalis*). In D.C. Zeiner, W. F. Laudenslayer Jr., K.E. Mayer, & M. White (Eds.), *California's Wildlife* (p. B512). Sacramento, CA: California Department of Fish and Wildlife.
32. Ormerod, S.J., & Watkinson, A.R. (2000). Editors' introduction: Birds and agriculture. *Journal Of Applied Ecology, 37*(5), 699-705. doi: 10.1046/j.1365-2664.2000.00576.
33. Ralph, C.J., Dunn, E.H., Peach, W.J., & Handel, C.M. (2004). Recommendations for the use of mist nets for inventory and monitoring of bird populations. *Studies in Avian Biology, 29*, 187-196.
34. Pyle, P. (1997). *Identification guide to North American birds, part I: Columbidae to Ploceidae* (p. ix-732). Bolinas, CA: Slate Creek Press.
35. Morishita, T.Y., Aye, P., Ley, E.C., & Harr, B.S. (1999). Survey of pathogens and blood parasites in free-living Passerines. *Avian Diseases, 43*(3), 549-552. doi: 10.2307/1592655
36. Valera, F., Hoi, H., & KrištÍn, A. (2005). Parasite pressure and its effects on blood parameters in a stable and dense population

of the endangered lesser grey shrike. *Biodiversity And Conservation, 15*(7), 2187-2195. doi: 10.1007/s10531-004-6902-z
37. Owen, J.C. (2011). Collecting, processing, and storing avian blood: A review. *Journal Of Field Ornithology, 82*(4), 339-354. doi: 10.1111/j.1557-9263.2011.00338.x
38. Eberhard, M.L., & Lammie, P.J. (1991). Laboratory diagnosis of filariasis. *Clinics In Laboratory Medicine, 11*(4), 977-1010. doi: 10.1016/s0272-2712(18)30531-6
39. Ciesla, B. (2012). *Hematology in practice* (pp. 1-348). Philadelphia, PA: F.A. Davis.
40. Godfrey, R.D., Fedynich, A.M., & Pence, D.B. (1987). Quantification of hematozoa in blood smears. *Journal Of Wildlife Diseases, 23*(4), 558-565. doi: 10.7589/0090-3558-23.4.558
41. Merino, S., Potti, J., & Fargallo, J. (1997). Blood parasites of passerine birds from Central Spain. *Journal Of Wildlife Diseases, 33*(3), 638-641. doi: 10.7589/0090-3558-33.3.638
42. Atkinson, C. T. (1991). Hemosporidiosis. In M. Friend & J. C. Franson (Eds.), *Field manual of wildlife diseases: General field procedures and diseases of birds* (pp. 193-200). Reston, VA: U.S. Geological Survey. doi: 10.3133/tm15
43. Bartlett, C. M. (2008). Filarioid nematodes. In C. T. Atkinson, N. J. Thomas, & D. B. Hunter (Eds.), *Parasitic diseases of wild birds* (pp. 439-462). Hoboken, NJ: John Wiley & Sons, Inc. doi: 10.1002/9780813804620
44. R Core Team. (2018). R: A language and environment for statistical computing (R package version 1.3.5) [Computer software]. Vienna, Austria: R Foundation for Statistical Computing.
45. Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal Of Statistical Software, 67*(1). doi: 10.18637/jss.v067.i01
46. Zeileis, A., & Hothorn, T. (2002). Diagnostic checking in regression relationships. *R News, 2*(3), 7-10.
47. Hlavac, M. (2015). Stargazer: Well-formatted regression and summary statistics table (R package version 1.3.5) [Computer software]. Retrieved August 30, 2019 from https://CRAN.R-project.org/package=stargazer
48. Wickham, H. (2016). Ggplot2: Elegant graphics for data analysis (R package version 1.3.5) [Computer software]. New York, NY: Springer-Verlag.
49. Sleeter, B., Wilson, T.S., Soulard, C.E., & Liu, J. (2011). Estimation of late twentieth century land-cover change in California. *Environmental Monitoring And Assessment, 173*(1-4), 251-266. doi: 10.1007/s10661-010-1385-8
50. Donald, P.F., Sanderson, F.J., Burfield, I.J., & van Bommel, F.P. (2006). Further evidence of continent-wide impacts of agricultural intensification on European farmland birds, 1990–2000. *Agriculture, Ecosystems & Environment, 116*(3), 189-196. doi: 10.1016/j.agee.2006.02.007
51. Jerrentrup, J.S., Dauber, J., Strohbach, M.W., Mecke, S., Mitschke, A., Ludwig, J., & Klimek, S. (2017). Impact of recent changes in agricultural land use on farmland bird trends. *Agriculture, Ecosystems & Environment, 239*, 334-341. doi: 10.1016/j.agee.2017.01.041
52. Moreira, F., Pinto, M. J., Henriques, I., & Marques, A. (2005). The importance of low-intensity farming systems for fauna, flora and habitats protected under the European birds and habitats directives: Is agriculture essential for preserving biodiversity in the Mediterranean region? In A. R. Burk (Ed.), *Trends in Biodiversity Research* (pp. 117-145). Hauppauge, NY: Nova Science Publishers.
53. Civantos, E., Monteiro, A.T., Gonçalves, J., Marcos, B., Alves, P., & Honrado, J.P. (2018). Patterns of landscape seasonality influence passerine diversity: Implications for conservation management under global change. *Ecological Complexity, 36*, 117-125. doi: 10.1016/j.ecocom.2018.07.001
54. Jeliazkov, A., Mimet, A., Chargé, R., Jiguet, F., Devictor, V., & Chiron, F. (2016). Impacts of agricultural intensification on bird communities: New insights from a multi-level and multi-facet approach of biodiversity. *Agriculture, Ecosystems & Environment, 216*, 9-22. doi: 10.1016/j.agee.2015.09.017
55. Marzluff, J., Clucas, B., Oleyar, M., & DeLap, J. (2015). The causal response of avian communities to suburban development: A quasi-experimental, longitudinal study. *Urban Ecosystems, 19*(4), 1597-1621. doi: 10.1007/s11252-015-0483-3
56. Hawkey, C.M., Hart, M.G., & Samour, H.J. (1985). Normal and clinical haematology of greater and lesser flamingos (*Phoenicopterus roseus* and *Phoeniconaias minor*). *Avian Pathology, 14*(4), 537-541. doi: 10.1080/03079458508436256
57. Maxwell, M.H., & Robertson, G.W. (1998). The avian heterophil leucocyte: A review. *World's Poultry Science Journal, 54*(2), 155-169. doi: 10.1079/wps19980012
58. Song, Z., Lou, Y., Hu, Y., Deng, Q., Gao, W., & Zhang, K. (2016). Local resource competition affects sex allocation in a bird: Experimental evidence. *Animal Behaviour, 121*, 157-162. doi: 10.1016/j.anbehav.2016.08.023
59. Paxton, E.H., Durst, S.L., Sogge, M.K., Koronkiewicz, T.J., & Paxton, K.L. (2017). Survivorship across the annual cycle of a migratory passerine, the willow flycatcher. *Journal Of Avian Biology, 48*(8), 1126-1131. doi: 10.1111/jav.01371
60. Guindre-Parker, S., & Rubenstein, D.R. (2018). No short-term physiological costs of offspring care in a cooperatively breeding bird. *The Journal Of Experimental Biology, 221*(21), jeb186569. doi: 10.1242/jeb.186569
61. Kight, C.R., & Swaddle, J.P. (2007). Associations of anthropogenic activity and disturbance with fitness metrics of eastern bluebirds (*Sialia sialis*). *Biological Conservation, 138*(1-2), 189-197. doi: 10.1016/j.biocon.2007.04.014
62. Dayananda, S. K., Goodale, E., Lee, M., Liu, J., Mammides, C., Quan, R., Slik, J.W., & Sreekar, R. et al. (2016). Effects of forest fragmentation on nocturnal Asian birds: A case study from Xishuangbanna, China. *Zoological Research, 37*(3), 151-158. doi: 10.13918/j.issn.2095-8137.2016.3.151
63. Grice, P., Evans, A., Osmond, J., & Brand-Hardy, R. (2004). Science into policy: The role of research in the development of a recovery plan for farmland birds in England. *Ibis, 146*(s2), 239-249. doi: 10.1111/j.1474-919x.2004.00359.x

Introducing an Anti-Terminator Paralog Gene to Induce Production of Natural Products in *Clostridium* Species

By: Alexander Leung; Research Sponsor (PI): Dr. Wenjun Zhang

ABSTRACT

LoaP is a class of proteins that has the potential to induce bacterial species into creating natural products such as antibiotics. LoaP can be placed as an insert into bacterial genomes, where it can potentially drive gene expressions and subsequently activate secondary biosynthetic pathways. Here, a construct was cloned by placing LoaP as an insert into the plasmid of *Clostridium beijerinckii*. However, this LoaP overexpression strain failed to produce a compound that its wild-type creates. LoaP appeared to decrease production rather than increase production of certain metabolites, as the wild-type strain gave rise to more natural products than the transformant LoaP strain.

Major, Year, and Department: Bioengineering, 2017, Department of Chemical and Biomolecular Engineering.

Keywords: B-598, nusG, LoaP, Pbdh, paralog, anti-terminator, secondary metabolites, XCMS, mass spectrometry, transformation.

INTRODUCTION

Bacteria have been studied for their ability to form natural products that can be useful for drug discovery and other applications, such as dyes and other synthetic materials. Some natural products come in the form of secondary metabolites, which have been observed from *Clostridium*.[1] The genomes of different *Clostridium* species have been studied for secondary metabolite sequences, such as polyketide synthase and non-ribosomal peptide synthetase genes, which are widespread amongst these species.[1] These genes are only activated by particular stimuli, such as environmental conditions found in aqueous soil extract.[1] Since the necessary stimulatory environmental conditions are difficult to recreate in a laboratory setting, bacteria are often incapable of producing secondary metabolites *in vitro* due to the dormancy of usual secondary metabolite biosynthetic pathways. In a previous study, overexpression of anti-terminator genes corresponding to biosynthetic gene clusters in *Clostridium celluloyticum* yielded novel natural products such as antibiotics known as closthioamides and related thioamides.[1]

In the former study, an anti-terminator gene was used to induce a secondary metabolite pathway. This class of proteins is referred to as LoaP. LoaP is a paralog from the *nusG* family of proteins, which tend to play an important role in transcription elongation.[2] Anti-terminator proteins prevent termination of transcription at terminator sites, and allow for expression in the sequences beyond them.[3] In the absence of an anti-terminator protein, RNA polymerase, or RNAP, will cease transcription at a terminator. In this context, the anti-terminator serves to increase the overall rate of transcription.[1] In RNAP complexes, *nusG* moves transcription forward past termination sites (known as p-dependent sites), which increases RNA polymerase processivity and helps with both making transcription efficient[1] and driving the creation of polycistronic mRNA.[1] Overall, this process helps the progression of protein production. We hypothesized that this may drive a different pattern of gene expression, which may in turn drive some biosynthetic gene pathways.

Overexpression of these anti-terminator genes in biosynthetic clusters may give rise to novel natural products produced by *Clostridium beijerinckii*, which is also known as B-598. In B-598, a paralog of *nusG* called LoaP was discovered.[2] If the B-598 paralog were to be excised from B-598's bacterial genome and placed as an insert into B-598's genome under a different constitutive promoter, it should activate a biosynthetic gene cluster and, potentially, create cryptic natural products. In our study, the LoaP protein was expressed under a constitutive promoter called Pbdh, which is from a different bacterial species known as *Clostridium saccharoperbutylacetonicum*.[2] Pbdh typically regulates butanol dehydrogenase. Promoters are sequences of DNA where transcription is initiated. Promoters are also important in regulating expression of downstream genes. There is a possibility that anti-terminator proteins will not be expressed in certain promoters. The promoter Pbdh was selected due to its constitutive nature, which ensures that it is continuously active and, therefore, does not need to be induced. Once the anti-terminator proteins are expressed under this promoter, a cryptic biosynthetic pathway can be driven in B-598 to yield novel natural products such as secondary metabolites. Figure 1 shows that the sequence used was associated with Cluster 10, which is a biosynthetic cluster that was hypothesized to be responsible for secondary metabolite activity. From this, it is possible that overexpression of LoaP in B-598 using a constitutive promoter may yield novel natural products.

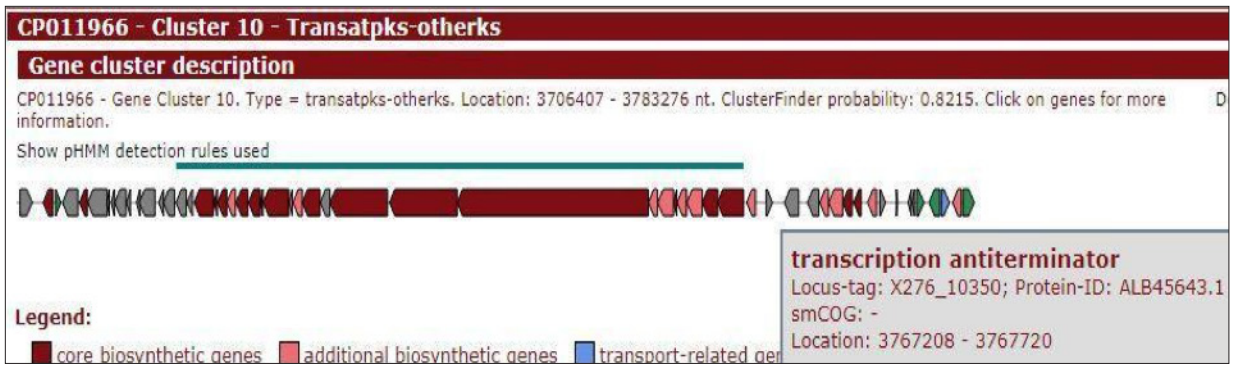


Figure 1. B-598 derived LoaP associated with Cluster 10 and used in the experiment.[8]

MATERIALS AND METHODS

I. Cloning and Gibson Assembly

Cloning was done by inserting the B-598 *nusG* into the backbone of B-598 under the Pbdh promoter. Lab restriction enzymes are used to cut out B-598's native *nusG* and replace it with B-598's LoaP under a different promoter. The backbone in this procedure is named PJL5. Its native *nusG* is cut out with BSP/MSSI, and generates sticky ends to facilitate insertion of the B-598's LoaP. B-598's paralog was also amplified via PCR to generate the plasmid DNA sequence of B-598. Both the PCR product and restriction digests were run in a gel electrophoresis procedure that separates out the bands so that they may be cut out of the gel and purified in a gel extraction procedure. These same restriction enzymes were also used to cut out LoaP from B-598's plasmid sequence to make it homologous to the cut-out site in PJL5. B-598's LoaP sequence was then ligated to the PJL5 plasmid, and the new plasmid was named PJL9.

However, it was discovered that restriction digest enzymes were not needed as the Gibson Assembly procedure helped to create sticky ends to avoid blunt-end DNA ligation. The Gibson Assembly procedure has the same principle as the restriction enzyme procedure and it was used in this experiment to generate the constructs. Without Gibson Assembly, in blunt-end DNA ligation, the LoaP insert may not have necessarily ligated to the correct site and may have ligated to any other available homologous site. Luckily, PJL9 was successfully constructed using Gibson Assembly. The theoretical construct created was seen in a model of the plasmid shown in Fig. 2. The sequence was also confirmed to be correct through a sequencing facility.

After PJL9 was generated, it was inserted into a bacterial colony via a transformation procedure. 5 µL of the plasmid's DNA was added to one 50 µL aliquot of *E. coli* XL-1 Blue competent cells. The mixture was then incubated on ice for 30 minutes, before undergoing heat shock. In heat shock, the mixture was placed in a 42°C

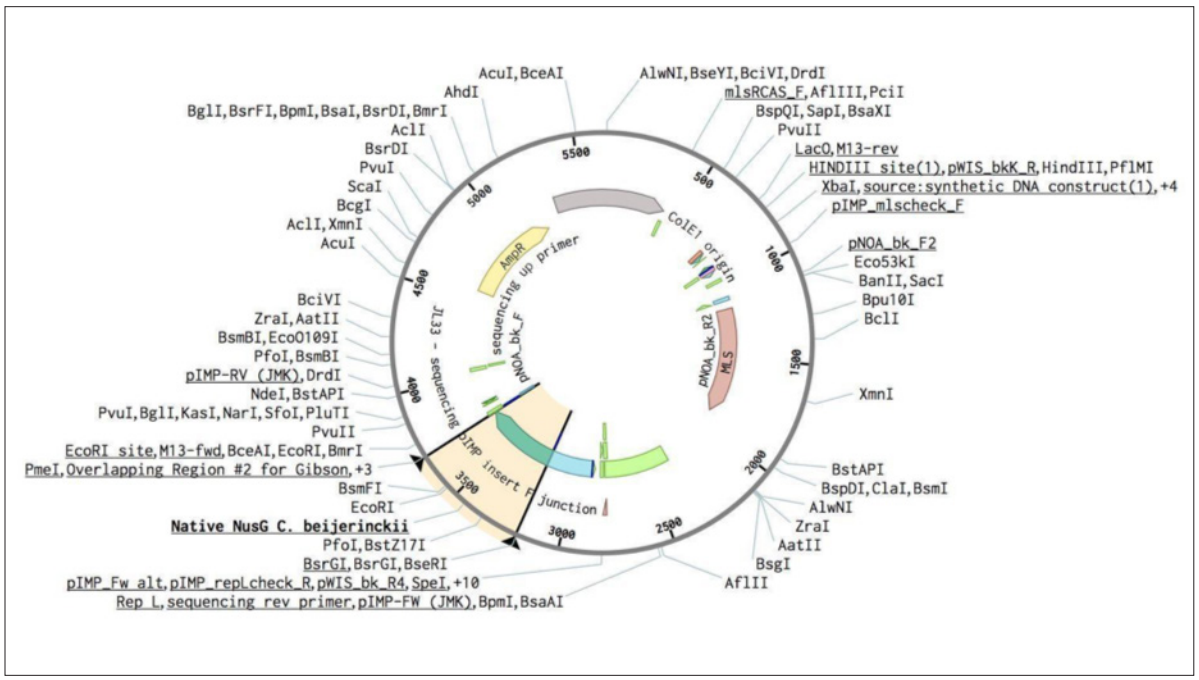


Figure 2.* nusG *B-598 insert (LoaP) placed into the backbone DNA sequence of B-598 to yield the PJL9 sequence.

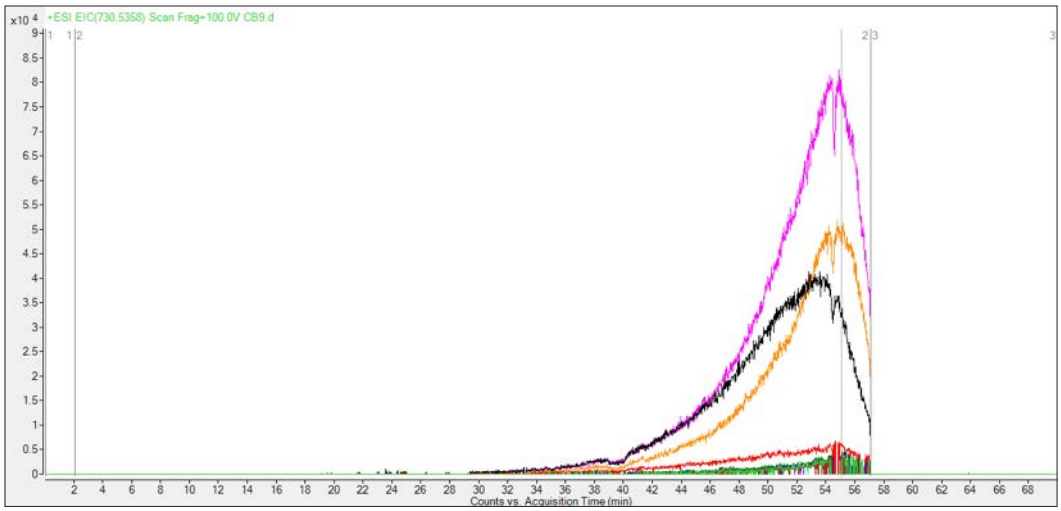


Figure 3. Chromatogram comparing B-598 LoaP overexpression strains versus wild-type. *Orange, black, and pink curves represent the wild-type bacteria, while red, blue, and green curves represented the transformants with the LoaP. The graph indicates that the 730.5358 m/z mass eluted from the column at fifty-five minutes.*

bath for 45 seconds. During the heat-shock step of the transformation procedure, *E. coli* XL-1 Blue competent cells incorporated the plasmid. The constructed plasmid was amplified as the colony was grown on an LB plate with a Carbenicillin antibiotic. Colonies on the media surface were picked and grown in liquid LB media in sterile tubes until the media showed cloudiness or bacterial growth. The DNA was extracted from the XL-1 Blue competent cells via a Zyppy Plasmid mini-prep procedure to obtain the sequence of the constructed plasmids. In order to confirm that the sequence is correct, test digests were run on the extracted DNA with a HindIII restriction enzyme. Constructs with the correct bands were prepared for sequencing and submitted to a sequencing facility to confirm if the insert had successfully ligated to the backbone.

II. Transformation into B-598

After extracting the DNA from *E. coli* cells, running test digests, and confirming the sequences are correct through the sequencing facility, the final PJL9 plasmid was incorporated into B-598 colonies. The lab had developed a specialized transformation procedure specifically that was implemented for this experiment.[4]

III. Fermentation and Chemical Extraction

The transformants were grown in TYA media with antibiotic erythromycin at 40 μg/mL for four days.[5] In addition, a wild-type bacterial control was grown alongside the transformants (also known as LoaP overexpression strains) in order to serve as a basis for comparison. Both colonies were grown in triplicate cultures, meaning there were three samples for the transformants and three samples for the wild-type bacteria in order to ensure that there are enough data to compare and demonstrate consistency. After 4 days, the transformants and wild-type bacteria were chemically extracted with EtOAc. EtOAc was added to the liquid cultures in a 1:1 ratio, before being vortexed. Afterwards, the mixture was spun down, the top layer was removed, then dried with nitrogen lines. The resulting extract was then suspended in HPLC-grade methanol and run in a quadrupole time-of-flight (QTOF) mass spectrometer for analysis.

IV. QTOF/XCMS

Compounds were separated out through MS-Reverse-Phase HPLC. The mass-to-charge ratio of both the transformants and

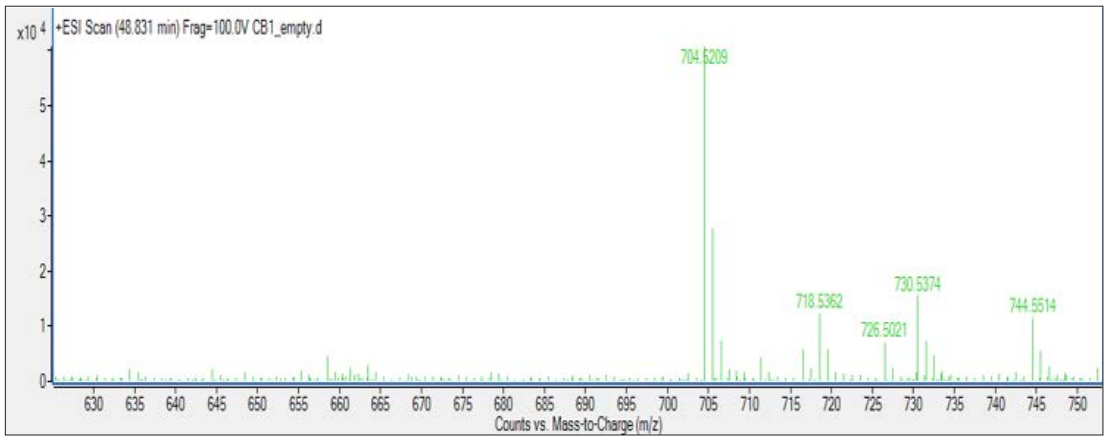


Figure 4. Counts versus mass charge confirming the presence of the 730.5358 m/z in the wild-type bacteria.

name	fold	log2fold	tstat	pvalue	qvalue	mzmed
M731T50	905.8176	-9.8230	-2.9431	0.0986	0.3131	730.5358

Table 1. Statistical variables from XCMS procedure corresponding to the 730.5358 m/z.

wild-type colonies were analyzed to see what compounds are being produced. However, in order to narrow down whether the transformants have a new compound or are missing compounds, an XCMS procedure was implemented. XCMS is a data-analysis approach for mass spectrometry data that examines spectrometry variables such as peak matching, peak detection, retention time alignment, etc.[7] In this study, XCMS generated a table with 1 row for every single mass peak detected. It then calculated the average size of that peak for each replicate and took the average of these values. The most significant variables in this study are median retention times, fold changes, and p-values. Median retention times signify how long the peak was observed in the sample. Since the machine has a 55-minute gradient, only a range of 2 to 57 minutes was considered.[7] Fold change reflects the difference between the average LoaP and control value and a 10-fold change is usually significant.[7] P-value indicates how statistically significant are the results, 0.01 or 0.05 being the cutoff. Using these three variables, the rows were narrowed down to obtain only significant results. Each row corresponds to a new peak, which then means that either a new mass was found or a typical mass is missing. After the rows were obtained, we returned to the raw data to analyze significant results.

RESULTS

The mass-to-charge ratio that was significant as identified from the XCMS method was 730.5358 m/z. This value satisfies the XCMS procedure by having a p-value larger than our cutoff of 0.01 and 0.05, while also having nearly a 10-fold change as indicated by Table 1. The chromatogram indicated a possible major significant difference between the LoaP overexpression and control samples as seen in Fig. 3. There are six curves in total to signify three wild-type samples and three LoaP samples. The LoaP curves were close to the baseline, indicating that the mass was not present in all three LoaP overexpression samples. Meanwhile, all wild-type curves showed high counts, indicating that the mass was present in a pretty high quantity as opposed to the LoaP overexpression samples. Another QTOF data file comparing counts to mass-to-charge ratio in Fig. 4 serves to confirm that the mass is present in the wild-type. Overall, the results indicate that a nonpolar compound was produced in the wild-type bacteria, but not the B-598 derived LoaP overexpression samples. This serves to show that the mutant strain was eliminating a product that would usually be created in the wild-type colonies.

DISCUSSION

The results obtained were unexpected, and contradicted the original hypothesis. Based on previous studies[2] we expected that inserting anti-terminator genes into bacterial genomes could change gene expression and activate biosynthetic pathways that can create new secondary metabolites. However, the results indicated that an unknown compound that was produced in the wild-type was not produced by the B-598 derived LoaP samples. In fact, it seemed as if the anti-terminator protein may have decreased production of one of the compounds usually produced. However, re-analysis of the B-598 genome indicated there are actually two LoaP sequences within the genome. The sequence used in the experiment was hypothesized to be associated with secondary metabolite biosynthetic gene clusters. Perhaps the loss of one product could be due to the fact that the sequence had some minor secondary effects that inhibited the creation of that one product. What exactly are the secondary effects is un-

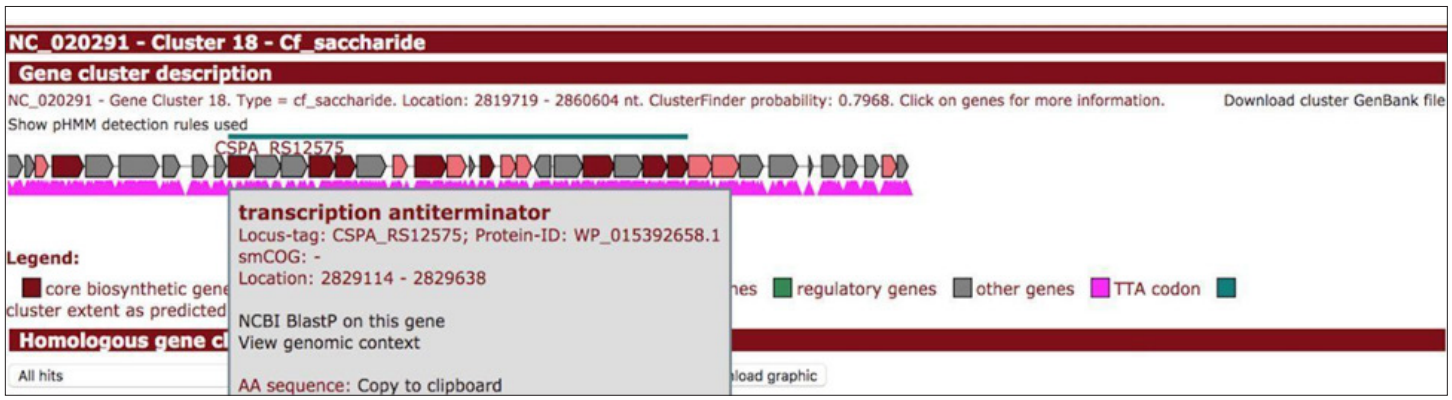


Figure 6. B-598 derived LoaP copy associated with Cluster 18, a saccharide biosynthetic cluster for future approaches.[8]

known at this time, although it can be theorized that LoaP served as a negative regulator for that product. Namely, it would prevent a certain gene from being expressed, thereby inhibiting the pathway that would otherwise create the product. In the future, the copy of LoaP found (associated with Cluster 18, Fig. 6), will be used instead to see if that induces the B-598 strain to create new natural products. Analysis of the cluster indicates that Cluster 18 might be a biosynthetic pathway for saccharides, which are primary metabolites. Regardless, overexpressing this copy might yield more fruitful results than the B-598 sequence in Cluster 10. In future work, the cloning procedure could be repeated for this homolog and inserted into B-598 to be overexpressed under the same promoter to test its ability to induce secondary metabolites. We hypothesize that a LoaP copy that is associated with a saccharide cluster could induce cryptic biosynthetic pathways of new metabolites when inserted into B-598. Another alternative is to search for other proteins often associated with heavy biosynthetic gene clusters (preferably anti-terminators) and undergo the same procedure to study secondary metabolite effects. Preferably, these proteins need to be positive regulators so that they simply can be expressed under the same promoters using the same cloning procedure mentioned above.

CONCLUSION

Theoretically, overexpression of the *nusG* vector in bacteria would help to induce bacteria into creating natural products such as antibiotics. However, the LoaP overexpression strain demonstrated the opposite of what was expected. In the chromatogram results, the wild-type bacteria were discovered to contain a peak with a mass that the transformant strain lacked. This runs contrary to the hypothesis that the insertion of the B-598 derived LoaP into B-598 would induce biosynthetic pathways in B-598, which could in turn create useful products like closthioamide.[2] The sequence inserted into B-598's decreased production in the bacteria. This could be due to the fact that the sequence used had some secondary effects that inhibited creation of the expected product. In the future, another LoaP sequence associated with a saccharide gene cluster can be inserted into the B-598 genome and studied for secondary metabolite activity. Other proteins similar to LoaP can also be used in a similar fashion to induce secondary metabolite activity.

ACKNOWLEDGEMENTS

I want to thank Professor Zhang and Jeffrey Li for allowing me to work under their facility and to conduct experiments. The Zhang Lab in University of California, Berkeley has offered me abundant resources and guidance in helping me to conduct scientific research of my own.

REFERENCES

1. Behnken, S., Lincke, T., Kloss, F., Ishida, K., & Hertweck, C. (2012). Antiterminator-Mediated Unveiling of Cryptic Polythioamides in an Anaerobic Bacterium. *Angewandte Chemie International Edition, 51*(10), 2425-2428. doi: 10.1002/anie.201108214
2. Goodson, J. R., Klupt, S., Zhang, C., Straight, P., & Winkler, W. C. (2017). LoaP is a broadly conserved antiterminator protein that regulates antibiotic gene clusters in Bacillus amyloliquefaciens. *Nature microbiology, 2*(5), 17003. doi: 10.1038/nmicrobiol.2017.3
3. Clark, D., & Pazdernik, N. (n.d.). Antitermination. *ScienceDirect*. Retrieved from https://www.sciencedirect.com/topics/biochemistry-genetics-and-molecular-biology/antitermination
4. Herman, N. A., Li, J., Bedi, R., Turchi, B., Liu, X., Miller, M. J., & Zhang, W. (2017). Development of a high-efficiency transformation method and implementation of rational metabolic engineering for the industrial butanol hyperproducer Clostridium saccharoperbutylacetonicum strain N1-4. *Appl. Environ. Microbiol., 83*(2), e02942-16. doi: 10.1128/AEM.02942-16
5. Chang, Y. Y., & Cronan, J. E. (1982). Mapping nonselectable genes of Escherichia coli by using transposon Tn10: location of a gene affecting pyruvate oxidase. *Journal of bacteriology, 151*(3), 1279-1289. doi: 10.1007/BF00328074
6. Division of Endocrinology, Metabolism and Lipid Research. (2013, Nov. 19). Time-of-Flight Fundamentals. *Biomedical Mass Spectrometry Resource*. Retrieved from https://msr.dom.wustl.edu/time-of-flight-fundamentals/
7. Smith, C. A., Want, E. J., O'Maille, G., Abagyan, R., & Siuzdak, G. (2006). XCMS: processing mass spectrometry data for metabolite profiling using nonlinear peak alignment, matching, and identification. *Analytical chemistry, 78*(3), 779-787. doi: 10.1021/ac051437y
8. Weber, T., Blin, K., Duddela, S., Krug, D., Kim, H. U., Bruccoleri, R., ... & Breitling, R. (2015). antiSMASH 3.0—a comprehensive resource for the genome mining of biosynthetic gene clusters. *Nucleic acids research, 43*(W1), W237-W243. doi: 10.1093/nar/gkv437